Quantum Mechanics III

Foundations

Prerequisites: *Linear Algebra for Physics*, *QM I: Finite Systems* and *QM II: Continuous Systems*.

The first two quantum mechanics courses supplied rules for predicting readings. This course examines the structure of those rules and the assumptions behind them. It then asks why a successful probability rule does not by itself explain an individual measurement outcome.

The sequence begins with states as expectation rules, including systems with infinitely many levels. Symmetry then constrains closed-system evolution. Composite systems lead to channels and open-system dynamics. Effects give probabilities for general apparatus readings; instruments also specify the state after a reading. These distinctions make the measurement problem precise before we compare proposed answers to it.

The lectures divide this argument by topic. They need not be completed in one sitting. Work through the examples and self-checks as they occur, then try the accompanying exercise sheet. Consult the separate worked answers after attempting the problems if you have that booklet. Public materials include study hints and self-check answers; the website does not publish the worked-answer booklet. The complete edition includes physics applications and self-check answers. The core edition omits those aids and is useful for reference after a first read. It is not the recommended first route through the course.

An untagged result is proved using the prerequisites and the stated standing facts. A result marked *sketched* gives the central argument with some steps omitted or with cited inputs; a result marked *cited* is used with a source but without its proof. These distinctions apply to the mathematical statements, not to whether an interpretation of quantum mechanics has been established. The final lectures separate proved consequences of a model from the physical and interpretive premises that model adds.

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Part I

States

A preparation supplies probabilities for readings. We first describe everything those probabilities say about a state, then distinguish the state from its possible preparations. Extending events from projections to effects reveals which probability assumptions recover the density-operator rule.

Lecture 1

What a state is

Objectives. *A state assigns expectations to observables. In a separable Hilbert space, positivity, normalisation and countable additivity on orthogonal events make this rule a density operator. This includes infinitely many populated levels, but excludes an exactly sharp position.*

- Prove that normal expectation rules are exactly density operators.
- Prove that no density operator assigns an exactly sharp position.
- Compute a thermal oscillator's level probabilities and mean energy.

The inner product $\langle \varphi, \psi \rangle$ is antilinear in its first argument and linear in its second. The space $\mathcal{H}$ is a Hilbert space, whereas H denotes a Hamiltonian. The expectation rule is ω; angular frequency is ν. We use E for an event projection and reserve P for momentum. Tensor-factor order will be stated when a composite system is introduced.

Facts of analysis used in this course. The standing facts S1–S4 are recalled from *QM II*, §0.4. We also accept S5 and S6 below without proof. These facts specify which analysis may be used in an otherwise complete proof.

S1 The Fourier transform is unitary on $L^2(\mathbb{R}^n)$ and has the Fourier inversion formula, with QM-II's normalisation. For general L^2 inputs, inversion is understood in the L^2 sense.

S2 The Fourier functions $e^{2\pi i k x/L}/\sqrt{L}$, $k \in \mathbb{Z}$, and the sine functions $\sqrt{2/L}\sin(n\pi x/L)$, $n \geq 1$, are complete orthonormal bases of $L^2(0, L)$, where $L > 0$.

S3 Products of functions from orthonormal bases of $L^2(\mathbb{R})$ form orthonormal bases of $L^2(\mathbb{R}^2)$ and $L^2(\mathbb{R}^3)$.

S4 Smooth functions of compact support are dense in L^2 of an interval and of $\mathbb{R}^n$. Schwartz functions are dense in $L^2(\mathbb{R}^n)$ as well.

S5 A compact self-adjoint operator on a separable Hilbert space has an orthonormal eigenbasis, including a basis of its kernel. Its nonzero eigenvalues have finite multiplicity and can accumulate only at zero.

S6 Every bounded positive operator B has a unique bounded positive square root $\sqrt{B}$, with $(\sqrt{B})^2 = B$.

1.1 A rule for expectations

In *QM I*, a preparation chose a unit vector ψ_k with probability p_k. Every bounded observable A then had mean $\sum_k p_k \langle \psi_k, A\psi_k \rangle$. The density matrix collected those means into $\mathrm{Tr}(\rho A)$ (Definition 3.3 and Theorem 3.4); different preparations could give the same matrix. *QM II* treated a particle on a line by unit vectors of $L^2(\mathbb{R})$ and closed with a question: what replaces the density matrix when the space has infinitely many dimensions?

Start with the means themselves. We work on a nonzero separable complex Hilbert space $\mathcal{H}$: it has a finite or countable orthonormal basis (ONB). Write $B(\mathcal{H})$ for its bounded

operators. A bounded observable is a self-adjoint member of this space. An event is an orthogonal projection E; its expectation is its probability. Even for an unbounded observable, such as position Q, its spectral projections $E_Q(I)$ belong to $B(\mathcal{H})$. Knowing their probabilities determines the distribution of its readings. Its mean may still be infinite or undefined.

It is convenient to assign a complex number to every bounded operator: write $A = C + \mathrm{i}D$ with $C = (A + A^\dagger)/2$, $D = (A - A^\dagger)/(2\mathrm{i})$ self-adjoint, and extend the expectation by complex linearity.

Definition 1.1 (An expectation rule)**.** An *algebraic state* is a complex-linear map $\omega : B(\mathcal{H}) \to \mathbb{C}$ such that $\omega(B) \geq 0$ for every positive operator B, and $\omega(\mathbb{1}) = 1$. Here $B \geq 0$ means B is bounded self-adjoint and $\langle \psi, B\psi \rangle \geq 0$ for every $\psi \in \mathcal{H}$.

The letter ω denotes an expectation rule, never a frequency; our angular frequency is ν. Positivity means that a nonnegative observable has a nonnegative mean. Linearity is a requirement on expectations, including those of noncommuting observables; it does not assert that they have simultaneous values. We will add one continuity requirement before calling such a rule a *state*.

Lemma 1.2 (Cauchy–Schwarz for an expectation rule)**.** *For any algebraic state and $A, B \in B(\mathcal{H})$,*

$$|\omega(B^\dagger A)|^2 \leq \omega(B^\dagger B)\omega(A^\dagger A), \qquad |\omega(A)| \leq \|A\|\,. \tag{1.1}$$

Proof. For self-adjoint C, the inequalities $-\|C\|\,\mathbb{1} \leq C \leq \|C\|\,\mathbb{1}$ show that $\omega(C)$ is real. Decomposing A into its self-adjoint real and imaginary parts gives $\omega(A^\dagger) = \overline{\omega(A)}$. Put $a = \omega(A^\dagger A)$, $c = \omega(B^\dagger A)$ and $d = \omega(B^\dagger B)$. Positivity for every $z \in \mathbb{C}$ gives

$$0 \leq \omega((A + zB)^\dagger (A + zB)) = a + 2\,\mathrm{Re}(\bar{z}c) + |z|^2 d.$$

If $d > 0$, choose $z = -c/d$. Then $0 \leq a - |c|^2/d$, so $|c|^2 \leq ad$, as claimed. If $d = 0$, choose $z = -tc$ with $t > 0$. The expression becomes $a - 2t|c|^2$, which cannot stay nonnegative for all t unless $c = 0$. Thus the inequality also holds when the denominator would vanish. Finally take $B = \mathbb{1}$ and use $0 \leq A^\dagger A \leq \|A\|^2\,\mathbb{1}$. ∎

Thus $0 \leq \omega(E) \leq 1$, and finite additivity of event probabilities already follows from linearity. Countable additivity will be a separate condition, not a consequence of this lemma.

1.2 Positive trace class

The finite-dimensional trace is a sum of diagonal entries. In infinite dimension a positive operator's sum may diverge: the identity has one on every diagonal entry. We first build the part of trace theory needed for states. The two standing facts used here are S5, the spectral theorem for compact self-adjoint operators, and S6, existence of a positive square root. They are stated in the standing-fact list at the start of this lecture.

Definition 1.3 (Positive trace class and density operators)**.** A positive bounded operator ρ is *positive trace class* if, for one ONB (e_k), $\sum_k \langle e_k, \rho e_k \rangle < \infty$. This sum is its *trace*, $\mathrm{Tr}\,\rho$; for a divergent positive sum write $\mathrm{Tr}\,\rho = +\infty$. A *density operator* is a positive trace-class operator with $\mathrm{Tr}\,\rho = 1$.

The words “for one ONB” are enough: the next proposition proves that the sum is independent of the basis. A positive operator of infinite trace is not trace class. The infinite value is basis independent too: a finite sum in another basis would, by the proposition, make every basis sum finite.

Proposition 1.4 (The trace and its spectral form)**.** *Every positive trace-class operator is compact. It has an orthonormal eigenbasis (φ_j) with eigenvalues $p_j \geq 0$, and*

$$\rho = \sum_j p_j \left|\varphi_j\right\rangle\!\left\langle\varphi_j\right|, \qquad \operatorname{Tr}\rho = \sum_j p_j = \sum_k \langle e_k, \rho e_k\rangle \tag{1.2}$$

for every ONB (e_k). The operator series converges in operator norm. For a density operator, the p_j sum to 1.

For a density operator, this decomposition supplies one possible preparation; it does not single out how the state was prepared. Lecture 2 studies that freedom. The proof first turns a small diagonal tail into an operator-norm approximation by finite-rank operators. Only after this compactness step may we use S5 to choose an eigenbasis. Parseval’s identity then compares the trace sums in different bases. This order avoids assuming that a general bounded positive operator already has an eigenbasis.

Proof. First choose the basis in the definition, and let $E_n = \sum_{k\leq n} |e_k\rangle\!\langle e_k|$. Put $R = \sqrt{\rho}$ (S6). For a finite sum $v = \sum_{k>n} c_k e_k$, the triangle inequality followed by the scalar Cauchy–Schwarz inequality gives

$$\|Rv\| \leq \sum_{k>n} |c_k|\,\|Re_k\| \leq \|v\| \left(\sum_{k>n} \langle e_k, \rho e_k\rangle\right)^{1/2}.$$

By continuity the same bound holds throughout $(\mathbb{1} - E_n)\mathcal{H}$. Also $\|Rw\|^2 = \langle w, \rho w\rangle \leq \|\rho\|\,\|w\|^2$. Since $\rho = R^2$, these two estimates give

$$\|\rho(\mathbb{1} - E_n)\|^2 \leq \|\rho\| \sum_{k>n} \langle e_k, \rho e_k\rangle \longrightarrow 0. \tag{1.3}$$

Thus ρ is a norm limit of the finite-rank operators ρE_n and is compact by the definition in *Linear Algebra*, Lecture 14: a compact operator is a norm limit of finite-rank operators.

S5 supplies an eigenbasis, including a basis of the kernel. Its eigenvalues are nonnegative and tend to zero if there are infinitely many nonzero ones, so the displayed spectral series converges in norm. For any ONB (e_k), Parseval gives

$$\sum_k \langle e_k, \rho e_k\rangle = \sum_k \sum_j p_j |\left\langle \varphi_j, e_k\right\rangle|^2 = \sum_j p_j \sum_k |\left\langle \varphi_j, e_k\right\rangle|^2 = \sum_j p_j.$$

All terms are nonnegative. Interchanging these sums means taking the supremum of the same finite partial sums; it requires no measure theory. In the initial basis the sum is finite, so all the others have that same finite value. ∎

For bounded A, define the expectation pairing by

$$\operatorname{Tr}(\rho A) := \sum_j p_j \left\langle \varphi_j, A\varphi_j\right\rangle. \tag{1.4}$$

It is absolutely convergent, with $|\operatorname{Tr}(\rho A)| \leq (\operatorname{Tr}\rho)\,\|A\|$. It really is a trace: in any ONB, expanding ρ gives

$$\sum_k \langle e_k, \rho A e_k\rangle = \sum_j p_j \sum_k \left\langle e_k, \varphi_j\right\rangle \left\langle \varphi_j, Ae_k\right\rangle = \sum_j p_j \left\langle \varphi_j, A\varphi_j\right\rangle.$$

For each j, Cauchy–Schwarz and Parseval bound the sum of the absolute values of the inner terms by $\|\varphi_j\| \, \|A^\dagger \varphi_j\| \le \|A\|$. The double sum is therefore absolutely convergent and can be reordered. In particular the pairing does not depend on which eigenbasis was chosen. It is linear in A, positive when $A \ge 0$, and equals 1 at $A = \mathbb{1}$ for a density operator. Every density operator thus gives an algebraic state.

For later lectures, $\mathcal{T}_1(\mathcal{H})$ denotes the complex span of the positive trace-class operators. Extend the trace linearly. This is well-defined: the absolutely convergent diagonal sums above compute each finite linear combination in any basis. No general trace-ideal theorem is needed in this lecture. This is the usual trace class, also characterised by $\operatorname{Tr}|T| < \infty$, where $|T| = \sqrt{T^\dagger T}$ (· *cited*; Hall, §19.2, pp. 421–422); the characterisation is not needed here.

1.3 Normal states are density operators

If (E_k) are mutually orthogonal event projections, their sum is an operator defined by its action: $\sum_k E_k\psi = \lim_{N\to\infty} \sum_{k\le N} E_k\psi$ for each ψ. Indeed the finite sums satisfy $\sum_{k\le N} \|E_k\psi\|^2 \le \|\psi\|^2$ by Pythagoras, so their vector tails tend to zero. The series converges in $\mathcal{H}$. Its limit acts as the identity on each range and as zero on their common orthogonal complement, hence is the projection onto their closed span. This is a *strong* sum. Convergence need not hold in operator norm: for basis truncations E_N, $E_N\psi \to \psi$ but $\|\mathbb{1} - E_N\| = 1$ in infinite dimension. The bound in Lemma 1.2 alone cannot pass to such a limit.

Definition 1.5 (Normality)**.** An algebraic state is *normal* if every countable orthogonal family of projections satisfies

$$\omega\Big(\sum_k E_k\Big) = \sum_k \omega(E_k). \tag{1.5}$$

In this course, *state* means a normal algebraic state.

Normality is countable additivity of probabilities for exclusive events. In particular, if an ONB exhausts the possible alternatives, their probabilities must sum to 1. It rules out assigning probability zero to every one of those alternatives while assigning probability one to their union. It is an assumption about the expectation rule, not about any particular preparation.

Example 1.6 (Recovering a matrix from an expectation rule)**.** In the orthonormal basis e_0, e_1 of $\mathbb{C}^2$, suppose a rule gives $\omega(|e_0\rangle\langle e_0|) = 3/4$, $\omega(|e_1\rangle\langle e_1|) = 1/4$ and $\omega(|e_1\rangle\langle e_0|) = 1/8$. Taking adjoints gives $\omega(|e_0\rangle\langle e_1|) = 1/8$ too. If a density matrix represents this rule, direct multiplication of the finite trace shows that its row j, column k entry is $\omega(|e_k\rangle\langle e_j|)$: the order of the two basis labels is reversed. Thus

$$\rho = \begin{pmatrix} 3/4 & 1/8 \\ 1/8 & 1/4 \end{pmatrix}, \qquad \operatorname{Tr}\big(\rho(|e_0\rangle\langle e_1| + |e_1\rangle\langle e_0|)\big) = \frac{1}{4}.$$

The last value also follows directly by adding the two given rank-one expectations. For $\psi_+ = (e_0 + e_1)/\sqrt{2}$, expanding its projection gives $\omega(|\psi_+\rangle\langle\psi_+|) = (3/4 + 1/4 + 1/8 + 1/8)/2 = 5/8$. The diagonal entries alone would not determine this probability: the off-diagonal expectations matter. The finite trace therefore determines a candidate matrix directly. In infinite dimension an additional bound is needed to show that the candidate coordinates define a bounded operator. The next proof supplies it.

We construct the density operator from rank-one tests, then recover all bounded expectations by a tail estimate. The steps and their inputs are shown in Figure 1.1.

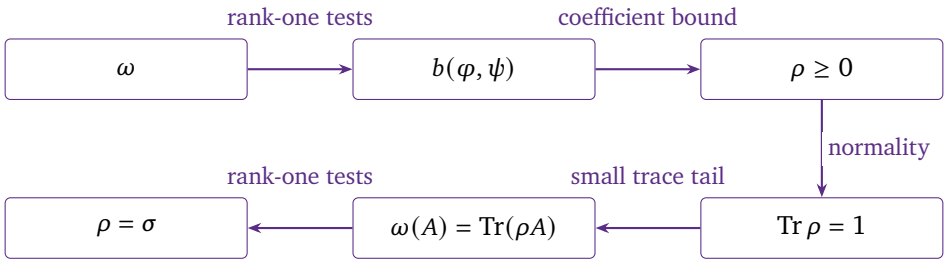


Figure 1.1: The construction from a normal expectation rule to its unique density operator. The form b collects rank-one expectations; normality makes the diagonal sum one, and the summable weights control every bounded A. Any density operator σ giving the same rule must equal ρ.

Theorem 1.7 (Normal-state representation)**.** *On a nonzero separable complex Hilbert space, a functional ω is a state if and only if there is a unique density operator ρ such that*

$$\omega(A) = \operatorname{Tr}(\rho A) \qquad \text{for every } A \in B(\mathcal{H}).$$

Proof. A density operator gives a normal state. Only normality remains to be checked. Let $E = \sum_k E_k$ be a strong orthogonal sum. In the eigenbasis of ρ, orthogonality gives

$$\operatorname{Tr}(\rho E) = \sum_j p_j \left\|E\varphi_j\right\|^2 = \sum_j p_j \sum_k \left\|E_k\varphi_j\right\|^2 = \sum_k \operatorname{Tr}(\rho E_k).$$

The interchange again uses nonnegative terms.

Construct the operator from a state. The rank-one operators reveal all its matrix elements. Set

$$b(\varphi,\psi) = \omega(|\psi\rangle\langle\varphi|).$$

This form is antilinear in φ, linear in ψ, and $|b(\varphi,\psi)| \leq \|\varphi\|\,\|\psi\|$ by Lemma 1.2, since $\||\psi\rangle\langle\varphi|\| = \|\psi\|\,\|\varphi\|$. Fix an ONB (e_j) and a vector ψ. For $u_N = \sum_{j\leq N} b(e_j,\psi)e_j$ we have

$$\|u_N\|^2 = \sum_{j\leq N} |b(e_j,\psi)|^2 = b(u_N,\psi) \leq \|u_N\|\,\|\psi\|\,.$$

Consequently these squared coefficients sum to at most $\|\psi\|^2$. The series

$$\rho\psi = \sum_j b(e_j,\psi)e_j \tag{1.6}$$

converges and defines a linear bounded operator with $\|\rho\| \leq 1$. The coefficients in (1.6) are candidates for the coordinates of $\rho\psi$. The bound on their squared partial sums ensures that they belong to the Hilbert space; merely assigning infinitely many matrix entries would not ensure this. Linearity follows because each coefficient $b(e_j,\psi)$ is linear in ψ. For a finite sum φ it satisfies $\langle\varphi,\rho\psi\rangle = b(\varphi,\psi)$; approximating an arbitrary φ by such sums proves the identity for all vectors. In particular, $\langle\psi,\rho\psi\rangle = \omega(|\psi\rangle\langle\psi|) \geq 0$. The identity $b(\varphi,\psi) = \overline{b(\psi,\varphi)}$, obtained by taking adjoints inside ω, shows that ρ is self-adjoint. Thus $\rho \geq 0$. This constructs ρ directly from an ONB, without an infinite-dimensional representation theorem for linear functionals.

Its trace is one. Normality on the rank-one basis projections gives

$$\sum_j \langle e_j, \rho e_j \rangle = \sum_j \omega(|e_j\rangle\langle e_j|) = \omega(\mathbb{1}) = 1.$$

By Proposition 1.4, ρ is a density operator.

Recover every bounded expectation. Now choose an eigenbasis (φ_j) of this ρ, including its kernel, and let $E_N = \sum_{j\le N} |\varphi_j\rangle\langle\varphi_j|$. Since $|\varphi_j\rangle\langle\varphi_j| A = |\varphi_j\rangle\langle A^\dagger \varphi_j|$, the construction gives

$$\omega(E_N A) = \sum_{j\le N} b(A^\dagger \varphi_j, \varphi_j) = \sum_{j\le N} p_j \langle \varphi_j, A\varphi_j \rangle .$$

Cauchy–Schwarz for ω, with $B = \mathbb{1} - E_N$, gives the crucial tail bound

$$|\omega((\mathbb{1} - E_N)A)|^2 \le \omega(\mathbb{1} - E_N)\omega(A^\dagger A) \le \Big(1 - \sum_{j\le N} p_j\Big) \|A\|^2 \longrightarrow 0. \tag{1.7}$$

Hence $\omega(A) = \lim_N \omega(E_N A) = \mathrm{Tr}(\rho A)$. For a finite-dimensional space the same argument terminates with $E_N = \mathbb{1}$.

Uniqueness. For any density operator σ, Equation (1.4) gives $\mathrm{Tr}(\sigma\,|\psi\rangle\langle\varphi|) = \langle\varphi, \sigma\psi\rangle$. If two density operators give the same rule, all their matrix elements agree, so they are equal. ∎

In $\mathbb{C}^n$, an orthogonal family has at most n nonzero projections, so linearity already implies normality. The theorem recovers *QM I*'s density matrices from rules for expectations, without first choosing an ensemble. A ray is included: $\rho = |\psi\rangle\langle\psi|$ for unit ψ gives $\omega(A) = \langle\psi, A\psi\rangle$. On an infinite-dimensional space the extra countable-additivity clause is essential. This is not peculiar to position space: every separable infinite-dimensional Hilbert space is unitarily equivalent to $L^2(\mathbb{R})$, by sending one ONB to another. The counterexample below therefore transfers to any such space.

For bounded A_N converging strongly to A, with a common bound $\|A_N\| \le M$, the pairing also gives $\mathrm{Tr}(\rho A_N) \to \mathrm{Tr}(\rho A)$. Indeed split its eigenbasis series into a finite head and a tail: each head term converges, while the tail of the difference is bounded by $2M \sum_{j>J} p_j$ (the strong limit also has norm at most M). This is a bounded version of the strong continuity used as an axiom by Hall (Def. 19.6, p. 423). We derived it from normality; its finite-head and small-tail argument also handles position below.

1.4 An oscillator at temperature T

The harmonic oscillator of *QM II*, Lecture 8, has the complete Hermite basis $(h_n)_{n\ge0}$ (Theorem 8.4) and energies $\hbar\nu(n + \frac{1}{2})$, with angular frequency $\nu > 0$. For $T > 0$, put $\beta = 1/(k_B T)$ and $q = e^{-\beta\hbar\nu}$, so $0 < q < 1$. The thermal weights are proportional to $e^{-\beta\hbar\nu(n+1/2)}$. Summing the geometric series gives

$$Z(\beta) = \sum_{n=0}^{\infty} e^{-\beta\hbar\nu(n+1/2)} = \frac{q^{1/2}}{1-q} = \frac{1}{2\sinh(\beta\hbar\nu/2)}, \qquad \rho_\beta = \sum_{n=0}^{\infty} (1-q)q^n\, |h_n\rangle\langle h_n| . \tag{1.8}$$

This is $e^{-\beta H}/Z(\beta)$, with the exponential defined on its eigenbasis. The series has nonnegative weights summing to 1, so it defines a density operator. It has infinitely many populated levels; finite rank is not required. The spectral event for energy $\hbar\nu(n + \frac{1}{2})$ is $|h_n\rangle\langle h_n|$, so the probability of level n is $(1-q)q^n$.

To sum its mean level number, let $S = \sum_{n\geq 0} nq^n$. The series is convergent for $q < 1$ (its successive nonzero terms have ratio tending to q). Subtracting qS from S yields $(1-q)S = \sum_{n\geq 1} q^n = q/(1-q)$. Therefore

$$\langle n\rangle = \frac{q}{1-q} = \frac{1}{e^{\beta\hbar\nu}-1}, \qquad \langle H\rangle = \hbar\nu\left(\frac{1}{2} + \frac{1}{e^{\beta\hbar\nu}-1}\right). \tag{1.9}$$

Here H is unbounded. Its mean is the mean of this energy-reading distribution. It is justified by the absolutely convergent series $\sum_n (1-q)q^n\hbar\nu(n+\frac{1}{2})$, not by inserting H into a theorem stated only for bounded A. As $T \downarrow 0$ the ground-state probability tends to 1 and the energy tends to $\hbar\nu/2$; for $k_BT \gg \hbar\nu$ the mean approaches k_BT to leading order. The free particle on the whole line has no normalised Gibbs density operator (an example reserved for Sheet 1); the geometric-series argument depends on the oscillator's levels.

1.5 What $|x_0\rangle$ cannot be

On $L^2(\mathbb{R})$, *QM II* already ruled out a normalised position eigenvector (its closing system card, §15.4). Could a mixed state nevertheless assign an exactly sharp position? That would require $\mathrm{Tr}(\rho f(Q)) = f(x_0)$ for every bounded continuous function f. The answer is still no.

Take continuous triangular bumps $f_n(x) = \max\{1 - n|x-x_0|/\ell, 0\}$, where $\ell > 0$ is a fixed length. They satisfy $0 \leq f_n \leq 1$ and $f_n(x_0) = 1$. The operator $f_n(Q)$ is multiplication by f_n. For any unit vector φ,

$$0 \leq \langle \varphi, f_n(Q)\varphi\rangle \leq \int_{|x-x_0|<\ell/n} |\varphi(x)|^2\,\mathrm{d}x \longrightarrow 0.$$

Here is the elementary shrinking-interval step without an integration limit theorem: approximate φ in L^2 by a bounded continuous function v of compact support, using QM-II's density fact S4. On an interval I, the integral of $|\varphi|^2$ is at most $2\,\|\varphi - v\|^2 + 2|I|\,\|v\|_\infty^2$. First make the first term small, then shrink I.

For $\rho = \sum_j p_j |\varphi_j\rangle\langle\varphi_j|$, split the expectation into a finite head and tail. The head tends to zero by the preceding estimate; the tail is bounded by $\sum_{j>J} p_j$, uniformly in n. Hence

$$\mathrm{Tr}(\rho f_n(Q)) \longrightarrow 0, \qquad \text{whereas} \qquad f_n(x_0) = 1.$$

No density operator has the proposed evaluation rule. To state the existence result precisely, start on the algebra $\{f(Q) : f \in C_b(\mathbb{R})\}$ of bounded continuous multiplication operators. The assignment $f(Q) \mapsto f(x_0)$ is well-defined, positive, unital and has norm one, since $\|f(Q)\| = \sup_x |f(x)|$ for continuous f. It has a positive normalised extension to $B(L^2(\mathbb{R}))$ (the positive Hahn–Banach extension argument, · *cited*; Landsman, proof of Prop. C.15, p. 652, with Lemma C.4, p. 646). Every such extension is non-normal by the argument just given; it is outside this course's state convention. Thus the missing assumption is normality, even after we admit mixed states.

What is a state? First answer

A state is a positive, normalised linear rule for bounded expectations, countably additive on orthogonal events. On $B(\mathcal{H})$ for separable $\mathcal{H}$, it is exactly $\omega(A) = \mathrm{Tr}(\rho A)$ for one density operator. A sharp event $E_A(I)$ has probability $\mathrm{Tr}(\rho E_A(I))$. Unbounded means require an additional convergence statement. Neither a preparation ensemble nor a measurement update is specified by this definition.

Summary. Finite positive trace forces compactness, and the trace is basis independent. Positivity and Cauchy–Schwarz reconstruct an operator from rank-one expectations; normality makes its trace one and recovers all bounded expectations. The thermal oscillator is an infinite-rank example. An exactly sharp position cannot be represented by any density operator. Next we ask how many preparations one state can have.

Exercises. Try Sheet 1 in the separate *Exercise sheets* booklet (English).

Sources. K. Landsman, *Foundations of Quantum Theory* (2017), Def. 4.11 and Thm 4.12, pp. 109–110; the extension argument in Prop. C.15, p. 652, and Lemma C.4, p. 646; B. C. Hall, *Quantum Theory for Mathematicians* (2013), Defs. 19.6–19.7 and Thm 19.9, pp. 423–425. Hall uses sequential strong continuity in place of our projection-additivity axiom. The proof above constructs the representing operator by its ONB coefficients and uses Landsman's Cauchy–Schwarz tail estimate.

Lecture 2

One state, many preparations

Objectives. *A density operator fixes expectations, not a preparation history. The extreme points of the set of density operators are exactly the rays. When* $\dim \mathcal{H} \geq 2$*, even a ray leaves some bounded observable uncertain. A unitary on padded preparation labels relates any two finite pure-state ensembles of the same state. For one optical mode, averaging the phase of a coherent state gives exactly a Poisson mixture of number states.*

- Prove that the extreme density operators have rank one, and find an uncertain event in any state when $\dim \mathcal{H} \geq 2$.
- Construct the unitary relating two finite ensembles of the same state.
- Derive the phase-averaged coherent-state identity and explain what it does and does not say about a laser.

2.1 What mixing forgets

Theorem 1.7 identified a state with the unique density operator that gives its bounded expectations. Uniqueness of the operator does not mean uniqueness of the preparation. In *QM I*, §3.6, half the spins prepared up and half down along z gave the same state as a preparation using the two x directions. The same distinction survives in infinite dimension. We keep the nonzero separable complex Hilbert space $\mathcal{H}$ of Lecture 1 and write $\mathcal{D}(\mathcal{H})$ for its density operators.

Definition 2.1 (Convexity and pure states)**.** A set is *convex* if it contains $t\rho_1 + (1-t)\rho_2$ whenever it contains ρ_1, ρ_2 and $0 \leq t \leq 1$. A point ρ in a convex set is *extreme* if, for ρ_1, ρ_2 belonging to that same set,

$$\rho = t\rho_1 + (1-t)\rho_2, \quad 0 < t < 1 \quad \Longrightarrow \quad \rho_1 = \rho_2 = \rho.$$

A *pure state* is an extreme point of $\mathcal{D}(\mathcal{H})$; a state that is not pure is *mixed*.

The density operators form a convex set: a convex combination is bounded and positive, and its diagonal sum in any ONB is $t \operatorname{Tr} \rho_1 + (1-t) \operatorname{Tr} \rho_2 = 1$. Its mean of a bounded A is $t \operatorname{Tr}(\rho_1 A) + (1-t) \operatorname{Tr}(\rho_2 A)$, by the basis expression for the trace proved in § 1.2. Physically, select one preparation with probability t and the other with probability $1-t$, and discard the selection record. Retaining that record is additional information about the preparation, not part of the density operator of this system alone.

The restriction $0 < t < 1$ matters. A pure state can occur in a mixture of two identical copies of itself; extremality only excludes a nontrivial mixture of *different* states. Nor does an extreme point have to be isolated: the pure qubit states form the boundary sphere of the qubit's Bloch ball. The question is whether a line segment joining different states can have this point strictly between its endpoints.

2.2 Pure means a ray

Theorem 2.2 (The extreme density operators). *For a density operator on $\mathcal{H}$, the following are equivalent: it is pure; it has rank one; it equals $|\psi\rangle\langle\psi|$ for a unit vector ψ. Two unit vectors give the same rank-one state exactly when they differ by a complex phase.*

Proof. A state of rank greater than one is not extreme. The spectral decomposition from Proposition 1.4 is $\rho = \sum_j p_j |e_j\rangle\langle e_j|$, with $p_j \geq 0$ and $\sum_j p_j = 1$. If more than one eigenvalue is positive, choose p_1 with $0 < p_1 < 1$ and set

$$\sigma = \frac{\rho - p_1 |e_1\rangle\langle e_1|}{1 - p_1} = \sum_{j \neq 1} \frac{p_j}{1 - p_1} |e_j\rangle\langle e_j| .$$

The last expression defines a positive bounded operator of trace one. Indeed, the positive partial sums converge in operator norm because their norm tails are bounded by tails of the summable weights; positivity passes to the limit. Its eigenbasis diagonal sum is $\sum_{j\neq 1} p_j/(1-p_1) = 1$. Thus $\rho = p_1 |e_1\rangle\langle e_1| + (1-p_1)\sigma$ is a convex decomposition into different states: $\sigma e_1 = 0$, whereas $|e_1\rangle\langle e_1|\, e_1 = e_1$. Consequently an extreme state has just one nonzero eigenvalue, which must be 1. It is $|\psi\rangle\langle\psi|$ for a unit eigenvector ψ.

A rank-one state is extreme. Suppose $|\psi\rangle\langle\psi| = t\rho_1 + (1-t)\rho_2$ with $0 < t < 1$. For any $v \perp \psi$,

$$0 = t \langle v, \rho_1 v\rangle + (1-t) \langle v, \rho_2 v\rangle .$$

Both terms are nonnegative, so each is zero. The positive square root (S6) gives $\left\|\sqrt{\rho_i} v\right\|^2 = 0$, hence $\rho_i v = 0$ for $i = 1, 2$. Self-adjointness then gives $\langle v, \rho_i \psi\rangle = \langle \rho_i v, \psi\rangle = 0$ for every $v \perp \psi$. Thus $\rho_i \psi = c_i \psi$ and $\rho_i = c_i |\psi\rangle\langle\psi|$; its trace is one, so $c_i = 1$. This proves extremality. A positive rank-one trace-one operator is necessarily such a projection, by its single nonzero eigenvalue.

Finally, if $|\psi\rangle\langle\psi| = |\varphi\rangle\langle\varphi|$, applying both sides to φ shows that $\varphi = \langle\psi, \varphi\rangle\, \psi$. The coefficient has modulus one because both vectors are unit vectors. The converse follows from $|c\psi\rangle\langle c\psi| = |c|^2 |\psi\rangle\langle\psi|$. ∎

This proof used the compact spectral theorem already supplied by L1, not a finite-dimensional restriction. An oscillator wave packet defines a pure state just as a spinor does. Its ray is now recovered from the convex structure of all normal states, rather than postulated as the only kind of state.

The familiar purity test also survives. The square ρ^2 is positive trace class: it has eigenvalues p_j^2, and $0 \leq p_j^2 \leq p_j$ makes their sum finite. Thus $\mathrm{Tr}(\rho^2) = \sum_j p_j^2 \leq \sum_j p_j = 1$. Equality holds precisely when $\sum_j p_j(1 - p_j) = 0$. Every term is nonnegative, so every p_j is either 0 or 1; exactly one is 1. Thus $\mathrm{Tr}(\rho^2) = 1$ if and only if the state is pure. In the qubit Bloch ball this is $\mathrm{Tr}(\rho^2) = (1 + |\mathbf{r}|^2)/2$, the boundary-sphere criterion from QM-I.

2.3 Pure does not mean sharp for everything

For bounded self-adjoint A, its variance in state ρ is

$$\mathrm{Var}_\rho(A) = \mathrm{Tr}(\rho A^2) - [\mathrm{Tr}(\rho A)]^2 = \mathrm{Tr}\big(\rho (A - \mathrm{Tr}(\rho A)\mathbb{1})^2\big) \geq 0.$$

It is the mean square deviation of its readings. Here A is bounded, so both expectations fall within L1's theorem and exist without any extra moment assumption. A state is *dispersion-free for A* when this variance vanishes, and *dispersion-free* when it vanishes for every bounded self-adjoint A.

Proposition 2.3 (No universally sharp quantum state)**.** *If* $\dim \mathcal{H} \geq 2$*, every density operator gives some rank-one projection* E *a strictly positive variance. No state is dispersion-free on all bounded observables.*

Proof. Since $E^2 = E$, put $w = \mathrm{Tr}(\rho E)$; then $\mathrm{Var}_\rho(E) = w(1-w)$. If ρ is mixed, Theorem 2.2 gives at least two positive eigenvalues, each less than one. Choose one, $p \in (0,1)$, and a unit eigenvector e. For $E = |e\rangle\langle e|$, $w = p$ lies strictly between 0 and 1.

If $\rho = |\psi\rangle\langle\psi|$ is pure, choose a unit $\varphi \perp \psi$, possible because $\dim \mathcal{H} \geq 2$. Let $\chi = (\psi+\varphi)/\sqrt{2}$ and $E = |\chi\rangle\langle\chi|$. Its probability is $w = |\langle \chi, \psi\rangle|^2 = 1/2$, so its variance is $1/4$. ∎

The order of quantifiers is important: *for every state there is an uncertain event*. It does not say that one fixed projection is uncertain in every state. A unit vector in its range gives it probability one; a unit vector in its kernel gives probability zero. On a one-dimensional space every bounded observable is a scalar and the unique state is dispersion-free, which explains the dimension hypothesis.

For example, a spin prepared in $|0\rangle$ has zero variance for σ_z and variance one for σ_x. Being pure excludes uncertainty arising from a nontrivial convex preparation of that state; it does not exclude uncertainty in a particular reading. This result concerns quantum states and their expectations. Whether an enlarged theory can attach further values to a system is a different question, taken up in Lecture 13. No claim about all such theories has been proved here.

2.4 The unitary freedom of finite ensembles

Definition 2.4 (An ensemble of a state)**.** An *ensemble of* ρ is a finite or countable list $\{(p_i, \psi_i)\}$ of probabilities and unit vectors such that $\sum_i p_i = 1$ and $\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|$. The vectors need not be orthogonal or distinct. Zero-probability entries may be omitted.

For a countable list this series converges in operator norm, since the norm of its tail is at most $\sum_{i>N} p_i$. Its limit is positive with trace one: summing its diagonal entries interchanges only nonnegative terms, as in L1. Every state has at least its spectral ensemble. A finite ensemble, however, has finite-rank density operator even when the ambient Hilbert space is infinite-dimensional. The next theorem concerns *finite ensembles only*.

Absorb each weight into its vector, writing $x_i = \sqrt{p_i}\psi_i$. Then $\rho = \sum_i |x_i\rangle\langle x_i|$, with $\sum_i \|x_i\|^2 = 1$. These vectors are not generally unit vectors. A zero-weight entry is the zero vector; recovering a unit vector from it is unnecessary.

Theorem 2.5 (Hughston–Jozsa–Wootters, finite-ensemble form)**.** *Let* $(x_i)_{i=1}^m$ *and* $(y_j)_{j=1}^m$ *be two finite lists in* $\mathcal{H}$ *with* $\sum_i \|x_i\|^2 = \sum_j \|y_j\|^2 = 1$*, padding the shorter original list with zero vectors to get equal lengths. They generate the same density operator if and only if there is an* $m \times m$ *unitary matrix* u *with*

$$x_i = \sum_{j=1}^{m} u_{ij} y_j \qquad (1 \leq i \leq m). \tag{2.1}$$

Proof. A unitary relation gives the same operator. Expand the rank-one sums:

$$\sum_i |x_i\rangle\langle x_i| = \sum_{j,k} \Big(\sum_i u_{ij} \overline{u_{ik}} \Big) |y_j\rangle\langle y_k| = \sum_j |y_j\rangle\langle y_j| .$$

The coefficient is δ_{jk} because the columns of u are orthonormal.

Equal operators give a unitary relation. Suppose both sums equal ρ. Its range is the span of either list. Indeed, for $\xi \in \mathcal{H}$, $\langle \xi, \rho\xi \rangle = \sum_i |\langle x_i, \xi \rangle|^2$ shows that $\ker \rho = \operatorname{span}\{x_i\}^\perp$. This finite span is closed. The formula $\rho\xi = \sum_i x_i \langle x_i, \xi \rangle$ shows that ρ maps the span into itself. Its restriction has zero kernel; an injective map of a finite-dimensional space to itself is surjective, so it is invertible there. This proves the range claim, also for the y list. Write this common range as K and let $e_1, \dots, e_r$ be an eigenbasis on it, with $\rho e_a = \lambda_a e_a$, $\lambda_a > 0$. Necessarily $r \le m$.

For each a make two vectors in the *label space* $\mathbb{C}^m$. Here a labels a physical direction in K, whereas i labels a preparation entry. The vector v_a collects the coefficients of that one physical direction across all entries. Dividing by $\sqrt{\lambda_a}$ will make these label vectors orthonormal:

$$v_a = \left(\frac{\langle e_a, x_i \rangle}{\sqrt{\lambda_a}} \right)_{i=1}^{m}, \qquad w_a = \left(\frac{\langle e_a, y_j \rangle}{\sqrt{\lambda_a}} \right)_{j=1}^{m}.$$

The eigenbasis matrix entries of ρ give

$$\sum_i \langle e_a, x_i \rangle \, \overline{\langle e_b, x_i \rangle} = \langle e_a, \rho e_b \rangle = \lambda_b \delta_{ab}.$$

Taking complex conjugates and dividing by $\sqrt{\lambda_a \lambda_b}$ shows $\langle v_a, v_b \rangle = \delta_{ab}$. The same calculation gives $\langle w_a, w_b \rangle = \delta_{ab}$. Thus each is an orthonormal r-frame in the same $\mathbb{C}^m$. Complete them to ONBs $(v_a)_{a=1}^m$, $(w_a)_{a=1}^m$ and set $u = \sum_{a=1}^m |v_a\rangle\langle w_a|$, the unitary sending each w_a to v_a. In coordinate i, $(v_a)_i = (u w_a)_i = \sum_j u_{ij} (w_a)_j$. Multiplying by $\sqrt{\lambda_a}$ gives, for $1 \le a \le r$,

$$\langle e_a, x_i \rangle = \sum_j u_{ij} \left\langle e_a, y_j \right\rangle .$$

All the x_i, y_j lie in K, so agreement of these r coordinates gives (2.1). Only finite-dimensional ONB completion was used; there is no inverse of ρ on the whole Hilbert space in this argument. ∎

The unitary acts on the list of preparation labels. It is not a unitary operation on the physical system sending the unit vectors of one ensemble to those of the other. Such an operation would preserve inner products; the new ensemble need not be orthogonal even if the original one was. The theorem classifies finite descriptions of a common state, not a transformation between distinct states.

The two decompositions in the following example appear as different segments through one Bloch point in Figure 2.1.

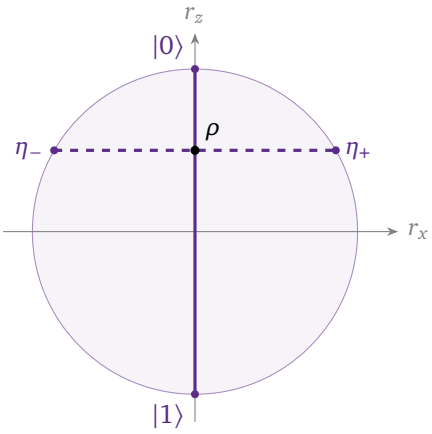


Figure 2.1: The section $r_y = 0$ of the qubit Bloch ball. The spectral endpoints with weights $3/4, 1/4$ and the nonorthogonal endpoints η_+, η_- with weights $1/2, 1/2$ give the same interior point $\mathbf{r} = (0, 0, 1/2)$; the segments describe mixtures, not physical trajectories.

For a concrete example (Nielsen–Chuang, p. 103), consider

$$\rho = \frac{3}{4}\,|0\rangle\langle 0| + \frac{1}{4}\,|1\rangle\langle 1| = \frac{1}{2}\,|\eta_+\rangle\langle\eta_+| + \frac{1}{2}\,|\eta_-\rangle\langle\eta_-|\,, \qquad \eta_\pm = \frac{\sqrt{3}\,|0\rangle \pm |1\rangle}{2}. \tag{2.2}$$

The vectors $\eta_\pm$ are unit vectors with overlap $1/2$, so they are not an eigenbasis. With $y_1 = \sqrt{3}\,|0\rangle/2$, $y_2 = |1\rangle/2$, the matrix

$$u = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

gives $x_1 = \eta_+/\sqrt{2}$, $x_2 = \eta_-/\sqrt{2}$. Their two outer products cancel the off-diagonal entries and recover ρ. Its unequal eigenvalues do not remove the ensemble freedom.

We can also *construct* this u using the converse proof. In the physical eigenbasis $|0\rangle, |1\rangle$, with eigenvalues $3/4, 1/4$, the spectral list y gives $w_1 = (1, 0)$, $w_2 = (0, 1)$. The other list gives $v_1 = (1, 1)/\sqrt{2}$, $v_2 = (1, -1)/\sqrt{2}$. These are already complete ONBs of the two-dimensional label space. Thus $u = \sum_{a=1}^{2} |v_a\rangle\langle w_a|$ has columns v_1, v_2, exactly the displayed matrix. On the physical system no unitary can turn the orthogonal pair $|0\rangle, |1\rangle$ into η_+, η_-, whose overlap is $1/2$.

Padding is substantive. The one-entry list (ψ) and the two-entry list $(\psi/\sqrt{2}, \psi/\sqrt{2})$ generate the same pure state. Pad the first to $(\psi, 0)$; the same 2×2 matrix gives the second list. There cannot be a rectangular isometry in both directions between label spaces of unequal dimensions. The zero padding makes the theorem's square unitary statement valid. The theorem above does not assert that the same completion argument works for two countably infinite lists.

2.5 A laser mode without a phase reference

Model a single optical mode as the oscillator of *QM II*, Lecture 8. Its Hermite basis vectors h_n are now called number states: a number reading gives n photons. The coherent states

from its Theorem 8.10 are

$$\psi_\alpha = e^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} h_n.$$

The phase of α is not a global phase multiplying the whole vector. Replacing α by $\alpha e^{i\phi}$ multiplies the coefficient of h_n by $e^{in\phi}$: the relative phases of different number components change. Multiplying the whole vector by one common phase would leave its pure state unchanged and could not produce a mixture. Fix $\alpha \in \mathbb{C}$ and average its phase uniformly. A precise meaning of this average is that every bounded A has expectation $\frac{1}{2\pi}\int_0^{2\pi} \langle \psi_{\alpha e^{i\phi}}, A\psi_{\alpha e^{i\phi}} \rangle \, d\phi$. This is an ordinary integral of a continuous scalar function; we will justify the limiting series directly.

Proposition 2.6 (The phase-averaged state)**.** *In the sense of equality of all bounded expectations,*

$$\frac{1}{2\pi}\int_0^{2\pi} |\psi_{\alpha e^{i\phi}}\rangle\langle\psi_{\alpha e^{i\phi}}| \, d\phi = \rho_{\mathrm{av}} := e^{-\bar{n}} \sum_{n=0}^{\infty} \frac{\bar{n}^n}{n!} |h_n\rangle\langle h_n|, \qquad \bar{n} = |\alpha|^2. \tag{2.3}$$

Proof. In a matrix element between h_m and h_n, the integrand is

$$e^{-|\alpha|^2} \frac{\alpha^m \bar{\alpha}^n}{\sqrt{m!n!}} e^{i(m-n)\phi}.$$

Its phase average vanishes for $m \neq n$ and is $e^{-\bar{n}}\bar{n}^n/n!$ for $m = n$. These diagonal entries are nonnegative and sum to one by the exponential series, so their sum defines a density operator by L1. For $\alpha = 0$ it is simply $|h_0\rangle\langle h_0|$.

To justify the average for every bounded A, truncate each coherent vector at level N, calling the result $s_N(\phi)$. Its squared error is

$$t_N = e^{-\bar{n}} \sum_{n>N} \frac{\bar{n}^n}{n!} \longrightarrow 0,$$

independently of ϕ, and $\|s_N(\phi)\| \leq 1$. Write $\psi = \psi_{\alpha e^{i\phi}}$. Adding and subtracting $\langle s_N, A\psi\rangle$ gives

$$\langle \psi, A\psi\rangle - \langle s_N, As_N\rangle = \langle \psi - s_N, A\psi\rangle + \langle s_N, A(\psi - s_N)\rangle .$$

Cauchy–Schwarz therefore bounds the absolute value by

$$|\langle \psi, A\psi\rangle - \langle s_N, As_N\rangle| \leq \|A\| \, (\|\psi\| + \|s_N\|) \, \|\psi - s_N\| \leq 2\,\|A\| \, \sqrt{t_N}.$$

Thus the expectations converge uniformly in phase from continuous finite sums; the scalar integral exists and its error obeys the same bound. For a finite sum the matrix-element computation above gives exactly $e^{-\bar{n}} \sum_{n\leq N} \bar{n}^n \langle h_n, Ah_n\rangle / n!$. The missing diagonal tail has absolute value at most $\|A\| \, t_N$. Taking $N \to \infty$ proves the identity for every bounded A. ∎

Definition 2.4 covers discrete lists; the phase integral uses a continuous family of pure states. Its identity is not an application of finite HJW, but a separate calculation with explicit convergence control. For $\bar{n} > 0$, every Poisson weight is positive, so ρ_{av} has infinite rank and is mixed by Theorem 2.2. At $\bar{n} = 0$ it remains the pure vacuum state. Its number probabilities are Poisson probabilities $p_n = e^{-\bar{n}}\bar{n}^n/n!$. Shifting the index in the convergent series gives $\sum_n np_n = \bar{n}$ and $\sum_n n(n-1)p_n = \bar{n}^2$, hence the number-reading variance is $\bar{n}$. These are finite moments of the discrete reading distribution; the number operator is unbounded, so they are not automatic substitutions into the bounded-observable variance theorem.

For example, at $\bar{n} = 4$ the probabilities of vacuum and one photon are $e^{-4} \simeq 0.0183$ and $4e^{-4} \simeq 0.0733$. A mixture of number states with those Poisson weights gives exactly the same probabilities and all other bounded expectations as the uniform phase ensemble. This does not make it the thermal state: at the same mean number, L1's thermal distribution has vacuum probability $1/(1+\bar{n}) = 0.2$.

Uniform averaging is an additional assumption about how phase is represented when the reference is unavailable. A fixed-phase coherent state and ρ_{av} are generally different states. For example, when $\alpha > 0$ is real, the bounded operator $A = |h_0\rangle\langle h_1| + |h_1\rangle\langle h_0|$ has mean $2e^{-\alpha^2}\alpha$ in ψ_α, but zero in ρ_{av}. Thus the lesson is not that phase can never matter. It is that the *phase-averaged* state has two ensemble descriptions. A reference or a retained preparation record changes what information is available and must be included in a full model.

What does a state say about its preparation?

A normal state fixes all bounded expectations. It is pure exactly when it is a ray, and mixed states can have many ensembles, even with distinct eigenvalues. Finite ensembles of the same state are related by a unitary on their padded label lists. No state on a space of dimension at least two is sharp for every bounded observable. The phase-averaged coherent model and a Poisson number mixture are one state; this does not select a preparation history or determine how an actual laser works.

Summary. Pure means extreme; in dimension at least two it does not mean dispersion-free. A spectral decomposition is one preparation description, not a uniquely privileged one. HJW makes the finite-ensemble freedom exact; the laser identity identifies a uniform phase family and a discrete number-state ensemble with one state by a separate calculation. Later, the same care about what a density operator does not specify will be essential when we discuss reduced states and decoherence. Next we ask whether probabilities assigned directly to events or effects already force the trace rule.

Exercises. Try Sheet 2 in the separate *Exercise sheets* booklet (English).

Sources. Busch–Lahti–Pellonpää–Ylinen, *Quantum Measurement* (2016), Thm 9.1, pp. 193–194; Example 9.1, p. 195; Thm 9.2, p. 196 (a more general decomposition statement, with its own kernel hypothesis). Nielsen–Chuang, *Quantum Computation and Quantum Information* (2010), Thm 2.6, pp. 103–104 (finite unitary freedom and zero padding). Landsman, *Foundations of Quantum Theory* (2017), pp. 46–48 (finite-dimensional version). K. Mølmer, *Phys. Rev. A* **55**, 3195–3203 (1997), especially pp. 3195–3197; reply, **58**, 4247 (1998). The coherent-state expansion is QM-II, Theorem 8.10.

Lecture 3

Probabilities on effects

Objectives. *Probabilities on all effects, additive whenever the effects sum to at most the identity, are exactly density operators. Probabilities on projections alone have the same characterisation in dimension at least three, but not for a qubit. The quadratic trace rule permits pair interference and forces an ideal three-path interference combination to vanish.*

- Prove Busch's theorem by extending an effect probability to a normal expectation rule.
- Exhibit a qubit probability assignment satisfying projection additivity but not the trace rule.
- Distinguish commuting event operations from the full projection lattice, and derive the ideal three-path identity with its assumptions.

3.1 From expectations to probabilities

Theorem 1.7 started with a positive linear expectation rule on $B(\mathcal{H})$ and obtained a density operator. Linearity already related the means of noncommuting operators. Could a rule assigning probabilities to possible readings force that linearity? The answer depends on the events to which probabilities are assigned and on which sums must be respected. Throughout, $\mathcal{H}$ is a nonzero separable complex Hilbert space, and the observable algebra under discussion is all of $B(\mathcal{H})$.

Definition 3.1 (Effect and generalised probability)**.** An *effect* is a bounded self-adjoint operator F with $0 \le F \le \mathbb{1}$. A *generalised probability* is a map $\mu : \{F : 0 \le F \le \mathbb{1}\} \to [0,1]$ such that $\mu(\mathbb{1}) = 1$ and

$$\mu\Big(\sum_k F_k\Big) = \sum_k \mu(F_k) \quad \text{whenever } F_k \ge 0, \quad \sum_k F_k \le \mathbb{1}. \tag{3.1}$$

The countable clause applies when the positive partial sums converge strongly to some $F \le \mathbb{1}$; it does not assume a general convergence theorem for arbitrary series. Each summand is then an effect, since $0 \le F_k \le F \le \mathbb{1}$. Finite sums are included.

Every projection is an effect, but an effect need not square to itself. For example, $F = \frac{1}{2}|0\rangle\langle 0|$ describes a yes outcome that occurs half the time for $|0\rangle$ and never for the orthogonal $|1\rangle$. One may realise this by an ideal test followed by independent random rejection of half its yes results. A general account of such readings comes in Lecture 9. At present an effect is the bounded operator to which a probability will be assigned, not an assertion that its only eigenvalues are zero and one.

The sum condition is stronger than additivity on orthogonal projections. Let $E_z = (\mathbb{1}+\sigma_z)/2$, $E_x = (\mathbb{1}+\sigma_x)/2$. Although they do not commute, the effects $F_1 = E_z/2$, $F_2 = E_x/2$ satisfy

$F_1 + F_2 \leq \mathbb{1}$, since $E_z, E_x \leq \mathbb{1}$. Thus our rule requires $\mu(F_1 + F_2) = \mu(F_1) + \mu(F_2)$. The remainder $\mathbb{1} - F_1 - F_2$ is also an effect. An operational model is to toss a fair coin, read σ_z on heads and σ_x on tails, and record outcome 1 for (heads, plus), outcome 2 for (tails, plus), and outcome 3 otherwise. In QM-I's language these outcomes have effects $F_1, F_2, \mathbb{1} - F_1 - F_2$. Assigning an effect the same probability whatever procedure realises it is precisely the extra consistency premise. This does not mean that the two sharp spin components have been read simultaneously. It states a consistency requirement for probabilities on these three effects, rather than simultaneous definite values of the two spin observables.

There is also a noncontextuality assumption in assigning one number $\mu(F)$ to each effect: its probability is independent of which complete family of outcomes contains it. Together with the sum rule, this is a substantive premise, not a consequence of the word "probability" alone. For a density operator the proposed assignment is $\mu(F) = \mathrm{Tr}(\rho F)$. The next theorem shows that these are all the assignments under the stated assumptions, including for a spin one half.

3.2 Busch's theorem

Theorem 3.2 (Probabilities on effects)**.** *For every generalised probability on the effects of a nonzero separable $\mathcal{H}$, there is a unique density operator ρ such that $\mu(F) = \mathrm{Tr}(\rho F)$ for every effect F. Conversely every density operator defines such a generalised probability.*

Proof. First obtain homogeneity on effects. Additivity gives $\mu(0) = 2\mu(0)$, hence $\mu(0) = 0$. If $0 \leq F \leq G \leq \mathbb{1}$, then $G = F + (G - F)$ is an allowed sum. Therefore $\mu(F) \leq \mu(G)$. For a positive integer n, adding n copies of F/n gives $\mu(F/n) = \mu(F)/n$. Adding $m \leq n$ of those copies gives

$$\mu\big((m/n)F\big) = (m/n)\mu(F).$$

For any real $t \in [0, 1]$, choose rational numbers $r_k \leq t \leq s_k$ in $[0, 1]$ tending to t. Monotonicity gives $r_k\mu(F) \leq \mu(tF) \leq s_k\mu(F)$, so $\mu(tF) = t\mu(F)$. This argument derives continuity in the scalar multiplier from positivity and additivity; it does not assume that μ is initially a linear functional.

Extend to every positive bounded operator. For $B \geq 0$ choose $c > 0$ with $c \geq \|B\|$ and put $f(B) = c\mu(B/c)$. This does not depend on c: if $d \geq c$, then $B/d = (c/d)(B/c)$ and scalar homogeneity gives $d\mu(B/d) = c\mu(B/c)$. Two arbitrary allowed scales can be compared through their larger one. In particular $f(F) = \mu(F)$ for an effect, by taking $c = 1$.

For $B, C \geq 0$, choose $c \geq \|B\| + \|C\|$, with $c > 0$. The effects $B/c, C/c$ have sum at most $\mathbb{1}$, so

$$f(B + C) = c\mu((B + C)/c) = f(B) + f(C).$$

Also $f(tB) = tf(B)$ for $t \geq 0$: for $t > 0$, use tc as the scale for tB when c is a scale for B; the zero case is immediate.

Extend to self-adjoint operators without ambiguity. Every bounded self-adjoint A can be written $B - C$ with $B, C \geq 0$, for example $B = A + \|A\|\,\mathbb{1}$, $C = \|A\|\,\mathbb{1}$. Define $g(A) = f(B) - f(C)$. If also $A = B' - C'$, then $B + C' = B' + C$. Positive additivity gives $f(B) + f(C') = f(B') + f(C)$, so the value is independent of the decomposition. Adding two differences and using positive additivity proves real additivity of g; positive scalar homogeneity and reversal of a difference prove real homogeneity, including negative scalars.

Extend to all bounded operators. Write uniquely $A = C + \mathrm{i}D$ with C, D self-adjoint, and set $\omega(A) = g(C) + \mathrm{i}g(D)$. Real linearity is already proved. Multiplication by i sends $C + \mathrm{i}D$ to

$-D + \mathrm{i}C$, so $\omega(\mathrm{i}A) = \mathrm{i}\omega(A)$; thus the extension is complex linear. On positives it is f, so it is positive and $\omega(\mathbb{1}) = 1$. On effects it is precisely μ.

Normality is where countable additivity enters. For orthogonal projections (E_k) the strong sum $E = \sum_k E_k$ is a projection at most $\mathbb{1}$, as established in § 1.3. The defining sum rule gives $\omega(E) = \mu(E) = \sum_k \mu(E_k) = \sum_k \omega(E_k)$. Thus ω is normal. By Theorem 1.7 it equals $\mathrm{Tr}(\rho\,\cdot)$ for a unique density operator. The extension was forced by μ, since positives are positive multiples of effects and every bounded operator is a complex combination of positives. This also proves uniqueness for the given effect probabilities.

Conversely, verify the whole sum rule. Write $\rho = \sum_j p_j \left|\varphi_j\right\rangle\!\left\langle\varphi_j\right|$ and let $F = \sum_k F_k \leq \mathbb{1}$ be an allowed strong sum of effects. Then

$$\mathrm{Tr}(\rho F) = \sum_j p_j \left\langle \varphi_j, F\varphi_j \right\rangle = \sum_j \sum_k p_j \left\langle \varphi_j, F_k\varphi_j \right\rangle = \sum_k \mathrm{Tr}(\rho F_k).$$

Strong convergence gives the middle equality for each vector; every summand is nonnegative, so the two scalar sums can be interchanged. Positivity, the bound by one and normalisation follow from L1. ∎

The extension steps required only finite additivity. In finite dimension this would already suffice, because every positive linear normalised functional is normal there. In infinite dimension the countable clause is essential for the conclusion to be a density operator: L1 cited the existence of non-normal algebraic states. Restricting such a functional to effects would preserve finite additivity, but not all the countable sums required here. The proof has therefore named exactly the additional assumption that makes its infinite-dimensional conclusion possible.

For a normal state and any ONB (e_k), the partial level events $E_N = \sum_{k\leq N} |e_k\rangle\!\langle e_k|$ must satisfy $\mu(E_N) \to 1$. There is no missing probability beyond all the levels. Finite additivity alone does not pass from E_N to its strong limit: $\|\mathbb{1} - E_N\| = 1$ in infinite dimension. Positive normalised functionals that vanish on every finite-rank operator do exist (· *cited*; Hall, Ex. 19.5, p. 439). Their projection assignments give $\mu(E_N) = 0$ for every N but $\mu(\mathbb{1}) = 1$. They are excluded here by the countable clause, not by an additional choice of basis or a finite-dimensional calculation.

Example 3.3 (Using the effect sum rule)**.** Suppose a qubit assignment gives $\mu(E_z) = 3/4$ and $\mu(E_x) = \mu(E_y) = 1/2$, where $E_a = (\mathbb{1} + \sigma_a)/2$. Homogeneity forces $\mu(\mathbb{1}/2) = 1/2$. Consider

$$F = \tfrac{1}{2}E_z + \tfrac{1}{4}E_x + \tfrac{1}{4}(\mathbb{1}/2) = \tfrac{1}{2}\mathbb{1} + \tfrac{1}{4}\sigma_z + \tfrac{1}{8}\sigma_x.$$

It is an effect because it is a convex combination of effects: both F and $\mathbb{1} - F$ are positive. Additivity applies to the three displayed positive summands even though the spin projections do not commute. It gives $\mu(F) = \frac{1}{2}(3/4) + \frac{1}{4}(1/2) + \frac{1}{4}(1/2) = 5/8$. The representing state has Bloch vector $(0, 0, 1/2)$ and density matrix $\mathrm{diag}(3/4, 1/4)$; multiplying matrices gives the same $\mathrm{Tr}(\rho F) = 5/8$.

By contrast, $B = 3F$ is not an effect. The positive extension gives $f(B) = 3\mu(F) = 15/8$, which is a possible expectation but cannot be an outcome probability. The construction distinguishes the units and range of an observable's mean from the dimensionless probability of an effect. It does not bound every positive observable's mean by one.

3.3 Gleason, and why a qubit escapes

Suppose probabilities are assigned only to projections, with $\mu(\mathbb{1}) = 1$ and countable additivity only on orthogonal families. This does not explicitly relate noncommuting effects. In most dimensions it still has a remarkable consequence, but the proof is much deeper than the extension argument just given.

Theorem 3.4 (Gleason · *cited*). *On a separable complex Hilbert space of dimension at least three, every countably additive probability on projections is $\mu(E) = \mathrm{Tr}(\rho E)$ for a unique density operator ρ.*

This is Landsman's Theorem 4.29, p. 120. We use its statement, not an unwritten proof. Its probability rule assigns the same number to a projection in every orthogonal resolution containing it. That shared assignment across different bases supplies much of the theorem's force.

In $\mathbb{C}^2$ every nonzero proper projection has the Bloch form $E(\mathbf{n}) = (\mathbb{1} + \mathbf{n}\cdot\boldsymbol{\sigma})/2$, with $|\mathbf{n}| = 1$. Its only nonzero orthogonal partner is $\mathbb{1} - E(\mathbf{n}) = E(-\mathbf{n})$. Hence any function $h : S^2 \to [0, 1]$ satisfying $h(-\mathbf{n}) = 1 - h(\mathbf{n})$ defines a projection probability: assign $\mu(E(\mathbf{n})) = h(\mathbf{n})$, $\mu(0) = 0$, $\mu(\mathbb{1}) = 1$. Every orthogonal sum is either trivial or this complementary pair, so all required additivity conditions hold.

For a continuous example take

$$h(\mathbf{n}) = \frac{1 + n_z^3}{2}.$$

If this were $\mathrm{Tr}(\rho E(\mathbf{n}))$ for a qubit density matrix with Bloch vector $\mathbf{r}$, it would equal $(1+\mathbf{r}\cdot\mathbf{n})/2$. The x and y axes force $r_x = r_y = 0$; the positive z axis forces $r_z = 1$. But at $\mathbf{n} = (\sqrt{3}/2, 0, 1/2)$ the prescribed value is $9/16$, whereas that density matrix gives $3/4$. This contradiction proves the dimension-two failure even for a continuous assignment.

A two-valued example will be useful in Lecture 13. For a unit $\mathbf{n}$, inspect (n_z, n_y, n_x) in that order and put $h_0(\mathbf{n}) = 1$ if its first nonzero entry is positive, and zero if that entry is negative. An antipodal pair has the same first nonzero position with opposite signs, so $h_0(-\mathbf{n}) = 1 - h_0(\mathbf{n})$. This assigns only zero or one to every qubit projection while respecting each orthogonal resolution. The equator and the two remaining axis points are included by the tie-breaking order.

Neither assignment is a quantum state missed by L1. By Busch, neither extends to an effect probability with the full sum property. In particular the two-valued assignment is not a positive linear expectation rule: L2 proves that every density operator has an event with probability strictly between zero and one. Different assumptions define different classes of assignments. This is the dimension-two escape that the Kochen–Specker discussion in Lecture 13 must retain.

3.4 Which events obey Boolean rules?

For projections, $E \wedge G$ denotes projection onto the intersection of their ranges; $E \vee G$ denotes projection onto their closed linear span. The complement is $\mathbb{1} - E$. The projections with these operations are called the *projection lattice*. The general existence of projections onto closed subspaces is the Hilbert projection theorem, cited in *Linear Algebra*, Lecture 13. Our computations below either construct the needed projection explicitly or take place in $\mathbb{C}^2$; they do not depend on that general cited theorem.

If $EG = GE$, then EG is self-adjoint and $(EG)^2 = EG$, with range exactly $\operatorname{ran} E \cap \operatorname{ran} G$. Likewise $E + G - EG = \mathbb{1} - (\mathbb{1} - E)(\mathbb{1} - G)$ is a projection. Its range is contained in the sum of the two ranges, and it fixes each of them, so it projects onto their sum. Thus

$$E \wedge G = EG, \qquad E \vee G = E + G - EG \quad (EG = GE).$$

For three pairwise commuting projections E, G, R, distributivity follows by multiplication:

$$E \wedge (G \vee R) = EG + ER - EGR = (E \wedge G) \vee (E \wedge R),$$

since $(EG)(ER) = EGR$. Complements, finite joins and finite meets of commuting projections still commute. These are the Boolean operations of ordinary events in one compatible family. QM-I's commutation criterion for compatibility explains their physical setting.

For instance, in a three-dimensional basis take $E = \operatorname{diag}(1, 1, 0)$ and $G = \operatorname{diag}(0, 1, 1)$. They overlap: $E \wedge G = \operatorname{diag}(0, 1, 0)$, while $E \vee G = \mathbb{1}$. In the state $\rho = \operatorname{diag}(1/2, 1/3, 1/6)$ their probabilities are $5/6$ and $1/2$, and the intersection probability is $1/3$. The ordinary inclusion–exclusion rule gives $5/6+1/2-1/3 = 1$ for the join. Simply adding the probabilities would double-count the shared event. Commutation gives a common Boolean description; it does not by itself say that two events are exclusive.

For incompatible events the same operations need not be distributive. In $\mathbb{C}^2$, let $E_1 = |0\rangle\langle 0|$, $E_2 = \big|(|0\rangle + |1\rangle)/\sqrt{2}\big\rangle\big\langle(|0\rangle + |1\rangle)/\sqrt{2}\big|$ and $E_3 = \big|(|0\rangle - |1\rangle)/\sqrt{2}\big\rangle\big\langle(|0\rangle - |1\rangle)/\sqrt{2}\big|$. Their ranges are shown in Figure 3.1.

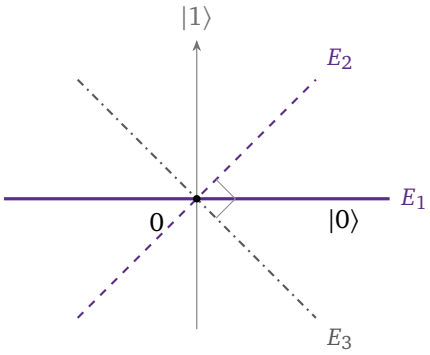


Figure 3.1: The real-coordinate section of the three complex rays. Each drawn line represents the complex span of its indicated vector: the orthogonal rays E_2, E_3 span $\mathbb{C}^2$, but each meets E_1 only at zero.

The last two rays span the plane, so $E_2 \vee E_3 = \mathbb{1}$. Each intersects the first ray only in zero. Consequently

$$E_1 \wedge (E_2 \vee E_3) = E_1 \neq 0 = (E_1 \wedge E_2) \vee (E_1 \wedge E_3).$$

These are subspace operations, not a claim that one can perform all three sharp readings without regard to their order. For instance $E_1 \vee E_2 = \mathbb{1}$, yet in the state $|0\rangle\langle 0|$ the sum of their probabilities is $1 + 1/2 = 3/2$, not one. A join of nonorthogonal rays is not generally a probability sum. The projection lattice does not provide one classical sample space for every incompatible reading at once.

3.5 Three paths, but only pair interference

Let E_A, E_B, E_C be mutually orthogonal path projections. Fix one unit preparation ψ, one subsequent unitary U and one detector effect $0 \leq D \leq \mathbb{1}$. Opening a subset S of ideal

paths means applying $E_S = \sum_{j\in S} E_j$. The common experimental arrangement is shown in Figure 3.2.

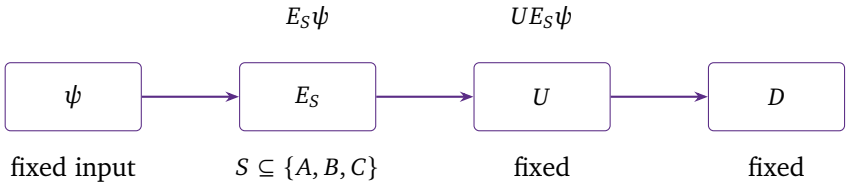


Figure 3.2: The ideal filter protocol. Only the subset S changes; the preparation, subsequent evolution and detector stay fixed. The transmitted vector $E_S\psi$ is not normalised again, and each detection count is divided by the same original-trial count to estimate its probability.

The detection probability per original trial is

$$p_S = \langle UE_S\psi, DUE_S\psi\rangle = \Big\|\sum_{j\in S}\eta_j\Big\|^2, \qquad \eta_j = \sqrt{D}\,UE_j\psi.$$

The square root is S6. Blocking paths discards amplitude; we do not renormalise the transmitted state. Since $D \le \mathbb{1}$ and E_S is a projection, $0 \le p_S \le 1$. There is no background in this ideal model, so $p_\varnothing = 0$.

Write $q_j = \|\eta_j\|^2$ and $b_{jk} = 2\,\mathrm{Re}\,\langle\eta_j, \eta_k\rangle$. Expanding the norm square gives $p_{jk} = q_j + q_k + b_{jk}$ and $p_{ABC} = p_A + p_B + p_C + b_{AB} + b_{AC} + b_{BC}$. Each pair contribution appears once in the triple and once among the pairs; each single appears once in the triple, twice among the pairs and once as a single. Thus

$$I_3 = p_{ABC} - p_{AB} - p_{AC} - p_{BC} + p_A + p_B + p_C = 0. \tag{3.2}$$

The ordinary two-path difference $p_{AB} - p_A - p_B = b_{AB}$ can be nonzero. The third-order combination, Sorkin's I_3, adds no new term because a norm square is quadratic. Three open paths can still interfere: p_{ABC} need not equal $p_A + p_B + p_C$, but its change is entirely a sum of pair contributions.

For a numerical ideal example take $\mathcal{H} = \mathbb{C}^3$, path basis e_A, e_B, e_C, $\psi = (e_A + e_B + e_C)/\sqrt{3}$, $U = \mathbb{1}$ and $D = |\psi\rangle\langle\psi|$. Then $p_S = |S|^2/9$: each single has $1/9$, each pair $4/9$, and the triple 1. Equation (3.2) reads $1 - 3(4/9) + 3(1/9) = 0$, whereas the two-path contribution is $2/9$. If each transmitted preparation were renormalised instead, its detection probability would be $|S|/3$ in this particular example. Here the conditioned I_3' happens still to vanish; the next detector shows why it need not. Such conditional probabilities describe a different protocol; they are not the quantities in the derivation. In general renormalisation destroys the common quadratic expression and its identity.

One can see a false residual using the same uniform preparation but the detector $D = |e_A\rangle\langle e_A|$. Before conditioning, $p_S = 1/3$ if $A \in S$ and zero otherwise, and $I_3 = 0$. The probability that any path passes is $|S|/3$. Dividing by that separate transmission probability gives $p_S' = 1/|S|$ when $A \in S$ and zero otherwise. These conditional numbers instead give $I_3' = 1/3 - 1/2 - 1/2 + 1 = 1/3$. All calculations obey Born's rule; the nonzero value comes from combining different conditioned preparations. This is why the common original trial count is a physical part of the proposed test, not merely a convention for writing p_S.

The assumptions making the mathematical test exact are now visible: the preparation, evolution and detector are held fixed; the filters add as orthogonal projections; and probabilities use the same original trial count. A different unitary for each slit configuration is

outside this identity. This is how a theorem becomes a proposed experimental comparison: its idealisation must be checked along with its numbers.

Probabilities and states

A countably additive probability on all effects, additive on every positive family summing to at most the identity, is exactly $\mathrm{Tr}(\rho\,\cdot)$. On projections alone, Gleason gives this conclusion in dimension at least three; a qubit admits other assignments. Commuting event projections obey Boolean rules; the full collection need not. An ideal self-adjoint observable's events are its spectral projections; together with those of an incompatible observable they need not form a Boolean algebra. The quadratic trace rule forces $I_3 = 0$ for the specified ideal path-filter protocol.

Summary. Busch derives linearity from the effect sum rule, then uses countable additivity to reach L1's normal states. Gleason reaches the same trace rule from fewer events but needs dimension at least three. Neither theorem derives its premises from probability alone. The event counterexample separates incompatible contexts, and the path calculation turns the probability law into a testable ideal identity. Next we ask which changes of rays preserve their transition probabilities.

Exercises. Try Sheet 3 in the separate *Exercise sheets* booklet (English).

Sources. Landsman, *Foundations of Quantum Theory*, Thm 2.44 pp. 73–74 (finite extension proof), §2.10 pp. 75–77, Thm 4.29 p. 120 (Gleason). P. Busch, *Phys. Rev. Lett.* **91**, 120403 (2003), arXiv:quant-ph/9909073. R. Sorkin, *Mod. Phys. Lett. A* **9**, 3119–3128 (1994), arXiv:gr-qc/9401003. U. Sinha et al., *Science* **329**, 418–421 (2010), arXiv:1007.4193. The infinite-dimensional normality step uses L1 directly, not a trace-duality theorem.

Part II

Symmetry and time

A symmetry preserves comparisons between pure states. We determine its possible mathematical forms, then add the group law and continuity needed for closed-system motion. The observable algebra also determines which relative phases can be detected.

Lecture 4

Symmetries are unitary or antiunitary

Objectives. *A bijection of all rays preserving transition probabilities comes from a unitary or an antiunitary operator. Its type is global, and its only remaining freedom is a phase. Complex conjugation supplies time reversal on the line; on a flux-threaded ring it reverses both momentum and flux. Every member of a one-parameter group of ray symmetries has unitary type.*

- Prove Wigner's theorem, including one global linear or antilinear choice and uniqueness up to phase.
- Compute time reversal on the line and a twisted ring, keeping operator domains and flux signs.
- Derive the odd ground-state current away from crossings, and explain why an antiunitary cannot be a member of a real one-parameter symmetry group.

4.1 What must a symmetry preserve?

QM-I called a unitary commuting with the Hamiltonian a symmetry (Definition 6.3). It also introduced an antiunitary time reversal (Definition 7.6). The two notions were not the same: time reversal is antilinear, so it was not a symmetry in the sense of Definition 6.3. Here we ask the kinematic question before specifying a Hamiltonian. Which changes of pure states preserve the probabilities for comparing them?

Definition 4.1 (Ray symmetry)**.** The *ray* $[\psi]$ of a nonzero vector is its one-dimensional complex span. For two rays their *transition probability* is

$$p([\psi],[\varphi]) = \frac{|\langle\psi,\varphi\rangle|^2}{\|\psi\|^2\,\|\varphi\|^2}.$$

A *ray symmetry* (or Wigner symmetry) is a bijection $\mathcal{S}$ of all rays preserving these transition probabilities.

Rescaling either vector cancels between numerator and denominator. For unit vectors the probability is the expectation of $|\varphi\rangle\langle\varphi|$ in $|\psi\rangle\langle\psi|$, as in QM-I and L2. Orthogonal rays have transition probability zero; identical rays have probability one. Both relations are therefore preserved by a symmetry and its inverse. The definition is about all rays of the specified space. Restrictions to coherent sectors will be stated separately in Lecture 5.

Definition 4.2 (Antiunitary)**.** An *antiunitary* $\mathcal{V} : \mathcal{H} \to \mathcal{H}$ is an onto antilinear map such that

$$\mathcal{V}(a\psi + b\varphi) = \bar{a}\mathcal{V}\psi + \bar{b}\mathcal{V}\varphi, \qquad \langle \mathcal{V}\psi, \mathcal{V}\varphi\rangle = \overline{\langle \psi, \varphi\rangle}.$$

A unitary is onto, linear and preserves the inner product itself.

Both kinds preserve norms and transition probabilities, and hence induce Wigner symmetries. Multiplying an implementation by a complex number of modulus one changes no image ray. It remains to prove the converse and to rule out a map that makes an independent linear/antilinear choice for each vector. That last consistency question is the main work.

4.2 Wigner's theorem: fixing an anchor ray

Theorem 4.3 (Wigner)**.** *On a separable complex Hilbert space with* $\dim \mathcal{H} \geq 2$*, every Wigner symmetry is induced by a unitary or antiunitary* $\mathcal{V}$*, meaning* $\mathcal{S}[\psi] = [\mathcal{V}\psi]$ *for every nonzero* ψ*. Its type is unique, and any two implementations differ by one complex phase.*

The proof has one organising idea: fix an output phase by comparison with an anchor, recover real inner products, and only then determine the complex type. Absolute overlaps alone do not fix arbitrary vector phases. The anchored superpositions supply the missing relative information.

Proof. 1. Choose an anchor and fix phases relative to it. Choose a unit vector e and a unit representative f of $\mathcal{S}[e]$. Put $K = e^{\perp}$ and $K' = f^{\perp}$. For nonzero $z \in K$, choose a unit representative g of $\mathcal{S}[z]$. Then $g \perp f$. The squared projections of $(e+z)/\sqrt{1+\|z\|^2}$ onto e and $z/\|z\|$ sum to one. The image ray has the same two probabilities with f, g, so a unit representative has no orthogonal remainder: Pythagoras puts it in $\operatorname{span}\{f, g\}$. Its f component is nonzero. Rescale uniquely to make that component equal to one, writing

$$\mathcal{S}[e+z] = [f + Wz], \qquad Wz \in \mathcal{S}[z], \quad Wz \perp f. \tag{4.1}$$

The transition with $[e]$ gives $1/(1+\|Wz\|^2) = 1/(1+\|z\|^2)$, so $\|Wz\| = \|z\|$. Put $W0 = 0$. This defines $W : K \to K'$ before any linearity assertion; the fixed anchor coefficient removes the otherwise arbitrary output phase.

2. Recover real inner products. For $z, w \in K$, transition preservation for their rays and their anchored rays gives

$$|\langle Wz, Ww\rangle| = |\langle z, w\rangle|, \qquad |1 + \langle Wz, Ww\rangle| = |1 + \langle z, w\rangle|.$$

The zero-vector cases hold directly. The normalising denominators cancel because norms were already shown equal. Subtract the squared equalities, using $|1+c|^2 = 1 + 2\operatorname{Re} c + |c|^2$, to obtain

$$\operatorname{Re}\langle Wz, Ww\rangle = \operatorname{Re}\langle z, w\rangle. \tag{4.2}$$

3. Real linearity follows from these Gram entries. Expand $\|W(z+w) - Wz - Ww\|^2$. Each norm and each real inner product in the expansion equals its input counterpart by (4.2). Thus the expansion is

$$\|z+w\|^2 + \|z\|^2 + \|w\|^2 - 2\operatorname{Re}\langle z+w, z\rangle - 2\operatorname{Re}\langle z+w, w\rangle + 2\operatorname{Re}\langle z, w\rangle = 0.$$

This is $\|(z+w)-z-w\|^2$, giving additivity. Equation (4.2) applies to every pair of inputs, including $z+w, z$ and $z+w, w$. For real r, the same argument applied to $\|W(rz)-rWz\|^2$ gives $W(rz)=rWz$. Explicitly, its terms are $r^2\|z\|^2+r^2\|z\|^2-2r\operatorname{Re}\langle rz,z\rangle=0$. In particular W is injective, since it preserves norms.

4. There is one global complex type. Since iz and z have the same ray, $W(iz)=\lambda Wz$ for some $|\lambda|=1$ when $z\neq 0$. Equation (4.2) gives $\operatorname{Re}\langle Wz, W(iz)\rangle=\operatorname{Re}\langle z, iz\rangle=0$, so $\operatorname{Re}\lambda=0$. Consequently $W(iz)=\sigma(z)iWz$, with $\sigma(z)$ either $+1$ or -1.

If z, w are real-linearly independent, real additivity gives

$$\sigma(z+w)i(Wz+Ww)=W(i(z+w))=\sigma(z)iWz+\sigma(w)iWw.$$

The vectors iWz, iWw are real-linearly independent: a real relation, after multiplication by $-i$, would contradict real linearity and injectivity of W. Comparing their coefficients therefore gives $\sigma(z)=\sigma(w)=\sigma(z+w)$. If instead $w=rz$ for nonzero real r, real homogeneity applied to $iw=r(iz)$ directly gives $\sigma(w)=\sigma(z)$. These cases cover every pair of nonzero vectors. Hence one sign works throughout K, even when K is one complex dimension.

In the plus case W is complex linear, since $W(iz)=iWz$; in the minus case it is antilinear. Combining real-inner-product preservation with $\operatorname{Re}\langle iz, w\rangle=\operatorname{Im}\langle z, w\rangle$ shows that W preserves the full inner product in the first case and conjugates it in the second.

5. Restore the anchor and all rays. Write $\chi(a)=a$ in the linear case, $\chi(a)=\bar{a}$ in the antilinear case, and define

$$\mathcal{V}(ae+z)=\chi(a)f+Wz \qquad (z\in K).$$

Orthogonality to the anchors proves the same norm and inner-product properties on all of $\mathcal{H}$. For $a=0$ the image-ray statement is already in the construction. For $a\neq 0$, use $[ae+z]=[e+a^{-1}z]$ in (4.1) and multiply the output representative by $\chi(a)$. Homogeneity of W then gives $\mathcal{S}[ae+z]=[\mathcal{V}(ae+z)]$. Its range is a complex subspace meeting every ray, by surjectivity of $\mathcal{S}$. It is therefore all of $\mathcal{H}$, making $\mathcal{V}$ unitary or antiunitary as claimed. This also covers vectors orthogonal to the anchor.

6. Prove uniqueness, including the type. If two implementations have the same type, their quotient is linear and fixes every ray. For independent x, y, write its actions as ax, by; fixing $[x+y]$ forces $a=b$. Collinear vectors have the same scalar by linearity, and comparison with an independent vector connects every ray. Thus the quotient is one scalar of modulus one. If the types differed, their quotient would be an antilinear map fixing all rays. On orthonormal e_1, e_2, fixing $[e_1+e_2]$ forces equal scalars on the two basis vectors. Antilinearity then maps $[e_1+ie_2]$ to $[e_1-ie_2]$, a different ray. This contradiction establishes type uniqueness for dimension at least two. ∎

The theorem is kinematic: it does not say that every such map commutes with the energy or is an available time evolution.

Example 4.4 (Why the complex type cannot vary with the vector)**.** On $\mathbb{C}^2$ take anchor $e=f=|0\rangle$ and write $z=\zeta|1\rangle$. Conjugation gives $W(\zeta|1\rangle)=\bar{\zeta}|1\rangle$ throughout the orthogonal line. A diagonal unitary $\operatorname{diag}(1, e^{i\beta})$ instead gives $W(\zeta|1\rangle)=e^{i\beta}\zeta|1\rangle$ throughout it. These are two consistent choices; the anchor proof does not choose the image of each individual coefficient independently.

To see what can go wrong, try using $W\zeta=\zeta$ when $\operatorname{Re}\zeta\geq 0$ but $W\zeta=\bar{\zeta}$ otherwise. For $z=(1+i)|1\rangle$ and $w=(-1+i)|1\rangle$, their inner product is $2i$. The attempted images have

inner product -2. The anchored vectors all have norm $\sqrt{3}$, but their transition probabilities would be

$$p([e+z],[e+w]) = \frac{|1+2i|^2}{9} = \frac{5}{9}, \qquad p([f+Wz],[f+Ww]) = \frac{|1-2|^2}{9} = \frac{1}{9}.$$

Thus the attempt already fails on ray probabilities, before questions about surjectivity arise. Rays lying in $e^{\perp}$ alone could not detect the error: that complex line has only one ray. The superpositions with the anchor reveal the relative phases that force consistency.

The mixed-state transformation sends ρ to $\mathcal{V}\rho\mathcal{V}^{-1}$. Each image is a complex-linear operator on $\mathcal{H}$, even when $\mathcal{V}$ is antilinear, because the two antilinear factors cancel. The transformation preserves convex combinations; on the full complex operator space it is conjugate-linear when $\mathcal{V}$ is antiunitary. Its spectral form is $\sum_j p_j \left|\mathcal{V}\varphi_j\right\rangle\!\left\langle\mathcal{V}\varphi_j\right|$; the eigenvalues and trace are unchanged. A related theorem of Kadison starts with the convex structure of mixed states rather than transition probabilities (named only; Moretti, Thm 12.14, p. 682).

4.3 Time reversal includes domains

On $L^2(\mathbb{R})$ let $(\Theta\psi)(x) = \overline{\psi(x)}$. Conjugation is an onto antilinear isometry, so it is antiunitary and $\Theta^2 = \mathbb{1}$. The position domain $D(Q) = \{\psi : x\psi \in L^2\}$ is invariant under it, and direct multiplication gives $\Theta Q\Theta^{-1} = Q$ with that domain.

For momentum use QM-II's Fourier characterisation, $P = \mathcal{F}^{-1}p\mathcal{F}$ on $\{\psi : p\mathcal{F}\psi \in L^2\}$. On Schwartz functions, the Fourier integral gives

$$\mathcal{F}(\Theta\psi)(p) = \overline{\mathcal{F}\psi(-p)}.$$

Both sides extend continuously to L^2 by S1 and S4. Reflection and conjugation preserve the condition $p\mathcal{F}\psi \in L^2$ and reverse the momentum multiplier's sign. Thus

$$\Theta D(P) = D(P), \qquad \Theta P\Theta^{-1} = -P.$$

This is an equality of operators with domains, not a derivative rule applied to arbitrary square-integrable functions. Positions are even and momenta odd under this time reversal.

If a self-adjoint H has a conjugation-invariant domain and $\Theta H\Theta^{-1} = H$, then $\Theta U_t\Theta^{-1} = U_{-t}$. Indeed the transformed group has Hamiltonian $-H$, equivalently generator $-H/\hbar$: antilinearity reverses the scalar i in its strong derivative on $D(H)$. QM-II, Proposition 7.9, on uniqueness of the unitary group with a given generator supplies the equality.

4.4 A ring's persistent current

Recall the twisted momentum of QM-II, Theorem 3.10 (self-adjointness sketched there): $P_\theta = -i\hbar\, d/dx$ on absolutely continuous functions with $\psi' \in L^2(0,L)$ and $\psi(L) = e^{i\theta}\psi(0)$. We allow real θ, with equivalent twists modulo 2π. Its normalised eigenvectors and eigenvalues are

$$e_{\theta,k}(x) = L^{-1/2}e^{i(2\pi k+\theta)x/L}, \qquad p_k(\theta) = \frac{\hbar}{L}(2\pi k+\theta), \quad k \in \mathbb{Z}.$$

The same conjugation Θ on $L^2(0,L)$ takes the boundary condition at θ into the one at $-\theta$. On these domains its derivative changes the sign of momentum:

$$\Theta D(P_\theta) = D(P_{-\theta}), \qquad \Theta P_\theta\Theta^{-1} = -P_{-\theta}, \qquad \Theta e_{\theta,k} = e_{-\theta,-k}.$$

For $H_\theta = P_\theta^2/(2m)$ the spectral levels are

$$E_k(\theta) = \frac{\hbar^2}{2mL^2}(2\pi k + \theta)^2, \qquad E_k(-\theta) = E_{-k}(\theta).$$

Thus $\Theta H_\theta \Theta^{-1} = H_{-\theta}$ with domains. If $H_\theta \psi = E\psi$, then $H_{-\theta}\Theta\psi = E\Theta\psi$, since E is real. The paired Hamiltonians have identical eigenvalues and multiplicities, so their ground energies are even in flux. Wherever the ground branch is differentiable, its flux derivative is odd. The square's domain is also carried to the opposite twist's domain, since both successive applications of momentum must exist. Equivalently, its eigenbasis domain is the square-summability condition with fourth powers of $2\pi k + \theta$, invariant under $(k, \theta) \mapsto (-k, -\theta)$.

To fix the physical sign, let the particle have charge $q \neq 0$ and define the twist by $\theta = -q\Phi/\hbar$, where Φ is the magnetic flux. As in QM-II, §3.4, this follows by removing the constant vector potential Φ/L from the periodic Hamiltonian $(P_0 - q\Phi/L)^2/(2m)$. Write the periodic wavefunction as $\psi_{\text{per}}(x) = e^{iq\Phi x/(\hbar L)}\varphi(x)$. Differentiation gives

$$(P_0 - q\Phi/L)\psi_{\text{per}} = e^{iq\Phi x/(\hbar L)}(-i\hbar\varphi'), \qquad \varphi(L) = e^{-iq\Phi/\hbar}\varphi(0).$$

The phase transformation is unitary and preserves absolute continuity and square-integrable derivatives. It moves the flux dependence from the differential rule into the boundary condition, with exactly the declared sign. Squaring the transformed momentum gives the same level formula and transformed square domain. Define the level current as charge times velocity divided by circumference, $I_k = qp_k(\theta)/(mL)$. It equals the flux derivative:

$$I_k = -\frac{\partial E_k}{\partial \Phi} = \frac{q}{\hbar}\frac{\partial E_k}{\partial \theta} = \frac{q\hbar}{mL^2}(2\pi k + \theta).$$

Differentiating with respect to a dimensionless twist without its flux conversion would miss this factor and could reverse the quoted sign.

For $-\pi < \theta < \pi$ the unique ground branch is $k = 0$, so

$$E_g(\Phi) = \frac{q^2\Phi^2}{2mL^2}, \qquad I_g(\Phi) = -\frac{q^2\Phi}{mL^2} \quad (|q\Phi/\hbar| < \pi).$$

The current is odd in flux and vanishes at zero flux. The spectrum and ground energy are periodic with flux period $2\pi\hbar/|q|$, by relabelling k. At odd multiples of π in twist, two lowest branches cross; the ground energy has a cusp. There is no single derivative or unique ground-state current there without specifying the state in the degenerate space. Time reversal pairs the opposite currents rather than selecting one.

The level crossings and their current branches are displayed in Figure 4.1. The plotted current ratio rises with twist on each branch; converting twist to flux gives the sign in the preceding formula.

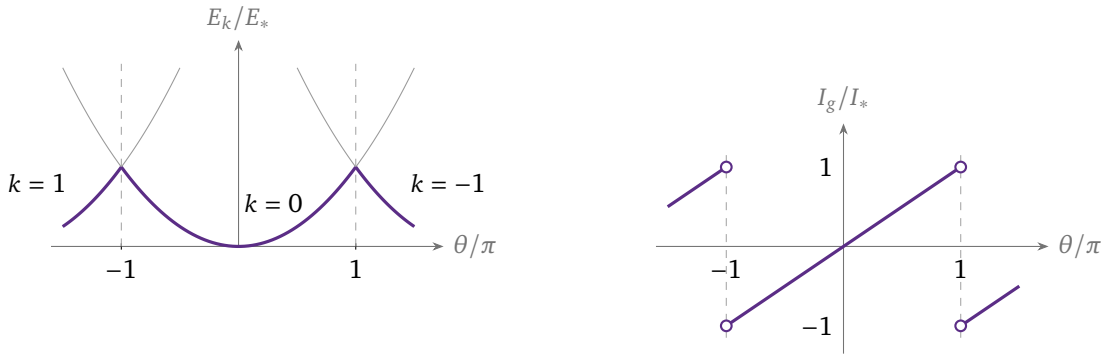


Figure 4.1: Ring levels and the ground-branch current, with $E_* = \hbar^2\pi^2/(2mL^2)$ and $I_* = q\hbar\pi/(mL^2)$, for $q \neq 0$. The thick energy curve is the lower envelope of the level parabolas; open current endpoints mark the two possible branch values at a crossing, not a uniquely selected current. The horizontal variable is twist, with $\theta = -q\Phi/\hbar$.

4.5 A group in time cannot contain an antiunitary

On $\dim\mathcal{H} \geq 2$, suppose ray symmetries satisfy $\mathcal{S}_{s+t} = \mathcal{S}_s\mathcal{S}_t$ for all real s, t. Each is implemented by Wigner's theorem, but

$$\mathcal{S}_t = (\mathcal{S}_{t/2})^2.$$

The square of a unitary is linear; the square of an antiunitary is also linear, because conjugation of scalar coefficients occurs twice. Both squares are unitary. Type uniqueness in Theorem 4.3 therefore forces every $\mathcal{S}_t$ to have unitary type. Continuity has not even been used in this argument; it will be needed for the generator in L5.

On a qubit, conjugation in the σ_z basis sends ρ to $\bar{\rho} = \rho^{\mathsf{T}}$ and its Bloch vector to $(r_x, -r_y, r_z)$. It preserves transition probabilities, but it has antiunitary type. It cannot occur as one member of such a real one-parameter group. This does not make time reversal unphysical: time reversal changes the orientation of time, whereas the group of time translations moves forward or backward within that orientation.

Integer time labels would be different: set the ray map to the identity at every even integer and to conjugation at every odd integer. Since conjugation squares to the identity, these maps form a group over $\mathbb{Z}$ and include an antiunitary type. Extending it to all real times would require a half-step whose square had that type, which Wigner forbids. The conclusion is about a group with arbitrarily divisible time labels, not a prohibition of every discrete antilinear transformation.

The remaining obstruction is phase. Choosing a unitary implementation separately at every time need not give $U_{s+t} = U_sU_t$; it gives that identity only up to a phase. Wigner alone has not derived the evolution postulate. Lecture 5 states the additional continuity premise and the lifting theorem that remove this obstruction.

For a concrete example of uncoordinated phases, take any genuine group U_t and a fixed time scale $\tau > 0$. The representatives $\mathcal{V}_t = e^{i(t/\tau)^2}U_t$ induce exactly the same ray maps, but

$$\mathcal{V}_s\mathcal{V}_t = e^{-2ist/\tau^2}\mathcal{V}_{s+t}.$$

This follows by subtracting $(s+t)^2$ from s^2+t^2. Each representative is unitary and depends continuously on time, yet the chosen family is not a group. Multiplying back by $e^{-i(t/\tau)^2}$ restores the group. Coordination of phases is therefore an extra step; continuity of an arbitrary choice of representatives does not by itself supply the group law.

Which changes of pure states are symmetries?

For dimension at least two, a bijection of all rays preserving transition probabilities has one unitary or antiunitary implementation, unique up to phase and with a unique type. A real one-parameter group contains only unitary-type maps, but its individual phases still have to be coordinated. Time reversal is antiunitary and reverses momenta and magnetic flux; its domain action is part of each operator statement.

Summary. Wigner's theorem derives the allowed linear and antilinear forms from ray probabilities. The anchor proof establishes one global type, even in infinite dimension. On a ring, time reversal pairs opposite twists, levels and currents; the current derivative needs a nondegenerate branch. The square argument excludes antiunitaries from a real translation group. The group law up to phase remains to be addressed.

Exercises. Try Sheet 4 in the separate *Exercise sheets* booklet (English).

Sources. Galindo–Pascual, *Quantum Mechanics I*, §7.2 pp. 262–266; Weinberg, *The Quantum Theory of Fields I*, App. 2.A pp. 91–96; Moretti, *Spectral Theory and Quantum Mechanics*, Thm 12.11 pp. 675–681. The real-linearity/global-sign completion of the anchor proof is supplied here. The ring operator and its eigenbasis are recalled from QM-II, Theorem 3.10; the line operators from Theorem 4.6.

Lecture 5

Time, and the symmetries that forbid superpositions

Objectives. *Time translations supply more structure than separate symmetries. Their phases can be coordinated, by a cited theorem, into the unitary group of QM-II. A lower energy bound then excludes an operator that translates energy by arbitrary amounts. Rotations raise a different question: which rays belong to one coherent sector, and which observables can compare them?*

- State the continuity and lifting premises behind $e^{-itH/\hbar}$, and track its phase freedom.
- Prove Pauli's obstruction with domains, and explain why a bare commutator is insufficient.
- Derive the rotation-sector restriction, represent the states of its observable algebra, and compute a neutron's 4π interference period.

5.1 From ray translations to a Hamiltonian

Lecture 4 classified each symmetry separately. A time evolution must also compose consistently: evolving for s, then t, is evolving for $s+t$. We assume a nonzero separable complex $\mathcal{H}$, with dimension at least two. The one-dimensional case has only one ray and no nontrivial ray dynamics.

Definition 5.1 (Continuous group of ray symmetries)**.** A family $\mathcal{S}_t$, $t \in \mathbb{R}$, consists of ray symmetries and obeys $\mathcal{S}_0 = \mathrm{id}$ and $\mathcal{S}_{s+t} = \mathcal{S}_s\mathcal{S}_t$. Continuity means that $t \mapsto p(\mathcal{S}_t[\psi], [\varphi])$ is continuous for every pair of rays, with the transition probability of Lecture 4.

This is continuity of every pure-state comparison probability, rather than differentiability of a wavefunction. By the half-time square argument, every member has unitary type. Choose representatives $\mathcal{V}_t$. Their products induce the same ray maps as $\mathcal{V}_{s+t}$, so uniqueness in Wigner's theorem gives

$$\mathcal{V}_s\mathcal{V}_t = c(s,t)\mathcal{V}_{s+t}, \qquad |c(s,t)| = 1.$$

This is a *projective* group law: the residual scalar is invisible on rays. Even continuous choices need not have $c = 1$, as Lecture 4's quadratic phase example showed. What is needed is a simultaneous choice of phases compatible with every addition of times.

Theorem 5.2 (One-parameter lift, cited). *Every continuous group of ray symmetries as above has a strongly continuous unitary group U_t implementing it: $\mathcal{S}_t[\psi] = [U_t\psi]$, $U_{s+t} = U_sU_t$, $U_0 = \mathbb{1}$, and $\|U_t\psi - U_{t_0}\psi\| \to 0$ as $t \to t_0$ for every ψ. Any two such group lifts differ by e^{ict} with one real constant c.*

This is Moretti, Theorem 12.45, pp. 714–715; continuity is his Definition 12.40, p. 708. Its proof uses tools beyond our prerequisites and is cited, not derived from Wigner alone. The stated result concerns the real one-parameter group.

QM-II proved that a strongly continuous unitary group has a self-adjoint generator (Theorem 7.8, Stone); its exponential correspondence was sketched there (Theorem 7.10). In that convention the generator is $K = H/\hbar$. Consequently, given the cited lift and that correspondence,

$$U_t = e^{-itH/\hbar}, \qquad H = H^\dagger \text{ on } D(H).$$

This deduction has those stated inputs; it is not a fresh proof of the lifting theorem or the general spectral exponential. Multiplying the lift by e^{ict} replaces H by $H - \hbar c\mathbb{1}$, with the same domain. The constant has units of inverse time, so $\hbar c$ is an energy shift. Ray probabilities cannot fix an absolute energy origin. They also do not select a particular Hamiltonian: the chosen ray dynamics are a premise.

There are three distinct assumptions in this chain. Transition probabilities determine the kind of each symmetry by Wigner. The group law relates distinct times, and probability continuity rules out abrupt ray motion. The lifting theorem then coordinates phases, after which Stone identifies the generator. Dropping the group law permits driven families whose composition depends on the starting time; dropping probability continuity removes the lift's hypothesis, and with it the strong continuity Stone needs. Neither closed-system unitarity nor a time-independent Hamiltonian describes every reduced system: tracing out a partner will lead to different evolution in L7. Thus the theorem explains the form of a specified closed-system dynamics under its premises, rather than assuming every time-indexed state change has this form.

For a density operator write $\rho = \sum_j p_j \left|\varphi_j\right\rangle\!\left\langle\varphi_j\right|$ as in L1. The evolved vectors are still an orthonormal family, so

$$\rho_t = U_t\rho U_t^\dagger = \sum_j p_j \left|U_t\varphi_j\right\rangle\!\left\langle U_t\varphi_j\right|$$

is positive and has trace one. For every bounded A, absolute convergence of these expectation series gives

$$\mathrm{Tr}(\rho_t A) = \sum_j p_j \left\langle U_t\varphi_j, AU_t\varphi_j\right\rangle = \mathrm{Tr}(\rho U_t^\dagger A U_t). \tag{5.1}$$

Thus the Schrödinger and Heisenberg descriptions of QM-I, Proposition 4.5, agree for all bounded readings. No cyclic trace rule for arbitrary unbounded products is needed. Differentiating this identity for an unbounded H requires extra domain conditions; the identity itself holds for every density operator.

Example 5.3 (Two energies and one energy origin). For $H = \mathrm{diag}(\varepsilon_0, \varepsilon_1)$ and $\psi = (|0\rangle + |1\rangle)/\sqrt{2}$, the evolved ray is represented by $(|0\rangle + e^{-it(\varepsilon_1-\varepsilon_0)/\hbar}\,|1\rangle)/\sqrt{2}$. Its probability for the initial ray is $[1 + \cos((\varepsilon_1 - \varepsilon_0)t/\hbar)]/2$. Adding $b\mathbb{1}$ to H changes only the removed common phase $e^{-itb/\hbar}$. The energy difference, and hence the oscillation period, is unchanged by any common energy shift.

5.2 What Pauli's obstruction actually excludes

A position translation shifts position by any real amount. Could a self-adjoint time operator similarly translate energy? Specify the proposed operation before using the formal expression $[T, H] = i\hbar\mathbb{1}$.

Theorem 5.4 (Pauli, energy-covariant form)**.** *Let H be self-adjoint on a nonzero $\mathcal{H}$ and bounded below: $\langle\psi, H\psi\rangle \geq E_{\min}\,\|\psi\|^2$ on $D(H)$. There is no self-adjoint T whose unitaries $W_s = e^{isT/\hbar}$ obey, for every real energy s,*

$$W_s D(H) = D(H), \qquad W_s H W_s^{-1} = H - s\mathbb{1}. \tag{5.2}$$

Proof. Choose a unit $\psi \in D(H)$; such a vector exists because a self-adjoint operator is densely defined. Domain invariance puts $\eta = W_s^{-1}\psi$ in $D(H)$, and unitarity gives $\|\eta\| = 1$. Thus

$$E_{\min} \leq \langle\eta, H\eta\rangle = \left\langle\psi, W_s H W_s^{-1}\psi\right\rangle = \langle\psi, H\psi\rangle - s.$$

The fixed expectation on the right is finite. Taking s larger than $\langle\psi, H\psi\rangle - E_{\min}$ contradicts the inequality. ∎

The argument actually excludes any family of energy-shifting unitaries with these properties. It does not need $\psi \in D(T)$ and never differentiates an unbounded product. Self-adjointness of T would supply unitaries for all real s; the lower bound prevents their proposed action. With $U_t = e^{-itH/\hbar}$, covariance would imply the Weyl relation $W_s U_t = e^{ist/\hbar} U_t W_s$. To justify this without differentiating a product on an unspecified common domain, fix s. The conjugated time group has Hamiltonian $W_s H W_s^{-1} = H - s\mathbb{1}$ by its strong derivative on $D(H)$. The group $e^{ist/\hbar}U_t$ has that same Hamiltonian and domain, so QM-II, Proposition 7.9, identifies them. The plus sign also follows from $e^{-it(H-s)/\hbar} = e^{ist/\hbar}e^{-itH/\hbar}$. Formally differentiating covariance at zero yields $[T, H] = i\hbar\mathbb{1}$, but reversing this implication needs hypotheses not contained in a commutator on a common test domain.

Here is a concrete warning. On $L^2(0, L)$ take the self-adjoint twisted momentum P_θ recalled in L4 and a constant $\alpha > 0$ with units energy per length. Set $H = \alpha Q$, $T = -P_\theta/\alpha$. The bounded multiplication operator H is nonnegative; T is self-adjoint conditional on that recalled momentum result (QM-II, Theorem 3.10, sketched there). For $\psi \in C_c^\infty(0, L)$ both products exist and differentiation gives

$$[T, H]\psi = i\hbar\psi.$$

Nevertheless (5.2) cannot hold. Here $D(H) = L^2(0, L)$, so its invariance is automatic. It is the energy-shift identity that fails: unitary conjugation preserves positivity, whereas $H - s\mathbb{1}$ has negative expectations for sufficiently large s. A further boundary warning concerns the different domain $D(T)$: multiplication by x need not preserve it, since $x\psi(x)$ has endpoint values $0, L\psi(L)$. The compact-support commutator controls neither the global energy-shift identity nor every operator domain.

This theorem is about a self-adjoint, globally energy-covariant time. It is not a prohibition of recording arrival times or waiting times. Time POVMs, a generalised reading introduced in L9, can be covariant; see Busch–Lahti–Pellonpää–Ylinen, p. 390 and Theorem 17.1, p. 395. The familiar short-time survival law of QM-II also refers to elapsed time as a parameter and requires no operator satisfying (5.2).

5.3 Rotations and coherent sectors

Time translations have a group lift. Rotations illustrate a different possibility. QM-I, Theorem 7.3, gives for each finite spin multiplet a rotation by 2π as $(-1)^{2j}\mathbb{1}$: plus for integer j, minus for half-integer j. Each is a common phase within its own multiplet, hence acts trivially on its rays, as the identity rotation in ordinary three-dimensional space must do.

Call a subspace *coherent for a projective rotation action* if all its rays are admitted as pure states and the ordinary rotation group $SO(3)$ acts on them with a well-defined projective action. Here $SO(3)$ means the ordinary proper rotations of space; "projective" again allows one common phase for each implementing operator, not a separate phase for each component of a vector. We assume that rotations about z are implemented, up to that common phase, by the spin operators $e^{-i\theta J_z/\hbar}$ of those multiplets. This connects the ray action to the stated physical angular momentum; an unrelated abstract rotation action would not suffice. Integer and half-integer blocks may be orthogonal sums of finite multiplets: their unitary rotations act blockwise and extend by norm limits of finite component sums. Thus the 2π signs remain $+1$ and -1 on the respective whole blocks.

Suppose one such subspace contained nonzero integer and half-integer components u, v. They are orthogonal spin sectors. Their sum transforms under 2π as $u + v \mapsto u - v$. These are different rays: if $u - v = a(u + v)$, comparison of the orthogonal components gives both $a = 1$ and $a = -1$. But 2π is the identity element of $SO(3)$ and must fix every ray. This proves the *univalence restriction*: one coherent sector carrying the stated projective action cannot contain both types.

The vector $u+v$ certainly exists in the direct-sum Hilbert space. The conclusion depends on which rays and which group action are physically admitted. Using the group of spin rotation operators instead changes the premise: in that group the 2π spin rotation is not the identity. A group name without its action on states is insufficient to establish a superselection rule. The spin rotation group $SU(2)$ distinguishes a 2π rotation from its identity; a 4π rotation is its identity. Admitting a projective action of the ordinary group $SO(3)$ on every physical ray is therefore a substantive premise. The action within each multiplet alone does not establish an empirical absence of coherence between the two spin types. That absence requires a specification of which preparations and observations are physically available. To describe the restricted observations explicitly, we now specify an algebra rather than silently discarding the Hilbert-space vectors.

5.4 A state belongs to an observable algebra

Let $\mathcal{H} = \mathcal{H}_{\text{int}} \oplus \mathcal{H}_{\text{half}}$, with both blocks nonzero and separable, and write their orthogonal projections as $E_{\text{int}}, E_{\text{half}}$. Define the bounded sign operator $\Gamma_s = E_{\text{int}} - E_{\text{half}}$, and specify the observable algebra

$$\mathcal{A} = \{A \in \mathcal{B}(\mathcal{H}) : A\Gamma_s = \Gamma_s A\} = \mathcal{B}(\mathcal{H}_{\text{int}}) \oplus \mathcal{B}(\mathcal{H}_{\text{half}}). \tag{5.3}$$

An *algebra* here is a collection containing the identity and closed under sums, complex scalar multiples, products and adjoints. These properties follow immediately from commutation with the self-adjoint Γ_s. Since $E_{\text{int}} = (\mathbb{1}+\Gamma_s)/2$ and $E_{\text{half}} = (\mathbb{1}-\Gamma_s)/2$, a commuting A preserves both blocks, proving the second equality; every block-diagonal A conversely commutes. Bounded observables are the self-adjoint members of $\mathcal{A}$. For an unbounded observable its spectral projections must belong to $\mathcal{A}$; merely a formal commutator is not its definition.

A state on $\mathcal{A}$ is positive, complex-linear, normalised and normal, as in L1, with orthogonal projection sums now taken in $\mathcal{A}$. There are no allowed cross-block observations in this specification. A pure state means an extreme point of the convex set of states of $\mathcal{A}$, as in L2 with this specified state set. That physical restriction is the *superselection rule*; positivity alone did not force us to choose this algebra.

Theorem 5.5 (Normal states of the two-block algebra)**.** *Every state of $\mathcal{A}$ is represented uniquely among block-diagonal density operators by*

$$\rho = w_{\text{int}}\sigma_{\text{int}} \oplus w_{\text{half}}\sigma_{\text{half}}, \qquad w_j \geq 0, \quad w_{\text{int}} + w_{\text{half}} = 1.$$

Here a σ_j is a density operator on block j when $w_j > 0$; its choice is immaterial when $w_j = 0$. Conversely every such operator defines a normal state on $\mathcal{A}$ by $\omega(A) = \operatorname{Tr}(\rho A)$. The pure normal states are precisely the rays in a single block.

Proof. Put $w_j = \omega(E_j)$. Positivity and $E_{\text{int}} + E_{\text{half}} = \mathbb{1}$ give the claimed weights. If $w_j > 0$, restrict to operators supported on that block and divide by w_j. This is a positive normalised functional on $\mathcal{B}(\mathcal{H}_j)$; orthogonal projection sums there are also such sums in $\mathcal{A}$, so it is normal. L1 supplies a unique σ_j. If $w_j = 0$, then for a positive operator B supported on that block, $0 \leq B \leq \|B\| E_j$ gives $0 \leq \omega(B) \leq 0$. Any bounded self-adjoint operator supported there is a difference of positive ones by adding a multiple of E_j; any bounded operator is a complex combination of two self-adjoint ones. Thus its expectation is zero. Splitting any A into its two blocks proves the representation, and the sum of the two positive traces is one.

Conversely the two density expectation rules are normal by L1, and their weighted sum is normal on $\mathcal{A}$. The weights are recoverable as $\omega(E_j)$, and the positive-weight restrictions determine σ_j; hence the block-diagonal ρ is unique. If both weights are positive, the state is a nontrivial convex mixture of two distinct sector states, so is not pure. If one weight is one, any convex decomposition into states of $\mathcal{A}$ must have zero weight in the other block, by positivity. Purity then reduces to purity on the surviving full block algebra, which L2 identifies with a ray. ∎

For unit $u \in \mathcal{H}_{\text{int}}$, $v \in \mathcal{H}_{\text{half}}$ and $0 < w < 1$, consider $\psi = \sqrt{w}\, u + e^{i\beta}\sqrt{1-w}\, v$. Every $A = A_{\text{int}} \oplus A_{\text{half}}$ has

$$\langle \psi, A\psi \rangle = w \langle u, A_{\text{int}} u \rangle + (1-w) \langle v, A_{\text{half}} v \rangle = \operatorname{Tr}((w\,|u\rangle\langle u| \oplus (1-w)\,|v\rangle\langle v|)A)\,.$$

The relative phase β disappears because all cross-block matrix elements vanish. The ray and the block mixture are the same state *on this algebra*. They are distinguishable on the full $\mathcal{B}(\mathcal{H})$, by the bounded self-adjoint operator $X_\beta = e^{-i\beta}\,|u\rangle\langle v| + e^{i\beta}\,|v\rangle\langle u|$. Its mean is $2\sqrt{w(1-w)}$ in the ray and zero in the mixture, for every stated phase; X_β is excluded from $\mathcal{A}$. L2's pure-ray theorem concerned the full algebra; restricting observations can turn its pure state into a mixed state. The pair "states and allowed observables" must travel together when specifying a quantum system.

Example 5.6 (A sector weight is not a probability amplitude)**.** Take $\mathcal{H}_{\text{int}} = \mathbb{C}$ and $\mathcal{H}_{\text{half}} = \mathbb{C}^2$. Specify $w_{\text{int}} = 1/3$, $\sigma_{\text{half}} = \mathbb{1}_2/2$. The unique block-diagonal representative is $\rho = \operatorname{diag}(1/3, 1/3, 1/3)$. For an allowed observable $A = a \oplus (b\mathbb{1}_2 + \mathbf{c}\cdot\boldsymbol{\sigma})$, with real coefficients, its expectation is $a/3 + 2b/3$. The Pauli terms average to zero because the state within the half-integer block is mixed. The event E_{half} has probability $2/3$; conditioned on that sector the normalised state is $\mathbb{1}_2/2$, not $(2/3)\mathbb{1}_2/2$. The weights describe classical uncertainty between sectors; the density operators describe the remaining uncertainty within them.

Replace the half-block state by $|{\uparrow}\rangle\langle{\uparrow}|$. The resulting state is still mixed on $\mathcal{A}$, even though each nonzero block state is pure: reading the sector projection distinguishes its two components with probabilities $1/3, 2/3$. It is the same restricted state as the Hilbert-space ray $\sqrt{1/3}\,u + e^{i\beta}\sqrt{2/3}\;\uparrow$, for every β. Changing the phase cannot change any allowed expectation. Removing the algebra restriction would restore observations sensitive to that phase, so it would describe a different physical system.

5.5 A relative 2π rotation is measurable

A neutron has spin one half. If a 2π rotation changes only a common phase, why do neutron interferometers detect the sign? The operation is relative to a second path. Use QM-I's finite tensor product, with path first and spin second. Prepare

$$\Psi_0 = \frac{|0\rangle + |1\rangle}{\sqrt{2}} \otimes |{\uparrow}\rangle\,, \qquad R_z(\theta) = e^{-i\theta\sigma_z/2}.$$

The separate path operations are shown in Figure 5.1.

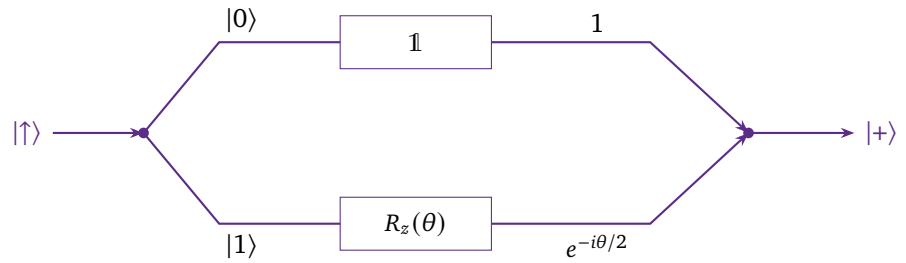


Figure 5.1: The ideal two-arm neutron experiment; only the plus output port is shown. The spin rotation acts only in path 1; the indicated factors are the relative path amplitudes before the common factor $1/\sqrt{2}$. Recombination compares them in the plus output, so a sign on one arm is observable even though a sign on both arms would be a global phase.

Rotate spin only in path 1, by the unitary $|0\rangle\langle 0| \otimes \mathbb{1} + |1\rangle\langle 1| \otimes R_z(\theta)$. It gives

$$\Psi_\theta = \frac{|0\rangle + e^{-i\theta/2}\,|1\rangle}{\sqrt{2}} \otimes |{\uparrow}\rangle\,.$$

Recombination tests the plus path $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$. The spin amplitude there is $(1 + e^{-i\theta/2})\,|{\uparrow}\rangle\,/2$, so

$$p_+(\theta) = \frac{1 + \cos(\theta/2)}{2}. \tag{5.4}$$

At 2π the paths cancel at this output; at 4π they add again. Rotating *both* paths by 2π instead multiplies the whole state by -1 and changes no probability. Both branches here have half-integer spin; this experiment does not superpose integer and half-integer sectors. The controlled one-arm operation is not the identity spatial rotation acting globally on the system.

Why unitary time evolution, and which superpositions?

A continuous real group of ray symmetries has, by the cited lift, a strongly continuous unitary group. With Stone's correspondence its action is $\rho \mapsto U_t \rho U_t^\dagger$, $U_t = e^{-itH/\hbar}$, with H self-adjoint and fixed up to an energy constant. A lower bound excludes a self-adjoint time that translates energy globally. A superselection rule specifies a smaller observable algebra; its states are expectation rules on that algebra, and its pure states need not be all Hilbert-space rays.

Summary. The group premise, probability continuity and cited lift explain the structure of time evolution. Pauli's proof needs global covariance with domains, not a formal commutator. The projective rotation premise separates integer and half-integer coherent sectors. Their block algebra has unique block-diagonal density representatives, and its pure states are sector rays. A neutron's relative rotation measures a 4π path fringe without comparing the two superselected spin types.

Exercises. Try Sheet 5 in the separate *Exercise sheets* booklet (English).

Sources. Moretti, *Spectral Theory and Quantum Mechanics*, Theorem 12.45 pp. 714–715, Definition 12.40 p. 708; Pauli discussion pp. 827–828 (our proof is the direct lower-bound argument, with the displayed course sign convention). Superselection: §7.7 pp. 378–379, §12.3.4 p. 782; Weinberg, *The Quantum Theory of Fields I*, §2.7 pp. 81–90. Busch–Lahti–Pellonpää–Ylinen, *Quantum Measurement*, pp. 390, 395. Experimental and mass-rule citations accompany the optional applications in the full edition.

Part III

Parts and open systems

A part of an entangled pair can have a mixed state even when the pair is pure. Ignoring a partner also changes the allowed form of evolution. We build channels from this requirement and then examine the additional assumption that open-system dynamics forgets its history.

Lecture 6

Composite systems in infinite dimension: EPR

Objectives. *A part of an entangled pure state can have a mixed state even with infinitely many levels. We construct the composite space and its Schmidt form, then define the partial trace by the observations made on one part. EPR's position–momentum correlations motivate a quantitative test: a separable preparation has a lower bound on two collective variances.*

- Prove the Schmidt decomposition for a unit vector of two separable Hilbert spaces and identify its reduced states.
- Construct the partial trace from bounded expectations, including mixed composite states.
- State EPR's premises and prove a separability bound; evaluate it on the specified squeezed covariances without treating a delta function as a state.

6.1 The composite as square-summable arrays

QM-I used a finite tensor product: coefficients c_{jk} specified a vector $\sum_{jk} c_{jk} e_j \otimes f_k$. Now let $\mathcal{H}_1, \mathcal{H}_2$ be nonzero separable complex Hilbert spaces, with fixed orthonormal bases (e_j), (f_k). Their *Hilbert tensor product* is the space of arrays with

$$\Psi = \sum_{j,k} c_{jk} e_j \otimes f_k, \qquad \|\Psi\|^2 = \sum_{j,k} |c_{jk}|^2 < \infty.$$

The inner product is the corresponding sum $\sum \bar{c}_{jk} d_{jk}$; Cauchy–Schwarz makes it convergent. Finite arrays are dense, by truncating the nonnegative norm sum. The product $u \otimes v$ has coefficients $u_j v_k$ and norm $\|u\| \, \|v\|$. For any other ONBs (u_ℓ), (v_m), their products form an ONB in this same space. Orthonormality follows from $\langle u \otimes v, u' \otimes v' \rangle = \langle u, u' \rangle \, \langle v, v' \rangle$, first on finite sums and then by Parseval. For completeness, if Ψ is orthogonal to every $u_\ell \otimes v_m$, fix ℓ. Its coefficient row $(\langle u_\ell \otimes f_k, \Psi \rangle)_k$ is orthogonal to every v_m, so is zero. Each old column is then orthogonal to every u_ℓ, and also zero. Thus $\Psi = 0$. The unitary identification between array descriptions sends each product $u \otimes v$ to that same product and preserves all inner products; its inverse is the reverse basis change.

For $L^2(\mathbb{R})$ factors, the product basis identifies this tensor product with $L^2(\mathbb{R}^2)$, by standing fact S3. Product functions become $(u \otimes v)(x_1, x_2) = u(x_1) v(x_2)$; general vectors need not factor. For a bounded A on part 1, define $A \otimes \mathbb{1}$ first on finite sums. Grouping coefficients by the second index gives

$$\|(A \otimes \mathbb{1})\Psi\|^2 = \sum_k \left\| A\Big(\sum_j c_{jk} e_j \Big) \right\|^2 \le \|A\|^2 \, \|\Psi\|^2 .$$

Thus it extends to the whole tensor product. Unit product vectors show its norm is exactly $\|A\|$. Its adjoint is $A^\dagger \otimes \mathbb{1}$ by testing finite tensors and continuity. If $A \geq 0$, S6 writes it as $(\sqrt{A})^\dagger \sqrt{A}$, so $A \otimes \mathbb{1} \geq 0$. These bounded operators represent all observations confined to part 1.

6.2 Schmidt's theorem without a finite-rank assumption

Theorem 6.1 (Schmidt decomposition). *Every unit $\Psi \in \mathcal{H}_1 \otimes \mathcal{H}_2$ has a finite or countable expansion*

$$\Psi = \sum_\ell \sqrt{p_\ell}\, u_\ell \otimes v_\ell, \quad p_\ell > 0, \quad \sum_\ell p_\ell = 1,$$

where the two displayed families are orthonormal. The sum converges in norm. Zero coefficients can be omitted; neither family need be a basis of its whole factor.

Proof. For the coefficient array above, form a *linear* operator $C : \mathcal{H}_2 \to \mathcal{H}_1$ by $(Cx)_j = \sum_k c_{jk} x_k$. Row-wise Cauchy–Schwarz gives $\|Cx\|^2 \leq \sum_{jk} |c_{jk}|^2 \, \|x\|^2$. Put $R = CC^\dagger$, the analogue of QM-I's coefficient formula for the first reduced state. It is bounded and positive. Its diagonal sum in (e_j) is

$$\sum_j \langle e_j, Re_j \rangle = \sum_j \left\| C^\dagger e_j \right\|^2 = \sum_{j,k} |c_{jk}|^2 = 1.$$

Proposition 1.4 gives compactness and an orthonormal eigenbasis, including the kernel; its nonzero eigenvalues are $p_\ell > 0$ with $\sum_\ell p_\ell = 1$. Let u_ℓ be their orthonormal eigenvectors. The coefficient of $u_\ell \otimes f_k$ in Ψ is $d_{\ell k}$:

$$d_{\ell k} = \langle u_\ell \otimes f_k, \Psi \rangle .$$

Thus define

$$d_{\ell k} = \sum_j \overline{(u_\ell)_j} c_{jk}, \qquad v_\ell = \frac{1}{\sqrt{p_\ell}} \sum_k d_{\ell k} f_k.$$

The components of $C^\dagger u_\ell$ are $\bar{d}_{\ell k}$, not $d_{\ell k}$. Consequently

$$\langle v_\ell, v_m \rangle = \frac{\langle C^\dagger u_m, C^\dagger u_\ell \rangle}{\sqrt{p_\ell p_m}} = \frac{\langle u_m, R u_\ell \rangle}{\sqrt{p_\ell p_m}} = \delta_{\ell m}.$$

For a basis vector u of $\ker R$, $\left\| C^\dagger u \right\|^2 = \langle u, Ru \rangle = 0$, so the corresponding coefficient row of Ψ is zero. Expand the first factor in this full eigenbasis of R. Its nonzero rows are precisely $\sqrt{p_\ell} v_\ell$, proving the stated tensor expansion. Parseval gives the squared tail $\sum_{\ell > N} p_\ell \to 0$, proving norm convergence as well as completeness of the expansion. ∎

The basis coefficient is conjugated because $\langle u_\ell, e_j \rangle = \overline{(u_\ell)_j}$. Thus C is an auxiliary linear matrix map, whereas the tensor coefficient formula contains $u_j v_k$, without conjugating either factor. The displayed component calculation keeps that convention explicit. Degenerate weights permit changes of Schmidt vectors within their eigenspaces; the resulting expansion need not be unique within degenerate subspaces.

For a product $c_{jk} = a_j b_k$, the range of C is one dimensional; this construction therefore gives exactly one nonzero Schmidt weight. Conversely a one-term Schmidt expansion is a

product. More than one weight therefore detects pure-state entanglement, just as in QM-I; infinite Schmidt rank is allowed. For instance, for $0 < r < 1$ the vector

$$\Psi = \sqrt{1-r}\sum_{n=0}^{\infty} r^{n/2} e_n \otimes f_n$$

has weights $(1-r)r^n$. If terms $0, \dots, N$ are retained, its omitted norm is $r^{(N+1)/2}$. Each finite truncation can approximate the vector, although none has its exact infinite-rank reduced state.

6.3 The state of a part

For a Schmidt vector, orthogonality of the v_ℓ gives, for bounded A,

$$\langle \Psi, (A \otimes \mathbb{1})\Psi \rangle = \sum_\ell p_\ell \, \langle u_\ell, A u_\ell \rangle \, .$$

One can obtain this by finite truncations: boundedness controls the norm limit and the expectation series is absolutely convergent. Thus $\rho_1 = \sum_\ell p_\ell \, |u_\ell\rangle\langle u_\ell|$ is the density operator of part 1; similarly $\rho_2 = \sum_\ell p_\ell \, |v_\ell\rangle\langle v_\ell|$. They have the same nonzero eigenvalues. Neither is generally pure even when the composite is. Every Schmidt expansion gives this bounded local expectation rule. L1's uniqueness therefore shows that its weights are the nonzero eigenvalues of ρ_1, including multiplicities, independent of the construction.

Theorem 6.2 (Partial trace). *For every density operator ρ on $\mathcal{H}_1 \otimes \mathcal{H}_2$, there is a unique density operator $\rho_1 = \mathrm{Tr}_2 \, \rho$ such that*

$$\mathrm{Tr}(\rho_1 A) = \mathrm{Tr}\left(\rho(A \otimes \mathbb{1})\right) \quad \textit{for every bounded } A \textit{ on } \mathcal{H}_1. \tag{6.1}$$

Proof. Define the right-hand side as a functional $\omega_1(A)$. It is linear, positive and normalised by L1 and the tensor properties above. To check normality, take orthogonal projections E_j on part 1 with strong sum E. Their tensor extensions have strong sum $E \otimes \mathbb{1}$: this is immediate on a product vector from the part-1 strong convergence, extends to finite sums, then to every vector by density and the uniform norm bound one for the partial projection sums. For any spectral density expansion $\rho = \sum_n p_n \, |\Psi_n\rangle\langle\Psi_n|$, it follows that

$$\omega_1(E) = \sum_n p_n \sum_j \left\langle \Psi_n, (E_j \otimes \mathbb{1})\Psi_n \right\rangle = \sum_j \omega_1(E_j).$$

All summands are nonnegative, so interchange is justified. L1's normal-state theorem supplies a unique ρ_1, which is exactly (6.1). ∎

This construction also proves affine dependence on ρ, by uniqueness. For mixtures, one may compute each pure reduced state and mix them with the same weights; the normal-state proof shows the result does not depend on a particular composite preparation. For density operators $\tau_1 = \sum_j a_j \, |u_j\rangle\langle u_j|$ and $\tau_2 = \sum_k b_k \, |v_k\rangle\langle v_k|$, define their product as $\tau_1 \otimes \tau_2 = \sum_{jk} a_j b_k \, |u_j \otimes v_k\rangle\langle u_j \otimes v_k|$. It is positive with trace one, and bounded product expectations factor. Its partial trace returns τ_1. A purification also exists: from $\rho_1 = \sum_n p_n \, |u_n\rangle\langle u_n|$ use $\sum_n \sqrt{p_n} u_n \otimes f_n$ with a large enough second space. The squared norm sum is $\sum_n p_n = 1$, and the bounded first-factor expectations give ρ_1 by the preceding Schmidt calculation. QM-I, Theorem 11.10, uses the conjugated eigenvectors as partner family; replacing them by any orthonormal (f_n) leaves these expectations unchanged.

For the geometric Schmidt example, $\rho_1 = (1-r)\sum_n r^n |e_n\rangle\langle e_n|$. A local event $|e_0\rangle\langle e_0|$ has probability $1-r$, although the composite vector remains pure. Tracing a part is a restriction of observations, not a random selection of one term of a Schmidt expansion. Compare L5: restriction to a smaller algebra can produce a mixed state, without changing the initial vector by any dynamics.

6.4 What EPR asked

Einstein, Podolsky and Rosen (1935) considered a pair with a fixed relative position and a fixed total momentum. Write $X_i = Q^{(i)}$ for position on part i in this section, and $P^{(i)}$ for momentum on part i. In the ideal description, reading X_1 would predict X_2 exactly; alternatively reading $P^{(1)}$ would predict $P^{(2)}$ exactly. The inference concerns the distant partner, not simultaneous sharp readings of X_1 and $P^{(1)}$ on the same particle.

Their criterion of reality says that certain prediction of a physical quantity, without disturbing the system, is sufficient for an element of reality. Their locality premise says that the choice of reading on part 1 does not change the physical reality of the separated part 2. They then assign both position and momentum reality to part 2, independent of which reading is chosen on part 1. Completeness would require the state description to represent all those elements of reality. A single wavefunction cannot give both local quantities sharp values. EPR therefore conclude that the wavefunction description is incomplete: otherwise the two inferred local quantities could not both be real. That argument depends on the reality and locality premises just stated; L13 examines locality more closely. Exact certainty also uses their ideal correlations; normalised preparations give finite inference errors.

The ideal expression $\delta(x_1 - x_2 - x_0)$ is not a normalised vector of $L^2(\mathbb{R}^2)$. L1's shrinking-bump test, applied to strips about $x_1 - x_2 = x_0$, excludes its exact position expectations; we give the strip estimate below. As in QM-II's plane wave, writing it formally does not create a state. Actual experiments have finite correlations. We can quantify one consequence without importing an ideal delta function into the state space.

On $\mathcal{S}(\mathbb{R}^2)$ direct differentiation gives

$$[X_1 - X_2, P^{(1)} + P^{(2)}] = [X_1, P^{(1)}] - [X_2, P^{(2)}] = 0.$$

There is therefore no local Heisenberg lower bound between these two collective quantities from their commutator. This does not remove the uncertainty relation for each local $X_i, P^{(i)}$. The next bound says what product preparations, and their mixtures, still cannot do.

6.5 A variance bound for separable preparations

A *finite separable preparation* is $\rho = \sum_{a=1}^{m} w_a |u_a \otimes v_a\rangle\langle u_a \otimes v_a|$, with $w_a \geq 0$, $\sum_a w_a = 1$, unit $u_a, v_a \in \mathcal{S}(\mathbb{R})$. All position and momentum moments used below are finite. For each collective B define $\langle B\rangle_\rho = \sum_a w_a \langle u_a \otimes v_a, B(u_a \otimes v_a)\rangle$, and similarly for B^2; variance is the second moment minus the square of the mean. These depend only on ρ: its finite-dimensional range lies in Schwartz space, so $B\rho$ and $B^2\rho$ are finite-rank bounded operators, whose finite matrix traces give exactly those sums. A general separable state is usually defined using a suitable closed convex hull; our proof is for this stated finite class, with the finite-moment extension named rather than hidden in a continuity claim for unbounded observables.

Proposition 6.3 (Duan-type bound, finite preparation form). *For the finite separable preparations above, and every length $s > 0$,*

$$\frac{\operatorname{Var}_\rho(X_1 - X_2)}{s^2} + \frac{s^2 \operatorname{Var}_\rho(P^{(1)} + P^{(2)})}{\hbar^2} \geq 2. \tag{6.2}$$

Proof. For a product vector, expectations factor, so the position covariance and momentum covariance between the parts vanish. Its left side is the sum of the two local contributions. If $\alpha_i = \Delta X_i / s$ and $\beta_i = s\Delta P^{(i)}/\hbar$, QM-II, Theorem 4.9, on Schwartz functions gives $\alpha_i \beta_i \geq 1/2$. Hence $\alpha_i^2 + \beta_i^2 \geq 2\alpha_i\beta_i \geq 1$, and summing the parts gives 2. For a finite mixture and either collective observable B, put $b_a = \langle B \rangle_a$, $\bar{b} = \sum_a w_a b_a$. Expanding its second moment gives the law of total variance

$$\operatorname{Var}_\rho B = \sum_a w_a \operatorname{Var}_a B + \sum_a w_a (b_a - \bar{b})^2 \geq \sum_a w_a \operatorname{Var}_a B.$$

Apply it to both collective observables and average the product bounds. ∎

The scale s converts position and momentum variances into dimensionless numbers. Changing it cannot allow a finite separable preparation to violate the inequality. The broader separable-state version with finite second moments is a standard continuous-variable entanglement criterion (cited: Duan et al., *PRL* **84**, 2722–2725 (2000)). Mere trace norm convergence of states does not justify continuity of unbounded second moments, which is why it is not used as a shortcut here.

Consider the normalised Gaussian family

$$\psi_r(x_1, x_2) \propto \exp\left[-\frac{e^{2r}(x_1 - x_2)^2}{4s^2} - \frac{e^{-2r}(x_1 + x_2)^2}{4s^2} \right], \quad r \geq 0.$$

For finite r this is a Schwartz vector. Both collective means vanish. Its covariances, supplied here for the test, are

$$\operatorname{Var}(X_1 - X_2) = s^2 e^{-2r}, \qquad \operatorname{Var}(P^{(1)} + P^{(2)}) = \frac{\hbar^2}{s^2} e^{-2r}.$$

Their Gaussian-integral derivation is a computation for Sheet 6; no general proof is deferred there. Inserting them gives $2e^{-2r}$, below 2 for every $r > 0$. Thus these vectors cannot be finite separable preparations; the cited broader criterion also certifies their entanglement among all separable states. The practical value of the test is that it uses only two second moments, including for noisy mixed preparations without a pure-state Schmidt form. For this pure example entanglement could also be checked by nonfactorisation. At $r = 0$ the exponent factors into independent Gaussians and the bound is saturated. At $r = (\log 2)/2$ the sum equals 1: the two terms are each $1/2$. The test concerns correlations, not either local variable becoming simultaneously sharp with its local conjugate.

The local uncertainties increase even though both tested collective variances decrease. To compute the local position variances, use

$$u = \frac{x_1 - x_2}{\sqrt{2}}, \qquad v = \frac{x_1 + x_2}{\sqrt{2}}.$$

The position density factors in these coordinates. The Gaussian variances are $s^2 e^{-2r}/2$ for u and $s^2 e^{2r}/2$ for v. The inverse relations are $x_1 = (u + v)/\sqrt{2}$ and $x_2 = (v - u)/\sqrt{2}$. Independence removes the covariance term, giving

$$\operatorname{Var} X_i = \frac{s^2}{4}(e^{-2r} + e^{2r}) = \frac{s^2}{2}\cosh(2r), \qquad i = 1, 2.$$

For a real Gaussian factor, QM-II's Gaussian calculation gives momentum variance $\hbar^2/(4\operatorname{Var} u)$, with u replaced by v for the other factor. The momentum variances of the two Gaussian factors are $\hbar^2 e^{2r}/(2s^2)$ and $\hbar^2 e^{-2r}/(2s^2)$. The same inverse coordinate change therefore gives

$$\operatorname{Var} P^{(i)} = \frac{\hbar^2}{2s^2}\cosh(2r), \qquad \Delta X_i \Delta P^{(i)} = \frac{\hbar}{2}\cosh(2r) \geq \frac{\hbar}{2}.$$

A narrow relative position is compatible with a broad distribution for each particle separately. Figure 6.1 shows this distinction using position probabilities alone.

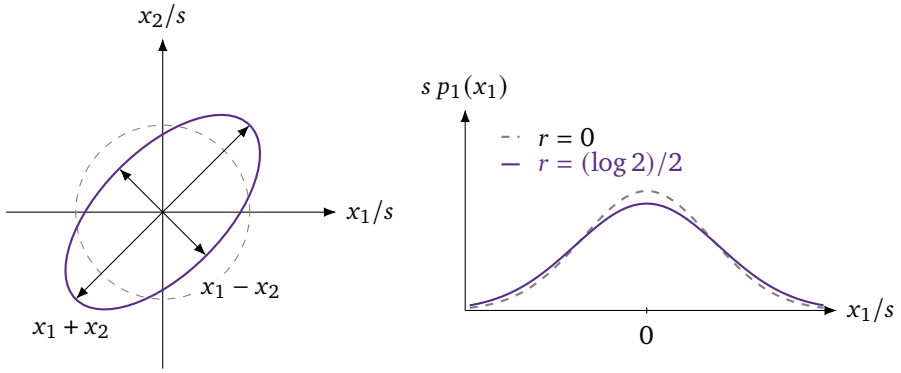


Figure 6.1: An equal-density contour and the normalised first-position marginal at $r = 0$ (dashed) and $r = (\log 2)/2$ (solid). Here p_1 is the first part's position density, and the vertical scale is $sp_1(x_1)$. The correlation narrows across the diagonal while the local marginal broadens; these are position plots, not joint position–momentum distributions.

As $r \to \infty$, the collective variances tend to zero, but the vectors have no norm limit. Here is the promised version of L1's bump test. Let f_N be continuous, $0 \leq f_N \leq 1$, equal to one at zero and zero outside $[-1/N, 1/N]$. For a smooth function v supported in $[-R, R]^2$,

$$\langle v, f_N(X_1 - X_2)v\rangle \leq \frac{4R}{N}\,\|v\|_\infty^2 \longrightarrow 0,$$

because the intersecting strip has width at most $2/N$ in each fixed x_2 slice. Approximation by such v (S4) and the multiplier bound one extend this limit to every L^2 vector. A finite spectral head and a small positive tail extend it to every density operator. Translating the strip gives the same result about x_0. Thus no normal state has the exact expectations $\langle f(X_1 - X_2)\rangle = f(0)$ for every bounded continuous f, as a delta would require. For fixed f, the squeezed expectations do tend to $f(0)$: outside a strip of width ϵ the probability is at most $\operatorname{Var}(X_1 - X_2)/\epsilon^2$, by integrating $\epsilon^2 1_{|x_1 - x_2| > \epsilon} \leq (x_1 - x_2)^2$; inside it use continuity at zero. A hypothetical norm limit would preserve bounded expectations and give the excluded delta rule. Increasingly sharp correlations do not supply a limiting state in the Hilbert space.

What is a composite, and what is a part?

Without superselection, a quantum system has a separable Hilbert space, its normal states and its bounded observables. The composite space is the Hilbert tensor product. The state of a part is the unique partial trace reproducing its bounded local expectations. Entanglement can make that state mixed even when the composite is pure. Superselection requires the specified observable algebra of L5 as well.

Summary. Square-summable arrays extend QM-I's composite rule. Compactness and S5 give Schmidt's theorem; L1 gives the mixed-state partial trace from local expectations. EPR adds premises about reality, locality and completeness. Normalised squeezed states replace its ideal delta expression by finite correlations, and their variance sum violates the product-mixture bound. Neither a reduced mixture nor a limiting delta selects a definite global alternative.

Exercises. Try Sheet 6 in the separate *Exercise sheets* booklet (English).

Sources. Hall, *Quantum Theory for Mathematicians*, Theorem 19.13 pp. 430–431; Moretti, *Spectral Theory and Quantum Mechanics*, §13.4 pp. 844–858. The matrix compactness and normal-state proofs above supply the needed prerequisites explicitly. Einstein–Podolsky–Rosen, *Phys. Rev.* **47**, 777 (1935); Duan et al., *PRL* **84**, 2722 (2000). The optical experiment and its reference are in the full edition.

Lecture 7

Channels: complete positivity and Stinespring

Objectives. *A part of a closed system need not evolve unitarily. Its state change must remain positive when the part is entangled with another system. In finite dimension, with trace preservation, that requirement is equivalent to a positive Choi matrix, a Kraus sum and a unitary evolution with an initially separate environment. We also construct the infinite-dimensional channels needed later.*

- Distinguish positivity from complete positivity using the transpose on half a singlet.
- Prove the finite Choi–Kraus–dilation equivalences, keeping tensor order and trace-preservation conditions explicit.
- Construct sufficient channels on a separable space and compute spontaneous-emission population and coherence changes.

7.1 Positivity must survive a spectator

A preparation with an ignored classical label gives a convex mixture. We assume a complex-linear evolution map on the span of density operators. In finite dimension that span is all matrices. In a separable $\mathcal{H}$, use L1's $\mathcal{T}_1(\mathcal{H})$, the complex span of positive trace-class operators. Complete positivity is a condition on that map, not on a single output state.

Definition 7.1 (Positive maps and channels)**.** A complex-linear map $\Phi : \mathcal{T}_1(\mathcal{H}) \to \mathcal{T}_1(\mathcal{H}')$ is *positive* if $\Phi(\rho) \geq 0$ for every positive trace-class ρ. For a finite spectator with basis $(|a\rangle)$, put $V_a\psi = \psi \otimes |a\rangle$ and write $Y_{ab} = V_a^\dagger Y V_b$. The *amplification* is

$$(\Phi \otimes \mathrm{id}_m)(Y) = \sum_{a,b=1}^{m} \Phi(Y_{ab}) \otimes |a\rangle\langle b| .$$

The map is *completely positive* (CP) if this amplification is positive for every finite m. A *channel* is CP and trace preserving (TP): $\operatorname{Tr}\Phi(X) = \operatorname{Tr} X$ for all $X \in \mathcal{T}_1$.

These blocks belong to $\mathcal{T}_1$: diagonal compressions of a positive trace-class operator are positive with finite trace, and off-diagonal blocks are complex combinations of compressions along $|j\rangle \pm |k\rangle$ and $|j\rangle \pm i\,|k\rangle$ by polarisation. General inputs follow from the positive-span definition. Reversing the tensor order gives the equivalent test $\mathrm{id}_m \otimes \Phi$: the swap of the two factors is unitary. Positivity together with trace preservation makes every density operator's output a density operator. CP further makes the local rule usable on arbitrary entangled preparations with a finite spectator, without negative probabilities.

Transposition $\tau(X) = X^{\mathsf{T}}$ in a fixed basis is positive: if $X \geq 0$, then $\langle z, X^{\mathsf{T}} z\rangle = \overline{\langle \bar{z}, X\bar{z}\rangle} \geq 0$. It is also trace preserving. Nevertheless take the singlet $\psi_- = (|01\rangle - |10\rangle)/\sqrt{2}$. In the ordered basis $00, 01, 10, 11$,

$$(\tau \otimes \mathrm{id}_2)(|\psi_-\rangle\langle\psi_-|) = \frac{1}{2}\begin{pmatrix} 0 & 0 & 0 & -1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -1 & 0 & 0 & 0 \end{pmatrix}.$$

The vector $(|00\rangle + |11\rangle)/\sqrt{2}$ has eigenvalue $-1/2$. Thus a positive map on isolated states need not be a physical local channel. A Wigner antiunitary on pure states does not provide a new complex-linear channel on one half of every entangled pair: its action on density operators is the transpose after a unitary change of basis, and the spectator test detects the obstruction.

7.2 One finite matrix tests complete positivity

For this section $\mathcal{H} = \mathcal{H}' = \mathbb{C}^n$. Write $\mathsf{e}_{ij} = |i\rangle\langle j|$ and $\Omega = \sum_{i=1}^{n} |i\rangle \otimes |i\rangle$, which is *not normalised*: $\|\Omega\|^2 = n$. Define the *Choi matrix*

$$C_\Phi = \sum_{i,j} \mathsf{e}_{ij} \otimes \Phi(\mathsf{e}_{ij}) = (\mathrm{id}_n \otimes \Phi)(|\Omega\rangle\langle\Omega|). \tag{7.1}$$

The first factor is a reference copy of the input; the second is output. This order is bookkeeping for C_Φ. In a physical dilation below, the system is first and environment second, as elsewhere in the course.

Theorem 7.2 (Finite Choi–Kraus theorem)**.** *For a complex-linear map on $n \times n$ matrices the following are equivalent: complete positivity; $C_\Phi \geq 0$; and a finite* Kraus form

$$\Phi(X) = \sum_{k=1}^{d} K_k X K_k^\dagger, \qquad d \leq n^2.$$

Such a map is trace preserving exactly when $\sum_k K_k^\dagger K_k = \mathbb{1}$.

Proof. CP implies $C_\Phi \geq 0$ by its definition and positivity of $|\Omega\rangle\langle\Omega|$. If $C_\Phi \geq 0$, its finite spectral decomposition can be written $C_\Phi = \sum_{k=1}^{d} |v_k\rangle\langle v_k|$, absorbing square roots of the nonzero eigenvalues into v_k, so the v_k need not be unit vectors. For the zero matrix take a zero Kraus operator. Define matrices by reshaping:

$$(K_k)_{ai} = (v_k)_{(i,a)}.$$

The Choi entry with row (i,a) and column (j,b) is then

$$[\Phi(\mathsf{e}_{ij})]_{ab} = \sum_k (v_k)_{(i,a)}\overline{(v_k)_{(j,b)}} = \sum_k (K_k)_{ai}\overline{(K_k)_{bj}} = \left[\sum_k K_k \mathsf{e}_{ij} K_k^\dagger\right]_{ab}.$$

The matrix units form a basis, so the Kraus formula follows for every X. Conversely each amplified Kraus map sends a positive Y to $\sum_k (K_k \otimes \mathbb{1}_m) Y (K_k^\dagger \otimes \mathbb{1}_m)$, which is positive: each summand's quadratic form is nonnegative. Thus it is CP. Finally finite matrix cyclicity gives $\mathrm{Tr}\,\Phi(X) = \mathrm{Tr}(X \sum_k K_k^\dagger K_k)$. Equality to $\mathrm{Tr}\,X$ for every matrix unit forces the sum to be $\mathbb{1}$, and that sum conversely gives TP. ∎

For a channel, $\operatorname{Tr} C_\Phi = n$, so C_Φ/n is a density matrix. It is the output from half the normalised maximally entangled input $\Omega/\sqrt{n}$. Normalising it at the start without changing the reshaping formula would multiply every Kraus operator by $1/\sqrt{n}$ and lose trace preservation. Also $\sum K_k K_k^\dagger = \mathbb{1}$ is a different condition: it says the map sends $\mathbb{1}$ to $\mathbb{1}$, called *unitality*. Channels need not be unital.

An off-diagonal Kraus operator makes the row and column order visible. In the two-qubit order $00, 01, 10, 11$, take

$$v = |01\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \qquad K = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} = |1\rangle\langle 0| .$$

The first label of v becomes the column of K. The second becomes its row, as shown in Figure 7.1. The positive Choi matrix $|v\rangle\langle v|$ therefore gives the CP operation $X \mapsto KXK^\dagger$. Its completeness operator is $K^\dagger K = |0\rangle\langle 0|$, so this operation is not TP. It sends the input $|0\rangle\langle 0|$ to $|1\rangle\langle 1|$ and the input $|1\rangle\langle 1|$ to zero. A positive Choi matrix determines a CP map; trace preservation remains a separate test.

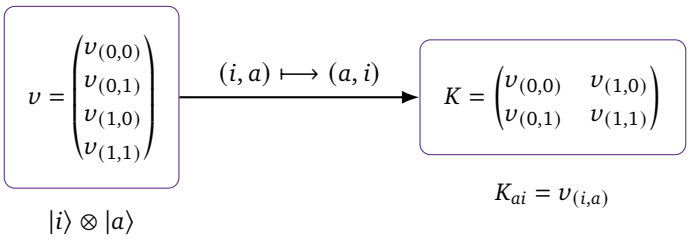


Figure 7.1: Reshaping one Choi vector in input-reference/output order. The input label i becomes a Kraus column and the output label a becomes a row; no component is complex conjugated.

For the identity channel, $C_{\mathrm{id}} = |\Omega\rangle\langle\Omega|$ has rank one; reshaping Ω gives the single Kraus operator $\mathbb{1}$. This provides a quick check of both tensor order and normalisation. Transposition's Choi matrix is the swap $\sum_{ij} \mathsf{e}_{ij} \otimes \mathsf{e}_{ji}$; its antisymmetric vectors have eigenvalue -1, so the theorem independently detects the failure found with the singlet.

7.3 A channel is a unitary with an ignored environment

Theorem 7.3 (Finite Stinespring and unitary dilation). *A finite-dimensional map is CP exactly when $\Phi(X) = \operatorname{Tr}_E(JXJ^\dagger)$ for a linear $J : \mathbb{C}^n \to \mathbb{C}^n \otimes \mathbb{C}^d$. It is TP exactly when J is an isometry. Every finite-dimensional channel can be realised as*

$$\Phi(\rho) = \operatorname{Tr}_E\left[U_E(\rho \otimes |e_\star\rangle\langle e_\star|)U_E^\dagger\right] \tag{7.2}$$

with a fixed unit environment vector $e_\star$ and a unitary U_E. In the finite Kraus basis one may choose $e_\star = e_1$.

Here the partial trace is extended linearly from density operators to all finite matrices. On rank-one products it is $\operatorname{Tr}_E(|a \otimes b\rangle\langle c \otimes d|) = \langle d, b\rangle |a\rangle\langle c|$ (QM-I, Theorem 11.1); bounded local pairings make the extension well-defined.

Proof. For Kraus operators set $J\psi = \sum_k K_k\psi \otimes e_k$, with an orthonormal environment basis. The partial trace gives $\mathrm{Tr}_E(JXJ^\dagger) = \sum_k K_k X K_k^\dagger$ on rank-one matrix units, hence on all matrices. Conversely expand any linear J in environment coordinates to obtain those K_k. Moreover $J^\dagger J = \sum_k K_k^\dagger K_k$, proving the isometry equivalence. For a channel the rule $\psi \otimes e_\star \mapsto J\psi$ is an isometry between two n-dimensional subspaces of the same nd-dimensional space. Their orthogonal complements both have dimension $n(d-1)$. Map an orthonormal basis of one complement onto one of the other, obtaining a unitary U_E with the required action. This proves (7.2). Conversely that unitary rule gives an isometry J and hence a channel. ∎

The environment is prepared in one fixed state *independently* of ρ. Ignoring it loses information and can make the system mixed. Different dilations and Kraus lists can give the same channel; unitary freedom after padding is related to L2's freedom of ensembles (Theorem 2.5) (named: Nielsen–Chuang, Theorem 8.2, p. 372). No preferred list of physical unobserved events follows from the channel alone.

Initial correlations require the actual joint preparation. Compare $|\Phi_+\rangle\langle\Phi_+|$, $\Phi_+ = (|00\rangle + |11\rangle)/\sqrt{2}$, with $\mathbb{1}_4/4$. Both system and environment marginals are $\mathbb{1}_2/2$ in both preparations. Apply the fixed unitary $|s, e\rangle \mapsto |s \oplus e, e\rangle$, a controlled NOT with environment controlling the system. The Bell vector becomes $|0\rangle \otimes |+\rangle$, whereas $\mathbb{1}_4/4$ stays unchanged. The system outputs are $|0\rangle\langle 0|$ and $\mathbb{1}_2/2$. Only the correlation differed, yet neither marginal recorded it. There is no single output determined by the system marginal for this preparation class. One needs a specified joint-state assignment; this does not say every correlated case forbids every restricted-domain map.

In infinite dimension the orthogonal-complement argument needs a further hypothesis or a larger environment: two isomorphic infinite-dimensional subspaces need not have isomorphic complements. The theorem just proved is finite-dimensional. We next prove sufficient infinite constructions; the general representation theorem will remain cited.

7.4 Sufficient constructions on a separable space

Let (K_k) be countably many bounded operators on a separable $\mathcal{H}$, with $\sum_k K_k^\dagger K_k = \mathbb{1}$ strongly. For positive trace-class $\rho = \sum_j p_j \left|\varphi_j\right\rangle\!\left\langle\varphi_j\right|$, the double rank-one family

$$\sum_{j,k} p_j \left|K_k\varphi_j\right\rangle\!\left\langle K_k\varphi_j\right| \quad \text{has total trace weight} \quad \sum_{j,k} p_j \left\|K_k\varphi_j\right\|^2 = \mathrm{Tr}\,\rho.$$

It defines a positive trace-class operator: its finite positive sums have operator-norm tails bounded by the omitted trace-weight tails; their norm limit is positive, and nonnegative diagonal sums give the stated trace. For every finite partial sum, its positive remainder has trace equal to the omitted weight, by the same nonnegative diagonal calculation. L1's bound $|\mathrm{Tr}(R_N A)| \le \|A\|\,\mathrm{Tr}\,R_N$ makes its bounded expectation tend to zero. Hence for bounded A the expectation is the series

$$\mathrm{Tr}(\Phi(\rho)A) = \sum_k \mathrm{Tr}(\rho K_k^\dagger A K_k). \tag{7.3}$$

The bound $\sum_k |\mathrm{Tr}(\rho K_k^\dagger A K_k)| \le \|A\|\,\mathrm{Tr}\,\rho$ follows termwise from the spectral expansion. This formula makes the output independent of the expansion and additive on positive inputs. Extend complex-linearly to L1's span $\mathcal{T}_1$. It is well-defined: any two decompositions have the same finite-sum pairings in (7.3), hence the same limits, and bounded expectations separate trace-class operators by their rank-one matrix elements.

Repeating the positive-series construction for $K_k \otimes \mathbb{1}_m$ proves CP: its (a,b) block is exactly $\sum_k K_k Y_{ab} K_k^\dagger = \Phi(Y_{ab})$, so it is the defined amplification. Their completeness follows first on finite spectator-coordinate vectors, then on all vectors, so the same total trace-weight identity proves TP. Thus the countable Kraus form defines a channel. In particular, $K_k = \sqrt{w_k}U_k$, with $\sum_k w_k = 1$ and unitary U_k, gives a discrete random-unitary channel.

For environment coordinates, the bounded map $\mathbb{1} \otimes \langle e_k|$ sends $\sum_\ell z_\ell \otimes e_\ell$ to z_k. An initially separate pure environment and an arbitrary unitary give $K_k\psi = (\mathbb{1} \otimes \langle e_k|)U_E(\psi \otimes e_\star)$. Parseval implies $\sum_k \|K_k\psi\|^2 = \|\psi\|^2$. The positive partial sums of $K_k^\dagger K_k$ are contractions; their quadratic tails tend to zero and therefore their strong tails do too, since $\|B\psi\|^2 \le \langle \psi, B\psi\rangle$ for $0 \le B \le \mathbb{1}$. This proves the required strong completeness. The partial trace in L6 gives the constructed channel: expand $U_E(\varphi_j \otimes e_\star) = \sum_k K_k\varphi_j \otimes e_k$; its bounded local expectation is $\sum_{j,k} p_j \left\langle K_k\varphi_j, AK_k\varphi_j \right\rangle$, which is (7.3). The general countable Kraus representation for the normal Heisenberg CP map associated with a trace-class channel is *cited*, not proved by a finite Choi matrix: Gustafson–Sigal, Theorem 19.21 p. 260 and Theorem 19.24 p. 261. An infinite unitary realisation may enlarge the ancilla to match complements.

The continuous construction below has a different input: a scalar average of expectations after unitary conjugation. First we check that this average is normal, so L1 constructs an output operator. Then the same construction with a finite spectator proves CP. Trace preservation follows from unitarity.

We will also need a continuous random kick. Here is a precise sufficient case, avoiding an unspecified operator integral. Suppose $\mathcal{W}(k)$ is a strongly continuous unitary family, $k \in \mathbb{R}$, and $w(k) \ge 0$ is a continuous probability density with improper integral one. For positive ρ define an expectation rule by scalar integrals:

$$\omega_\rho(A) = \int_{\mathbb{R}} w(k) \operatorname{Tr}(\rho \mathcal{W}(k)^\dagger A \mathcal{W}(k))\, dk. \tag{7.4}$$

The integrand without the density w is continuous and bounded by $\|A\| \operatorname{Tr}\rho$: finite spectral sums are continuous and their tails are uniformly bounded by $\|A\| \sum_{j>N} p_j$. Hence the integral exists. Positivity is immediate and $\omega_\rho(\mathbb{1}) = \operatorname{Tr}\rho$. After normality is checked below, for $r = \operatorname{Tr}\rho > 0$ apply L1 to ω_ρ/r and multiply the resulting density operator by r; the zero input has zero output. We check normality explicitly.

For orthogonal projections with strong sum $E = \sum_j E_j$, put $B_N = E - \sum_{j\le N} E_j$. The continuous functions $h_N(k) = \operatorname{Tr}(\rho \mathcal{W}(k)^\dagger B_N \mathcal{W}(k))$ decrease to zero pointwise, by L1, and are nonnegative. They converge uniformly on every compact interval: for a chosen positive tolerance, continuity supplies a neighbourhood where one h_N is small at each point; finitely many such neighbourhoods cover the closed interval, and monotonicity gives one index working on the whole interval. Outside it the integral is bounded by $\operatorname{Tr}\rho$ times the probability-density tail. First choosing that tail small, then N large, proves $\int w h_N \to 0$. Thus (7.4) is normal and L1 constructs its positive trace-class output. Linearity and well-defined extension follow from the bounded pairings as above. Applying the same argument to finite spectators with $\mathcal{W}(k) \otimes \mathbb{1}_m$ proves CP: its blocks are exactly the integrated $\Phi(Y_{ab})$. Trace preservation follows from $\mathcal{W}(k)^\dagger \mathcal{W}(k) = \mathbb{1}$. This proves this continuous mixture is a channel; L11 uses a Gaussian density and $\mathcal{W}(k) = V(\hbar k) = e^{ikQ}$, whose strong continuity follows by a spatial-tail cutoff and uniform convergence on bounded positions.

7.5 Spontaneous emission: population and coherence

Use QM-I's convention $H = (\hbar\nu_0/2)\sigma_z$, $\nu_0 > 0$: $|0\rangle = |\uparrow\rangle$ is excited and $|1\rangle$ is ground. An ideal zero-temperature emission channel with decay probability $0 \le \gamma \le 1$ has

$$K_0 = \sqrt{1-\gamma}\,|0\rangle\langle 0| + |1\rangle\langle 1|\,, \qquad K_1 = \sqrt{\gamma}\,|1\rangle\langle 0|\,.$$

Their completeness sum is $\mathbb{1}$, so they define a channel. We work in the frame rotating with H, omitting its free coherence phase; a thermal bath would also allow absorption. For $\rho = \begin{pmatrix} a & c \\ \bar{c} & 1-a \end{pmatrix}$,

$$\Phi_\gamma(\rho) = \begin{pmatrix} (1-\gamma)a & \sqrt{1-\gamma}\,c \\ \sqrt{1-\gamma}\,\bar{c} & 1-(1-\gamma)a \end{pmatrix}. \tag{7.5}$$

The excited population loses a factor $1-\gamma$; the coherence is multiplied by $\sqrt{1-\gamma}$. At $\gamma = 1$ every state becomes the ground state, so the channel is not invertible and is not a unitary conjugation. For $\gamma > 0$ it is not unital: the maximally mixed state moves toward the ground state.

A dilation makes the lost information explicit. With environment ready $|0_E\rangle$, take

$$J\,|0\rangle = \sqrt{1-\gamma}\,|0, 0_E\rangle + \sqrt{\gamma}\,|1, 1_E\rangle\,, \qquad J\,|1\rangle = |1, 0_E\rangle\,.$$

The images are orthonormal, so this is an isometry; completing a finite basis supplies U_E. The two environment coordinates give exactly the displayed Kraus operators. For an initially excited system, the joint state stays pure; its system marginal becomes mixed for $0 < \gamma < 1$. Interpreting the environment labels as a detector's observed outcomes would add a reading procedure, studied in L10. Ignoring those labels already determines the channel.

If $\gamma(t) = 1 - e^{-t/T_1}$, the population decays as e^{-t/T_1} and coherence as $e^{-t/(2T_1)}$. Thus the pure-decay coherence time is $T_2 = 2T_1$. This exponential time dependence is a model assumption, not a consequence of the fixed-time Kraus representation; L8 identifies the semigroup assumption and adds dephasing.

What can happen to an open system?

A channel is a complex-linear, completely positive, trace-preserving state map. In finite dimension it is exactly a part of a unitary evolution with an independently prepared environment. Countable Kraus sums and the specified continuous unitary mixtures also define channels on separable spaces. An initially correlated preparation needs its own joint-state specification; a system marginal alone need not determine its future.

Summary. The spectator test excludes transpose as a local channel. A positive finite Choi matrix yields Kraus operators by explicit reshaping; their completeness sum makes the dilation an isometry and permits a finite unitary extension. Infinite sufficient constructions need convergence and normality, not a formal replacement of finite sums by integrals. Spontaneous emission reduces population and coherence by different factors while its system and environment together may remain pure.

Exercises. Try Sheet 7 in the separate *Exercise sheets* booklet (English).

Sources. Watrous, *The Theory of Quantum Information*, Theorem 2.22 pp. 82–84 (opposite Choi factor order); Heinosaari–Ziman, *The Mathematical Language of Quantum Theory*, pp. 186–189,

207–209 (also an output-first convention in Theorem 4.48); Nielsen–Chuang, §8.2 pp. 360–376; Gustafson–Sigal, *Mathematical Concepts of Quantum Mechanics*, 3rd ed. (2020), Theorems 19.21, 19.24 pp. 260–265. The explicit infinite-series and continuous-expectation arguments above use L1 and L6.

Lecture 8

Open-system dynamics: the GKSL equation

Objectives. *A channel describes one change of state. Modelling changes by a continuous, time-homogeneous semigroup imposes more structure. In finite dimension its generator has the GKSL form. We prove that this form generates channels, sketch why it is necessary, and derive the relation between population relaxation and coherence decay for a qubit.*

- Identify the semigroup and continuity assumptions, and prove that a given finite GKSL generator yields channels.
- Explain conditional Choi positivity in the converse sketch without claiming that the generator itself is CP.
- Solve the qubit Bloch equations and distinguish positivity from CP in the T_1, T_2 constraint.

8.1 A model with no remembered starting time

Throughout the main theorems $\mathcal{H} = \mathbb{C}^n$. A *quantum dynamical semigroup* is a family of channels Φ_t, $t \geq 0$, satisfying

$$\Phi_0 = \mathrm{id}, \qquad \Phi_{s+t} = \Phi_s \Phi_t, \qquad \|\Phi_t - \mathrm{id}\| \longrightarrow 0 \quad (t \downarrow 0).$$

The norm is an operator norm on the finite-dimensional matrix space; all such norms give the same convergence. Continuity at zero implies it at every time: small-time maps have a common finite norm bound, and the semigroup law bounds the maps on any finite time interval by splitting it into small steps. Then $\Phi_{t+h} - \Phi_t = \Phi_t(\Phi_h - \mathrm{id})$ and $\Phi_t - \Phi_{t-h} = \Phi_{t-h}(\Phi_h - \mathrm{id})$ give right and left norm continuity. Composition depends only on the elapsed times, rather than on when the experiment starts. The same rule applies to every intermediate state without retaining system–environment correlations or environmental memory as extra input. There is no requirement that Φ_t have a positive inverse, and in general negative times are not channels.

A continuous semigroup on a finite-dimensional vector space has a bounded linear generator $\mathcal{L}$ with $\Phi_t = e^{t\mathcal{L}}$ and $\dot{\rho} = \mathcal{L}(\rho)$ (cited: Rivas–Huelga, Theorem 1.3.1 pp. 5–6 and Corollary 1.3.1 p. 7). This exponential uses a matrix power series on the vector space of matrices. The Hamiltonian exponential acts on vectors instead; the two spaces and their generators differ. Our forward proof starts with a specified $\mathcal{L}$ and proves its channel property directly, without using this cited converse-to-continuity step as an input.

L7's decay channels illustrate the semigroup condition. Their excited population multiplier is $a(t) = 1 - \gamma(t)$, and composition multiplies $a(s)a(t)$. Taking $a(t) = e^{-t/T_1}$ supplies the semigroup law; an arbitrary $\gamma(t)$ need not. For a fixed initially separate environment, each

map $\Phi_{t,0}(\rho) = \mathrm{Tr}_E[U_t(\rho \otimes \sigma_E)U_t^\dagger]$ is a channel. At an intermediate time s, the joint state may be correlated, and L7's correlation example shows why the marginal need not determine its future. A semigroup additionally posits a channel $\Phi_{t,s} = \Phi_{t-s}$ satisfying $\rho(t) = \Phi_{t,s}(\rho(s))$. Factorisation is not automatically restored.

8.2 Why GKSL form produces channels

For matrices A, B, write $\{A, B\} = AB + BA$.

Theorem 8.1 (Finite GKSL, forward direction). *Let $H = H^\dagger$ be an $n \times n$ matrix and let $L_1, \ldots, L_d$ be a finite list of matrices. Then*

$$\mathcal{L}(X) = -\frac{i}{\hbar}[H, X] + \sum_{k=1}^{d}\left(L_k X L_k^\dagger - \frac{1}{2}\{L_k^\dagger L_k, X\}\right) \tag{8.1}$$

generates a semigroup of channels $\Phi_t = e^{t\mathcal{L}}$ for $t \geq 0$.

Proof. Set $R = \sum_k L_k^\dagger L_k$ and $A_0 = -iH/\hbar - R/2$. For a time step $h \geq 0$ the map

$$\Psi_h(X) = (\mathbb{1} + hA_0)X(\mathbb{1} + hA_0)^\dagger + h\sum_k L_k X L_k^\dagger$$

is CP by its Kraus form. Expanding gives exactly $\Psi_h = \mathrm{id} + h\mathcal{L} + h^2\mathcal{R}_2$, where $\mathcal{R}_2(X) = A_0 X A_0^\dagger$. This step need not be TP: its Kraus completeness sum is $\mathbb{1} + h^2 A_0^\dagger A_0$. We use it only to approximate a CP limit.

Choose a submultiplicative operator norm for maps on the matrix space. The exponential power series and the exact expansion imply, for small h,

$$\left\|\Psi_h - e^{h\mathcal{L}}\right\| \leq c_1 h^2, \qquad \|\Psi_h\|, \ \left\|e^{h\mathcal{L}}\right\| \leq e^{Mh}$$

with fixed finite constants c_1, M. For example, $\|\Psi_h\| \leq 1 + h\,\|\mathcal{L}\| + h^2\,\|\mathcal{R}_2\|$; the exponential remainder is bounded by $h^2\,\|\mathcal{L}\|^2\, e^{h\|\mathcal{L}\|}/2$. For any two maps A, B, the telescoping identity $A^N - B^N = \sum_{j=0}^{N-1} A^{N-1-j}(A - B)B^j$ holds without commutation. Taking $h = t/N$ therefore gives

$$\left\|\Psi_{t/N}^N - e^{t\mathcal{L}}\right\| \leq N c_1 (t/N)^2 e^{Mt} = \frac{c_1 t^2 e^{Mt}}{N} \longrightarrow 0.$$

Every finite composition is CP: products of its Kraus operators form a Kraus list. Its Choi matrix is positive by L7; entrywise convergence passes every nonnegative quadratic form to the limit, so the limiting Choi matrix is positive. L7 implies $e^{t\mathcal{L}}$ is CP.

Independently, finite trace cyclicity in (8.1) gives $\mathrm{Tr}\,\mathcal{L}(X) = 0$ for every X. Differentiating the convergent power series yields $\frac{d}{dt}\mathrm{Tr}(e^{t\mathcal{L}}X) = \mathrm{Tr}\,\mathcal{L}(e^{t\mathcal{L}}X) = 0$. The trace is therefore its initial value, so the limit is TP. The exponential has the semigroup law and norm continuity by its power series, completing the proof. ∎

The approximation does not interpret every short step as a normalised physical experiment. Trace preservation is proved for the exact limit, not borrowed from the approximants. Nor does the proof assume that a matrix generator is positive or CP. It is the exponential that must map states to states, just as a Hamiltonian itself is not an evolution operator.

The L_k are called *Lindblad operators*; each has units time$^{-1/2}$. Their list and the split into Hamiltonian and noise terms are not unique. A different representation can describe

the same generator, so identifying an individual term with a measured event requires an additional experimental model. The finite theorem proves no statement about arbitrary unbounded L_k on L^2.

Example 8.2 (One generator, an explicit channel)**.** For $H = 0$ and $L_1 = \sqrt{\kappa/2}\sigma_z$, $\kappa \geq 0$, the generator is $\mathcal{L}(X) = \kappa(\sigma_z X \sigma_z - X)/2$. Diagonal entries are constant and off-diagonal entries acquire the factor $e^{-\kappa t}$. Thus its exact solution is

$$\Phi_t(X) = \frac{1 + e^{-\kappa t}}{2} X + \frac{1 - e^{-\kappa t}}{2} \sigma_z X \sigma_z.$$

Both coefficients are nonnegative and sum to one, so L7 independently recognises the solution as a random-unitary channel. It leaves the two σ_z rays fixed, while suppressing their relative coherence. Multiplying the off-diagonal factors verifies the semigroup law directly. The same formula can describe classical phase noise or an ignored quantum record. The reduced state alone does not determine which environment produced it, a distinction needed for the measurement problem in L12. If κ were negative, the off-diagonal factor would exceed one and $|+x\rangle\langle+x|$ would leave the Bloch ball. The nonnegative-rate restriction is visible here without solving a general matrix exponential.

8.3 Why every finite channel semigroup has this form

The converse is *sketched*; see Rivas–Huelga, Theorem 4.2.1 pp. 35–39, with constant coefficients for a semigroup. The source treats more general time-dependent divisible dynamics as well. Here the generator is time independent and the space finite dimensional.

Use L7's unnormalised $\Omega = \sum_i |ii\rangle$ and its Choi convention. For a differentiable channel family,

$$C_{\Phi_t} = |\Omega\rangle\langle\Omega| + tC_{\mathcal{L}} + o(t).$$

Let $E_\perp = \mathbb{1} - |\Omega\rangle\langle\Omega|/n$. For $v \perp \Omega$, positivity for $t > 0$ implies $0 \leq \langle v, C_{\Phi_t} v\rangle = t\,\langle v, C_{\mathcal{L}} v\rangle + o(t)$. Dividing by t and taking the limit yields

$$E_\perp C_{\mathcal{L}} E_\perp \geq 0. \tag{8.2}$$

It does not give $C_{\mathcal{L}} \geq 0$. The nonnegative part lives on the complement of the maximally entangled vector.

The Choi derivative $C_{\mathcal{L}}$ is Hermitian because each C_{Φ_t} is. Factor that compressed matrix by finite spectral theory. Reshape its vectors as in L7 to get a CP map $\mathcal{N}(X) = \sum_k L_k X L_k^\dagger$ whose Choi matrix is exactly the compression. The remaining Hermitian Choi matrix has zero compression onto $\Omega^\perp$. In a basis beginning with $\Omega/\sqrt{n}$, it has only a first row and column. It can therefore be written $|g\rangle\langle\Omega| + |\Omega\rangle\langle g|$ for a vector g. The first-row/column assertion can be checked directly. Write the residual matrix as D in the decomposition $\mathbb{C}^n \otimes \mathbb{C}^n = \mathrm{span}\{\Omega/\sqrt{n}\} \oplus \Omega^\perp$. For a real scalar d and a vector $v \in \Omega^\perp$, it has block form

$$D = \begin{pmatrix} d & v^\dagger \\ v & 0 \end{pmatrix}, \qquad g = \frac{d}{2n}\Omega + \frac{1}{\sqrt{n}} v.$$

The coordinates of Ω are $(\sqrt{n}, 0)$ in this decomposition. Multiplying the two rank-one matrices gives

$$|g\rangle\langle\Omega| + |\Omega\rangle\langle g| = \begin{pmatrix} d/2 & 0 \\ v & 0 \end{pmatrix} + \begin{pmatrix} d/2 & v^\dagger \\ 0 & 0 \end{pmatrix} = D.$$

Reshape by $g = (\mathbb{1} \otimes A_0)\Omega$, so $g_{(i,a)} = (A_0)_{ai}$. Indeed $\sum_{ij} \mathsf{e}_{ij} \otimes A_0 \mathsf{e}_{ij} = |g\rangle\langle\Omega|$, and the adjoint expression gives $|\Omega\rangle\langle g|$. Thus the residual map is $A_0 X + X A_0^\dagger$. This is the algebraic step underlying the converse; the source gives the full coefficient argument.

Differentiating $\operatorname{Tr}\Phi_t(X) = \operatorname{Tr} X$ at zero gives $\operatorname{Tr}\mathcal{L}(X) = 0$ for every X, forcing the resulting decomposition to satisfy $A_0 + A_0^\dagger = -\sum_k L_k^\dagger L_k = -R$. Its anti-self-adjoint part has the form $-iH/\hbar$ with $H = H^\dagger$, so $A_0 = -R/2 - iH/\hbar$. Substitution gives (8.1). Together with the forward theorem, this describes all continuous finite channel semigroups. The sketch is a finite-matrix statement; it does not establish a general unbounded infinite-dimensional GKSL theorem.

8.4 The damped qubit and its two times

Use $|0\rangle$ excited, $|1\rangle$ ground, and

$$H = \frac{\hbar\nu_0}{2}\sigma_z, \qquad L_1 = \frac{|1\rangle\langle 0|}{\sqrt{T_1}}, \qquad L_2 = \sqrt{\frac{\gamma_\phi}{2}}\sigma_z, \quad T_1 > 0, \quad \gamma_\phi \geq 0.$$

The first noise term loses excited population. The second leaves populations fixed but changes the sign of an off-diagonal entry inside $\sigma_z\rho\sigma_z$, so its net contribution there is $-\gamma_\phi\rho_{01}$. For $\rho = \begin{pmatrix} a & c \\ \bar{c} & 1-a \end{pmatrix}$ the two dissipators are

$$\mathcal{N}_1(\rho) = \frac{1}{T_1}\begin{pmatrix} -a & -c/2 \\ -\bar{c}/2 & a \end{pmatrix}, \qquad \mathcal{N}_2(\rho) = \begin{pmatrix} 0 & -\gamma_\phi c \\ -\gamma_\phi \bar{c} & 0 \end{pmatrix}.$$

The commutator contributes $-i\nu_0 c$ in the upper off-diagonal entry and its conjugate below. Adding these matrices gives

$$\dot{\rho}_{00} = -\rho_{00}/T_1, \tag{8.3}$$

$$\dot{\rho}_{01} = -\left(i\nu_0 + \frac{1}{2T_1} + \gamma_\phi\right)\rho_{01}. \tag{8.4}$$

Trace and Hermiticity give the other two entries. Define T_2 by the coherence decay rate:

$$\frac{1}{T_2} = \frac{1}{2T_1} + \gamma_\phi, \qquad T_2 \leq 2T_1. \tag{8.5}$$

T_1 and T_2 are *times*; γ_ϕ is a rate. The factor $1/2$ in the pure-dephasing Lindblad operator is needed to produce precisely this rate, rather than twice it.

For initial $|{+x}\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$, integration gives

$$\rho_{00}(t) = \tfrac{1}{2}e^{-t/T_1}, \qquad \rho_{01}(t) = \tfrac{1}{2}e^{-t/T_2}e^{-i\nu_0 t}.$$

With QM-I's $\rho = (\mathbb{1} + \mathbf{r}\cdot\boldsymbol{\sigma})/2$, this is

$$r_x = e^{-t/T_2}\cos(\nu_0 t), \quad r_y = e^{-t/T_2}\sin(\nu_0 t), \quad r_z = e^{-t/T_1} - 1. \tag{8.6}$$

The Bloch vector spirals toward the ground pole $r_z = -1$. The probability of finding $+x$ is $[1 + e^{-t/T_2}\cos(\nu_0 t)]/2$, whereas the excited population is $e^{-t/T_1}/2$. These different observables separate the two times. In a frame rotating at ν_0 the oscillation is removed and the transverse radius simply decays. If $\gamma_\phi = 0$, L7's emission model is recovered and $T_2 = 2T_1$; extra dephasing shortens T_2 without altering the population decay.

The decay times can be compared without the free rotation. Divide the excited population by its initial value and the coherence magnitude by its initial magnitude. For this input the resulting curves are $a = e^{-t/T_1}$ and $b = e^{-t/T_2}$. Figure 8.1 compares pure emission with an additional dephasing rate. The population curve stays fixed while the coherence curve changes.

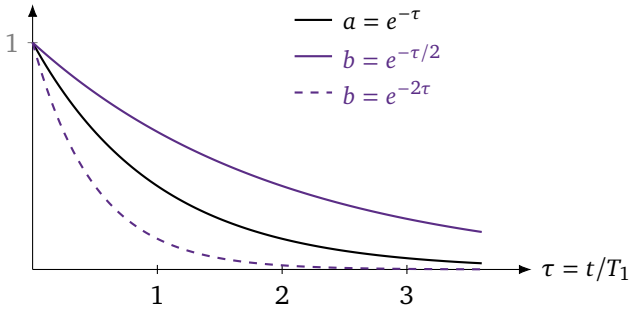


Figure 8.1: Normalised population a and coherence magnitude b for the initial $+x$ state. Solid violet gives $T_2 = 2T_1$; dashed violet gives $T_2 = T_1/2$, while the black population curve is the same in both models.

8.5 Where complete positivity adds a constraint

Consider independently specified population and coherence multipliers for relaxation to the pure ground state: the population is multiplied by a and the coherence by b. Positivity alone then imposes $T_2 \le 2T_1$ when they have the exponential parametrisation below. To see this, remove the irrelevant phase rotation, write $a = e^{-t/T_1}$, $b = e^{-t/T_2}$, and start with a pure state having excited population ϵ and coherence $\sqrt{\epsilon(1-\epsilon)}$. The output determinant is

$$\epsilon(a - b^2) + \epsilon^2(b^2 - a^2).$$

Nonnegativity for arbitrarily small $\epsilon > 0$ requires $b^2 \le a$. Thus this model alone cannot demonstrate the difference between positive and completely positive dynamics.

Use instead the *unital* Bloch map $(r_x, r_y, r_z) \mapsto (br_x, br_y, ar_z)$, with $0 \le a, b \le 1$. It describes the symmetry of equal upward and downward relaxation rates, with a maximally mixed equilibrium (an infinite-temperature idealisation) when the semigroup is CP. Independently of CP, the displayed map contracts the Bloch ball and is therefore positive. Its Choi matrix is

$$C = \frac{1}{2}\begin{pmatrix} 1+a & 0 & 0 & 2b \\ 0 & 1-a & 0 & 0 \\ 0 & 0 & 1-a & 0 \\ 2b & 0 & 0 & 1+a \end{pmatrix}.$$

Its eigenvalues are $(1-a)/2$ twice and $(1+a)/2 \pm b$. Complete positivity requires $2b \le 1+a$. With the exponential a, b above, a small-time expansion of the lowest eigenvalue gives

$$\frac{1+a}{2} - b = t\left(\frac{1}{T_2} - \frac{1}{2T_1}\right) + O(t^2).$$

If $T_2 > 2T_1$, these maps form a positive continuous semigroup but fail CP at small positive times. Conversely $T_2 \le 2T_1$ gives $b \le \sqrt{a} \le (1+a)/2$, hence CP at every time. The extra

constraint has now been demonstrated in a model where ordinary positivity does not already impose it.

For a numerical spectator test, take $a = 0.8$, $b = 0.95$. Every isolated Bloch vector contracts, yet the lowest Choi eigenvalue is $(1+0.8)/2 - 0.95 = -0.05$. Acting on half of the normalised maximally entangled input produces $C/2$, with negative eigenvalue -0.025 along $(|00\rangle - |11\rangle)/\sqrt{2}$. A putative reading of that projection would have a negative probability. Testing only population and coherence bounds on single-system states would have missed it.

Finally, a semigroup is an assumption about the physical regime. Born, Markov and often secular or coarse-graining approximations enter microscopic reservoir derivations; none is proved by writing a GKSL expression. In a weak-coupling Born approximation, the reservoir state is kept close to a specified reference state when computing the reduced evolution. The Markov approximation neglects the dependence of the reduced rate on earlier system states over the reservoir's memory time. A secular approximation averages terms oscillating at sufficiently different transition frequencies. Coarse-graining averages over a time window short compared with relaxation but long enough to suppress the reservoir correlations being neglected. These are separate assumptions about coupling strengths and timescales. QM-II's short-time law (Proposition 12.8) requires a unit vector in $D(H)$; its cited Khalfin theorem (Theorem 12.9) also requires a lower energy bound. In QM-II's Friedrichs decay model the excited population is exactly that closed-system survival probability: the initial excited vector is the only excited-system sector. It starts quadratically, rather than with a nonzero exponential slope, and cannot remain exponentially bounded at arbitrarily late times. Thus e^{-t/T_1} is an effective intermediate-time model there, with $1/T_1 \approx \Gamma$ from the cited golden rule (QM-II, Theorem 12.13). A generic reduced population is not automatically a survival probability; its relation to the closed dynamics must be checked, not used to evade the decay model's short- and long-time restrictions. For infinite systems later lectures build their channels explicitly; any differential generator statements have their own domain qualifications.

How does an open system evolve in time?

A finite-dimensional open system modelled by a continuous time-homogeneous semigroup of channels obeys $\dot{\rho} = \mathcal{L}(\rho)$ with a GKSL generator. The forward implication is proved here and the converse sketched. The semigroup is a physical modelling assumption. Closed evolution is the case with no Lindblad terms. Infinite-dimensional generators require additional hypotheses beyond this finite theorem.

Summary. CP short steps converge in norm to the finite GKSL exponential; trace preservation is checked separately. Conditional Choi positivity explains the converse structure. Population and coherence decays give $1/T_2 = 1/(2T_1) + \gamma_\phi$. The unital qubit shows why CP is stronger than positivity. Exact subsystem semigroup behaviour and exact closed-system survival are different claims.

Exercises. Try Sheet 8 in the separate *Exercise sheets* booklet (English).

Sources. Rivas–Huelga, *Open Quantum Systems*, pp. 5–7, Theorem 4.2.1 pp. 35–39; Weinberg, *Lectures on Quantum Mechanics*, §6.9 pp. 237–243; Nielsen–Chuang, p. 386 (their Lindblad normalisation differs by $\sqrt{2}$); Schlosshauer, *Decoherence and the Quantum-to-Classical Transition*, pp. 155–167. The forward proof includes its own finite product estimate.

Part IV

Observation

An apparatus specifies both probabilities for its readings and changes to the measured system. We distinguish these two pieces of information, then examine how an environment suppresses interference. The remaining question is whether unitary evolution also accounts for one definite global outcome.

Lecture 9

Readings beyond spectral measures: POVMs

Objectives. *A pointer can have noisy statistics that are not a spectral measure of the system alone. Effects describe those probabilities. We prove that every countable family of such effects can be obtained by a sharp reading on a larger system, derive the effects of a position pointer, and identify the obstruction to a joint sharp position–momentum reading.*

- Define a POVM and prove its countable Naimark dilation, including a valid enlarged unitary implementation.
- Compute the calibrated pointer's effect, keeping the reflection of its response density.
- Prove the joint-sharp-reading commutation criterion and evaluate the coherent-state phase-space reading.

9.1 Probabilities need effects, not necessarily projections

QM-I's probe (Example 11.11) already formed $F_j = K_j^\dagger K_j$ and called its family a POVM. These are the effects defined generally in Lecture 3. No requirement made $F_j^2 = F_j$. For a whole reading procedure, its events must fit together into one normalised probability assignment for every state.

For elementary real-outcome calculations, events are intervals and their finite disjoint unions. In phase space use rectangles and finite disjoint unions of rectangles. An additivity statement assumes its union belongs to this event class. The general cited theorems below use *Borel sets*, the larger class generated by countable unions and complements; their full event-space framework is a cited input, not a measure-extension construction supplied here.

Definition 9.1 (Positive operator-valued measure)**.** For countably many outcomes, a *POVM* is a family of effects $F_j \geq 0$ with $\sum_j F_j = \mathbb{1}$ strongly. Its outcome probabilities are $p_j = \mathrm{Tr}(\rho F_j)$. For real or phase-space outcomes it assigns effects $F(B)$ to event sets, with $F(\varnothing) = 0$, $F(\text{whole space}) = \mathbb{1}$, and, for every disjoint countable event family,

$$\left\langle \psi, F\Big(\bigcup_j B_j\Big)\psi \right\rangle = \sum_j \left\langle \psi, F(B_j)\psi \right\rangle .$$

A POVM is *sharp* when its effects form a projection-valued measure (PVM), with $F(B\cap C) = F(B)F(C)$; otherwise it is unsharp.

This is QM-II's interval spectral-measure structure with multiplicativity dropped. Finite additivity extends to finite unions, and countable additivity is required whenever the union

is still an admissible event.

Positivity gives $0 \le F(B) \le \mathbb{1}$ by its complement. The countable axiom is deliberately weaker than multiplicativity. L1's spectral density expansion and interchange of nonnegative sums show it gives countably additive scalar probabilities for every density operator, not just rays. A joint phase-space POVM has marginals $F_1(I) = F(I \times \mathbb{R})$ and $F_2(J) = F(\mathbb{R} \times J)$.

The mean operator alone loses information. For $0 < \eta < 1$, a noisy qubit reading with numerical outputs ± 1 has effects $F_\pm = (\mathbb{1} \pm \eta\sigma_z)/2$ and first moment $F_+ - F_- = \eta\sigma_z$. The sharp operator $A = \eta\sigma_z$ has that same mean operator, but its outputs are $\pm\eta$ with spectral projections $(\mathbb{1} \pm \sigma_z)/2$. In $|0\rangle$ the noisy reading gives ± 1 with probabilities $(1 \pm \eta)/2$, while the sharp one gives $+\eta$ with certainty. Their second moment operators are $\mathbb{1}$ and $\eta^2\mathbb{1}$ respectively. Thus specifying a first moment does not specify the reading's statistics. The identification of an observable with one self-adjoint operator includes the additional assertion that its statistics are sharp.

9.2 Naimark: a sharp reading on a larger space

Theorem 9.2 (Countable Naimark dilation)**.** *For a countable POVM on nonzero separable $\mathcal{H}$, pad a finite list with zero effects if necessary and let (e_j) be the basis of ℓ^2. There is an isometry $J : \mathcal{H} \to \mathcal{H} \otimes \ell^2$ with*

$$J\psi = \sum_j \sqrt{F_j}\psi \otimes e_j, \qquad F_j = J^\dagger(\mathbb{1} \otimes |e_j\rangle\langle e_j|)J.$$

With an additional two-dimensional factor this is implemented by a unitary and fixed ready ancilla, with $U_M(\psi \otimes e_0 \otimes f_0) = J\psi \otimes f_0$ and pointer $\Pi_j = \mathbb{1} \otimes |e_j\rangle\langle e_j| \otimes \mathbb{1}_2$. The same probabilities are a sharp pointer reading there.

Proof. S6 supplies each positive square root. Orthogonality of the register coordinates gives

$$\|J\psi\|^2 = \sum_j \left\|\sqrt{F_j}\psi\right\|^2 = \sum_j \langle \psi, F_j\psi\rangle = \|\psi\|^2 .$$

The series therefore converges in the L6 tensor norm. It is linear by limits of finite sums; polarization gives $J^\dagger J = \mathbb{1}$. Pairing arbitrary φ, ψ with the jth register projection gives $\langle \sqrt{F_j}\varphi, \sqrt{F_j}\psi\rangle = \langle \varphi, F_j\psi\rangle$, proving the compressed-effect formula.

Write $\mathcal{K}_0 = \mathcal{H} \otimes \ell^2$ and $\mathcal{K} = \mathcal{K}_0 \otimes \mathbb{C}^2$, with last-factor basis f_0, f_1. Define an isometry between the input subspace and image by

$$\psi \otimes e_0 \otimes f_0 \longmapsto J\psi \otimes f_0.$$

Both subspaces are closed: the input has its explicit product projection, and the image projection is $J'J'^\dagger$, where $J'\psi = J\psi \otimes f_0$. Both orthogonal complements contain $\mathcal{K}_0 \otimes f_1$, so both have countably infinite Hilbert dimension. Complement ONBs can be constructed by applying the complementary projections to a countable basis of $\mathcal{K}$ and using Gram–Schmidt on their dense spans. Map these ONBs onto one another, extending the displayed isometry to an onto unitary U_M. With ready ancilla $e_0 \otimes f_0$, read the pointer projections $\Pi_j = \mathbb{1} \otimes |e_j\rangle\langle e_j| \otimes \mathbb{1}_2$. Their probabilities are exactly $\mathrm{Tr}(\rho F_j)$, first on rays by the compression and then on density operators by their positive spectral sums. ∎

The spare factor makes both complements visibly infinite without computing the complement of $J(\mathcal{H})$, even for infinite outcome lists. An arbitrary infinite isometry need not extend

on the original space without it. This construction proves existence of a reading model; it does not specify a unique apparatus or post-reading state. L10 will distinguish procedures with the same POVM.

For real Borel outcomes the general Naimark theorem is *cited*: every normalised POVM is a compression $F(B) = J^\dagger E(B) J$ of a PVM on a larger space (Busch–Lahti–Pellonpää–Ylinen, abbreviated BLPY below, Theorem 7.7 p. 146). Its proof is not the replacement of the countable sum above by an unexplained continuous basis of eigenvectors.

9.3 A calibrated position pointer

Take system coordinate x and meter coordinate y, a normalised $\varphi \in \mathcal{S}(\mathbb{R})$ and coupling strength $g > 0$. Define the coupling first by the shear

$$(U_g \Psi)(x, y) = \Psi(x, y - gx).$$

A change of y variable proves norm preservation for smooth compactly supported functions; S4 extends it to an isometry of $L^2(\mathbb{R}^2)$, and U_{-g} is its inverse. In the meter's partial Fourier representation it is multiplication by $e^{-igxp_M/\hbar}$. The notation $e^{-igQ\otimes P^{(2)}/\hbar}$ denotes this unitary; the shear definition keeps us from assuming a domain for a formal unbounded operator product. For a Schwartz system input it gives $\psi(x)\varphi(y - gx)$.

Calibrate the meter reading as $\hat{x} = y/g$. The probability that it lies in I is

$$\begin{aligned} p_\psi(I) &= \int |\psi(x)|^2 \int_{gI} |\varphi(y - gx)|^2 \, dy \, dx \\ &= \int |\psi(x)|^2 f_I(x) \, dx, \qquad f_I(x) = \int_I g|\varphi(g(z - x))|^2 \, dz. \end{aligned}$$

All integrands are nonnegative and the total is one. Thus $0 \le f_I \le 1$, and $F(I) = f_I(Q)$ is an effect. The interval effect also has the bounded compression form

$$F(I) = J_\varphi^\dagger U_g^\dagger (\mathbb{1} \otimes E_{Q^{(2)}}(gI)) U_g J_\varphi, \qquad J_\varphi \psi = \psi \otimes \varphi.$$

The known meter interval PVM is strongly additive; bounded compression therefore proves additivity for admissible unions and finite unions. Both sides of the probability identity are continuous quadratic forms in ψ, so S4 extends the Schwartz calculation to every L^2 vector. Positive spectral sums then give the same effect for every density operator. No position eigenvector was an input state.

With convolution defined as $(f * e)(x) = \int f(z) e(x - z) \, dz$, the result is

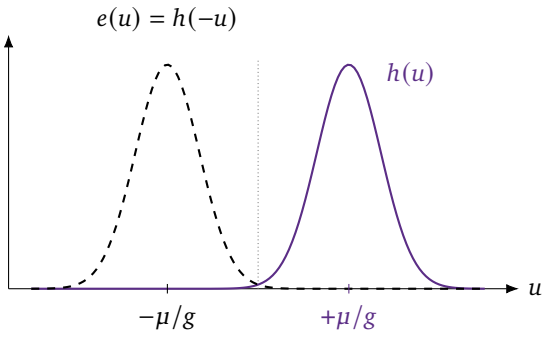
$$F(I) = (1_I * e)(Q), \qquad e(u) = g|\varphi(-gu)|^2. \tag{9.1}$$

The reflection is essential. The response's *actual reading error* $\hat{x} - x$ has density $h(u) = g|\varphi(gu)|^2 = e(-u)$; e is the kernel used in the effect convolution, not that error density. A meter initially centred at μ shifts the readout by $+\mu/g$, although e is centred at $-\mu/g$. Known bias can be removed in calibration. For a centred even Gaussian the two densities coincide, which can conceal a wrong sign.

The same response density has two roles in this formula. For a fixed input coordinate x, the density of the calibrated output z is $h(z - x)$. The interval probability at that coordinate is therefore

$$f_I(x) = \int_I h(z - x) \, dz.$$

The convolution convention uses the argument $x - z$ instead. It consequently requires $e(x - z) = h(z - x)$. Figure 9.1 shows why a displaced response makes the sign visible. Here x is an integration coordinate, not a proposed exactly sharp preparation.



$$f_I(x) = \int_I h(z - x)\, dz = (1_I * e)(x)$$

Figure 9.1: The reading-error density h (solid) and the reflected effect kernel e (dashed) for a Gaussian meter with positive bias. The widths agree, but their centres have opposite signs; the effect integrates the output density over the chosen interval.

For example, with a Gaussian meter and a bounded interval of positive length, both its interval and complement integrals are strictly positive. Thus $0 < f_I(x) < 1$ for every x. An interval effect is generally not a projection: where $0 < f_I(x) < 1$, its square multiplies by $f_I(x)^2 \neq f_I(x)$. Finite meter width produces blurred edges instead of a sharp interval event. This implements QM-II's promised unsharp position reading with a concrete response function. For multiplication operators, equality *almost everywhere* means equality outside a set of positions of length zero. Changing values on such a set does not change the represented L^2 function or multiplication operator. Later we will compute its state change and the momentum spread added by ignoring the meter; neither is specified by (9.1) alone.

9.4 Joint sharp readings must commute

Lemma 9.3 (An effect dominated by a projection)**.** *If $0 \le F \le E$ and E is a projection, then $F = EFE$, so $EF = FE = F$.*

Proof. For $v \in \ker E$, $0 \le \langle v, Fv\rangle \le \langle v, Ev\rangle = 0$. S6 implies $\left\|\sqrt{F}v\right\|^2 = 0$, hence $Fv = 0$. Self-adjointness kills all cross terms with $\ker E$ as well, giving $F = EFE$. ∎

Proposition 9.4 (Joint sharp readings)**.** *Two countable PVMs have a joint POVM exactly when all their projections commute. If only the first marginal is sharp, the second marginal's effects must commute with it.*

Proof. Suppose a countable joint POVM (M_{jk}) has sharp first marginal $E_j = \sum_k M_{jk}$ and second marginal $F_k = \sum_j M_{jk}$. Since $M_{jk} \le E_j$, the lemma gives $E_j M_{jk} = M_{jk}$. For a different first label ℓ, $E_j M_{\ell k} = 0$, because $M_{\ell k}$ is supported in the orthogonal range of E_ℓ. Taking the strong marginal sum therefore yields

$$E_j F_k = M_{jk} = F_k E_j.$$

Thus a joint reading with one sharp marginal already forces the other marginal's effects to commute with it. In particular, two countable PVMs can have a joint POVM only if every pair of their projections commutes.

Conversely, for commuting PVMs (E_j), (F_k) set $M_{jk} = E_j F_k$. These are projections, mutually orthogonal for different pairs. For every vector, nonnegative sums give $\sum_{jk} \langle \psi, M_{jk}\psi \rangle = \sum_j \|E_j\psi\|^2 = \|\psi\|^2$, using completeness of the second PVM within each first-sector vector. Thus their sum is strongly $\mathbb{1}$ and their marginals are the prescribed PVMs. This proves the countable joint-sharp-reading equivalence. ∎

For real outcomes, a joint POVM M with sharp marginals E_1, E_2 gives the same necessity using the four events $I \times J$, $I \times J^c$, $I^c \times J$, $I^c \times J^c$ and the lemma; complements of intervals are finite unions of intervals, so these events remain in the elementary class. It yields $E_1(I)E_2(J) = M(I \times J) = E_2(J)E_1(I)$ for every I, J. The general continuous converse uses a joint spectral construction and is *cited*, BLPY, Theorem 11.3 pp. 269–270, for its real Borel value spaces.

For Q, P the relevant noncommutation is spectral, not merely a formal commutator on a test domain. Here is the spectral argument, *sketched* using QM-II's cited spectral calculus. If every position projection commuted with every momentum projection, it would commute with $U(a) = e^{-iaP/\hbar}$. But this unitary translates a wave packet from an interval I to $I + a$. Choose a nonzero smooth packet supported in I and a translation with disjoint intervals: projecting before or after translation gives different results. Thus some spectral projections do not commute. There is no joint POVM with both exact Q and exact P as marginals. Joint unsharp readings remain possible.

9.5 A coherent-state phase-space reading

Recall QM-II's coherent vectors ψ_α with oscillator length s, $\alpha = (q/s + isp/\hbar)/\sqrt{2}$. The vacuum-generated phase-space POVM has density

$$F(d^2\alpha) = \frac{1}{\pi} |\psi_\alpha\rangle\langle\psi_\alpha| \, d^2\alpha. \tag{9.2}$$

Its normalisation and marginals are *cited*, BLPY, equation (8.14) p. 174 and Proposition 8.9 p. 175. Its position marginal is position blurred by a Gaussian of variance $s^2/2$; its momentum marginal is momentum blurred by one of variance $\hbar^2/(2s^2)$. The coordinates satisfy $dq\,dp = 2\hbar\, d^2\alpha$, so this is equivalently $|\psi_\alpha\rangle\langle\psi_\alpha| \, dq\,dp/(2\pi\hbar)$.

For coherent input ψ_β, the known number-basis expansion gives

$$\langle \psi_\alpha, \psi_\beta \rangle = e^{-(|\alpha|^2+|\beta|^2)/2} \sum_{n=0}^{\infty} \frac{(\bar{\alpha}\beta)^n}{n!} = e^{-(|\alpha|^2+|\beta|^2)/2+\bar{\alpha}\beta}.$$

The reading density is therefore

$$p_\beta(\alpha) = \frac{1}{\pi} e^{-|\alpha-\beta|^2}. \tag{9.3}$$

The two real components have variance $1/2$, and the density integrates to one by the ordinary Gaussian integral. In physical coordinates the observed variances are s^2 and $\hbar^2/s^2$: each contains the coherent input's vacuum variance *plus* the reading's vacuum blur. The added standard deviations have product $\hbar/2$, whereas the total observed standard deviations have product $\hbar$. There is no exact joint reading hidden in a notation that assigns both numbers to one outcome. It is a specified noisy joint POVM.

What does a pointer read?

A reading's statistics are a POVM. Every countable POVM is a projective reading of the system with an ancilla; real-outcome dilation is cited. Sharp statistics are the spectral measures of self-adjoint operators. Joint sharp marginals must commute, so exact Q and P cannot both be marginals of one reading. A calibrated pointer and the specified eight-port detection instead supply explicit unsharp effects. A POVM specifies probabilities; the state after a recorded outcome needs an instrument.

Summary. Effects extend sharp spectral readings. Square roots yield countable Naimark dilation, with an enlarged space ensuring unitary completion. The shear pointer gives a reflected convolution kernel. Projection domination proves the sharp joint-reading constraint. The coherent-state POVM gives a Gaussian phase-space output, with separate preparation and reading contributions to its variance.

Exercises. Try Sheet 9 in the separate *Exercise sheets* booklet (English).

Sources. Busch–Lahti–Pellonpää–Ylinen, *Quantum Measurement*, pp. 71, 146, 168–175, 269–270, 443–447; Busch–Lahti–Mittelstaedt, *The Quantum Theory of Measurement*, §III.2.6 pp. 41–43; Heinosaari–Ziman, Proposition 3.25 pp. 123–124. QM-II's cited spectral theorem is the input to the explicitly sketched Q, P argument; coherent vectors are its Theorem 8.10.

Lecture 10

Instruments: the state after a reading

Objectives. *A POVM predicts a reading's probabilities but does not determine the state left for the next experiment. Lüders' update follows from a precise conditioning premise on all sub-events. An instrument specifies both statistics and state changes. We construct it from a pointer, compute the position pointer's added spread, and compare two full sequences with the same first POVM.*

- Prove Lüders' uniqueness from sub-event conditioning on a separable Hilbert space.
- Define an instrument and derive the instrument induced by a pointer model.
- Compute the Gaussian pointer's resolution and unread momentum spread, and the joint probabilities and final states of the qubit capstone.

10.1 Lüders as conditioning on every sub-event

QM-I, Postulate 4.1, postulated the ideal state update for a sharp reading. We now ask which probability-conditioning premise characterises it. Let ρ be a density operator on nonzero separable $\mathcal{H}$, and E a projection with $p = \mathrm{Tr}(\rho E) > 0$.

Theorem 10.1 (Lüders conditioning). *There is exactly one density operator τ satisfying*

$$\mathrm{Tr}(\tau E) = 1, \qquad \mathrm{Tr}(\tau E') = \frac{\mathrm{Tr}(\rho E')}{p} \quad \text{for every projection } E' \le E.$$

It is $\tau = E\rho E/p$. The first condition is also the case $E' = E$ of the second.

Proof. Expand $\rho = \sum_j r_j \left|\varphi_j\right\rangle\!\left\langle\varphi_j\right|$ by L1. The positive series $E\rho E = \sum_j r_j \left|E\varphi_j\right\rangle\!\left\langle E\varphi_j\right|$ has trace $\sum_j r_j \left\|E\varphi_j\right\|^2 = \mathrm{Tr}(\rho E) = p$, so its normalisation is a density operator. It has support in E. For $E' \le E$, pairing its series with E' gives the required conditional probability, since $EE' = E'E = E'$. Conversely expand a candidate $\tau = \sum_j t_j \left|\chi_j\right\rangle\!\left\langle\chi_j\right|$ in positive spectral terms. The identity $\mathrm{Tr}(\tau(\mathbb{1} - E)) = 0$ says that the sum of the nonnegative weights times $\left\|(\mathbb{1} - E)\chi_j\right\|^2$ vanishes. Every nonzero weighted χ_j lies in $\mathrm{ran}\, E$, hence $\tau = E\tau E$. For every unit $u \in \mathrm{ran}\, E$, use $E' = \left|u\right\rangle\!\left\langle u\right|$ to obtain $\langle u, \tau u\rangle = \langle u, \rho u\rangle / p$. Homogeneity extends equality of these quadratic forms to every vector in that subspace. Complex polarisation determines all their matrix elements there, and both candidate operators vanish on the perpendicular complement. Thus $\tau = E\rho E/p$. ∎

The premise concerns every subprojection, not just the elements of one chosen basis. Merely asking that the outcome E become certain permits many other states in its range when $\mathrm{rank}\, E \ge 2$. The theorem also does not assert that every real apparatus implements this conditioning rule, or explain a dynamical physical collapse. It characterises a rule once

the premise is imposed. At $p = 0$ it supplies no normalised conditional state. The proof works for a qubit and infinite-rank E; it uses no Gleason dimension restriction (compare BLPY pp. 211–212).

10.2 An instrument carries probabilities and outputs

Definition 10.2 (Instrument)**.** For countable outcomes an *instrument* is a family of CP maps $\mathcal{I}_j$ on $\mathcal{T}_1(\mathcal{H})$ whose sum, convergent in every bounded expectation pairing, is a channel. For real Borel outcomes it is a family $B \mapsto \mathcal{I}(B)$ of CP maps, with total map $\mathcal{I}(\mathbb{R})$ a channel and countable additivity in every bounded pairing $\mathrm{Tr}(A\mathcal{I}(B)(\rho))$. For a density input, the outcome probability is $p(B) = \mathrm{Tr}\,\mathcal{I}(B)(\rho)$; if it is positive, the conditional state is $\mathcal{I}(B)(\rho)/p(B)$.

For explicit position computations we use the intervals and finite unions of L9. A full Borel position-pointer model uses the cited Borel spectral framework recalled in QM-II, §5.1; no measure-extension proof is claimed here. The general model argument below starts with a PVM already given. The sum or total map describes the *unread* experiment. Each outcome operation is generally trace decreasing on positive inputs, not a channel by itself. Normalising it makes the conditional rule nonlinear; its unnormalised CP map is what composes simply with the next operation.

The effect must depend on the operation, rather than on a chosen Kraus list. We reconstruct it from the operation's outcome trace. Here is why its probabilities form a POVM even without assuming a Kraus list. For an operation write $b(\varphi, \psi) = \mathrm{Tr}\,\mathcal{I}(B)(|\psi\rangle\langle\varphi|)$. Linearity gives a sesquilinear form, positive on equal arguments, with $0 \le b(\psi,\psi) \le \|\psi\|^2$, because the total instrument preserves trace and the complementary operation is positive. The quadratic-form Cauchy–Schwarz argument gives $|b(\varphi,\psi)| \le \|\varphi\|\,\|\psi\|$. As in L1, set $u_N = \sum_{n\le N} b(e_n,\psi)e_n$ in one ONB. Then $\|u_N\|^2 = b(u_N,\psi) \le \|u_N\|\,\|\psi\|$, so its coefficient series converges. Defining $F(B)\psi = \lim_N u_N$ yields a bounded linear operator with $\langle\varphi, F(B)\psi\rangle = b(\varphi,\psi)$ and $0 \le F(B) \le \mathbb{1}$. For a density's finite spectral head the pairing is $\mathrm{Tr}(\rho F(B))$; the remaining positive tail contributes at most its trace, proving the same equality for the full density. Instrument additivity and normalisation on rays now give the POVM axioms.

For a countable Kraus description the formula is simpler:

$$\mathcal{I}_j(\rho) = \sum_r K_{jr}\rho K_{jr}^\dagger, \qquad F_j = \sum_r K_{jr}^\dagger K_{jr}, \qquad \sum_{j,r} K_{jr}^\dagger K_{jr} = \mathbb{1} \tag{10.1}$$

with strong sums. L7 proves the total channel property; each positive outcome sum has the same trace-tail construction. The POVM retains the F_j while discarding the information about how each operation changes the state. That information matters for subsequent readings.

A *measurement model* specifies a meter space, one fixed ready density τ_M, a unitary U_M and a sharp pointer PVM $\Pi(B)$. Prepare $\rho \otimes \tau_M$ independently. The induced operation is

$$\mathcal{I}(B)(\rho) = \mathrm{Tr}_M\Big[(\mathbb{1}\otimes\Pi(B))U_M(\rho\otimes\tau_M)U_M^\dagger(\mathbb{1}\otimes\Pi(B))\Big]. \tag{10.2}$$

To prove CP explicitly, write $\tau_M = \sum_\ell s_\ell\,|m_\ell\rangle\langle m_\ell|$, choose a meter ONB (e_k) and set $P_B = \mathbb{1}\otimes\Pi(B)$, $J_\ell\psi = \psi\otimes m_\ell$. The coordinate operators

$$K_{\ell k}\psi = \sqrt{s_\ell}(\mathbb{1}\otimes\langle e_k|)P_B U_M J_\ell\psi$$

are bounded. Expanding product vectors in meter coordinates, as in L7, gives $\mathcal{I}(B)(\rho) = \sum_{\ell,k} K_{\ell k}\rho K_{\ell k}^\dagger$. Parseval gives their strong completeness sum $\sum_{\ell,k} K_{\ell k}^\dagger K_{\ell k} = \sum_\ell s_\ell J_\ell^\dagger U_M^\dagger P_B U_M J_\ell \le \mathbb{1}$; the latter convex sum converges in norm. L7's positive-series proof applies with total trace weight at most $\operatorname{Tr}\rho$, and with every finite spectator, proving CP. The partial trace of a positive output of trace below one means its trace times the partial trace of its normalisation (zero stays zero); complex-linear extension is well-defined by bounded local pairings. The total map is trace preserving. For disjoint pointer events, $A \otimes \mathbb{1}$ commutes with the pointer projections, so cross terms between their ranges vanish in every bounded local pairing. Strong convergence of orthogonal projection sums (norm convergence on each fixed vector) and the positive spectral-tail bound prove countable additivity. Thus this model induces an instrument, not merely a list of separate CP maps. A one-sided projection formula gives the same partial trace, but the sandwich makes the CP argument explicit.

Conversely every instrument on the countable or real Borel spaces used here has such a pointer model (*cited*: BLPY, Theorem 7.14 p. 160). BLPY, Theorem 10.1 p. 227, implements a POVM; the instrument theorem also fixes the output operations. Its general proof is not our finite Choi theorem. In finite dimension, adding an outcome register to L7's Kraus isometry gives the familiar construction. This establishes possible models, not uniqueness of an apparatus behind its statistics.

10.3 The position pointer's instrument

Use L9's shear $\psi(x)\varphi(y - gx)$ in the impulsive limit, neglecting system and meter free evolution during the interaction. Take a normalised Schwartz meter φ and calibrated reading $z = y/g$, $g > 0$. The operation is

$$\mathcal{I}(I)(\rho) = \int_I K_z \rho K_z^\dagger \, dz, \qquad (K_z\psi)(x) = \sqrt{g}\,\varphi(g(z - x))\psi(x). \tag{10.3}$$

Each K_z is bounded since φ is bounded. The integral is first identified by its rank-one matrix-element pairings, which uniquely determine the trace-class output of the pointer model. Its bounded expectation pairings then follow for that operator. For a pure input, tracing the meter coordinates in gI gives the kernel

$$\psi(x)\overline{\psi(x')} \int_I g\varphi(g(z - x))\overline{\varphi(g(z - x'))}\, dz,$$

which is exactly the pairing of (10.3). Positive spectral sums and their probability tails extend it to every density operator. CP and additivity were already proved by the model, so no unintroduced operator-integration theorem is being assumed. The effect density is $K_z^\dagger K_z = g|\varphi(g(z - Q))|^2$, reproducing L9's reflected convolution after integration.

$K_z\rho K_z^\dagger$ is an operator *density* in the reading z; its trace is a probability density, not a dimensionless event probability. For a finite interval one integrates before normalising. Where the density trace is positive, normalising it gives the usual posterior density rule for a resolved continuous readout; it is not division by the probability of the zero-width event $\{z\}$.

The same sharp Q PVM does not admit an exactly repeatable real-outcome instrument (*cited*: BLPY, Theorem 10.4 p. 248). This leaves legitimate Lüders operations for individual interval events. In particular, treating $I \mapsto E_Q(I)\rho E_Q(I)$ as one additive instrument fails: splitting an interval removes cross terms between its two subintervals. Approximate or finite-resolution repeatability is a different condition.

10.4 Resolution and unread momentum spread

In meter momentum coordinates the shear state is $\psi(x)\widetilde{\varphi}(p_M)e^{-igxp_M/\hbar}$, using the Fourier convention of QM-II. Ignoring the meter is independent of which meter basis is used for the partial trace. For a fixed p_M the system receives the phase $e^{-igp_MQ/\hbar}$, hence the momentum kick $-gp_M$. For a pure input the unread output momentum density is therefore

$$w_{\mathrm{out}}(p) = \int |\widetilde{\varphi}(p_M)|^2 w_{\mathrm{in}}(p + gp_M)\, dp_M = (w_{\mathrm{in}} * \kappa)(p), \qquad \kappa(q) = \frac{1}{g}\,|\widetilde{\varphi}(-q/g)|^2 . \quad (10.4)$$

The Fourier shift rule verifies the sign; a positive mean p_M produces a negative system momentum shift. The same relation holds for density operators by their positive spectral sums. This is the unread channel; a *selected* reading multiplies the position amplitude by K_z and Fourier-transforms that product, rather than convolving its momentum density by κ.

For a centred Gaussian meter, K_z weights the position amplitude by a Gaussian centred at the indicated coordinate. The resulting momentum density depends on the phase and shape of that particular filtered amplitude. When the outcome is ignored, the meter momentum supplies the independent kick distribution instead. The two calculations answer different experimental questions even when they come from the same coupling.

Define the apparatus *resolution* as the standard deviation of its reading-error density, $\epsilon_Q = \Delta Q^{(2)}/g$, and its *added momentum spread* as that of the unread kick density, $\eta_P = g\Delta P^{(2)}$. Known means are calibrated or tracked separately. If the input momentum variance is finite, the ordinary variance rule for independent convolution gives $\mathrm{Var}_{\mathrm{out}}\, P = \mathrm{Var}_{\mathrm{in}}\, P + \eta_P^2$; the mean shifts by $-g\langle P^{(2)}\rangle$. These are statements about this model's reading error and added noise, not a replacement for the input state's own spreads.

For the centred unchirped Gaussian

$$\varphi(y) = (2\pi\sigma_M^2)^{-1/4} e^{-y^2/(4\sigma_M^2)},$$

$|\varphi|^2$ has variance σ_M^2, and its Fourier density has variance $\hbar^2/(4\sigma_M^2)$. The two densities in question are Gaussians, with

$$\epsilon_Q = \frac{\sigma_M}{g}, \qquad \eta_P = \frac{\hbar g}{2\sigma_M}, \qquad \epsilon_Q\eta_P = \frac{\hbar}{2}. \quad (10.5)$$

Sharper position resolution increases the unread added momentum spread. For any normalised Schwartz meter, QM-II's proved uncertainty relation (Theorem 4.9) gives $\epsilon_Q\eta_P \geq \hbar/2$. More general finite-spread meters require its cited extension on $D(Q^{(2)}) \cap D(P^{(2)})$. Equality is not asserted for a chirped Gaussian. If one defines uncentred root mean square errors instead of standard deviations, biases also enter; they vanish in the centred Gaussian used for (10.5). This is a model-specific apparatus relation, distinct from a universal error–disturbance theorem and from the preparation inequality $\Delta Q\Delta P \geq \hbar/2$ for the system input.

10.5 Capstone: one first POVM, two sequences

Fix the input $\rho = |+x\rangle\langle+x|$, where $|+x\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$. First read σ_z with a CNOT meter initially in $|0_M\rangle$: $|j\rangle\,|0_M\rangle \mapsto |j\rangle\,|j_M\rangle$, followed by the meter reading. The meter coordinate operators are $(\mathbb{1}\otimes\langle k_M|)(\mathbb{1}\otimes|j_M\rangle\langle j_M|)U_M(\psi\otimes|0_M\rangle) = \delta_{kj}E_j\psi$. Thus $\mathcal{I}_j(\rho) = E_j\rho E_j$, $E_j = |j\rangle\langle j|$, $j = 0, 1$. Each first outcome has probability $1/2$. Next rotate the system by

$$U = e^{-i\theta\sigma_y/2} = \begin{pmatrix} \cos(\theta/2) & -\sin(\theta/2) \\ \sin(\theta/2) & \cos(\theta/2) \end{pmatrix}, \qquad \theta = \pi/3.$$

Finally read the noisy POVM $F_k = (\mathbb{1} + k\eta\sigma_z)/2$, $k = \pm 1$, with visibility $\eta = 1/2$ (distinct from added spread η_P) and specified operations $\mathcal{I}'_k(X) = M_k X M_k^\dagger$, where

$$M_+ = \mathrm{diag}(\sqrt{3}/2, 1/2), \qquad M_- = \mathrm{diag}(1/2, \sqrt{3}/2).$$

These positive square roots have $\sum_k M_k^\dagger M_k = \mathbb{1}$. The instrument, rather than just these effects, fixes the final states. The joint probabilities are

$$p(j,k) = \mathrm{Tr}\left(\mathcal{I}'_k(U\mathcal{I}_j(\rho)U^\dagger)\right) = \frac{1}{4}\left(1 + k(-1)^j \eta\cos\theta\right). \tag{10.6}$$

Indeed after the first reading and rotation the unit vectors are $\chi_0 = (\sqrt{3}\,|0\rangle + |1\rangle)/2$ and $\chi_1 = (-|0\rangle + \sqrt{3}\,|1\rangle)/2$. Applying M_k and normalising gives the complete table:

(j,k)	$p(j,k)$	final unit vector
$(0,+)$	$5/16$	$(3\,\|0\rangle + \|1\rangle)/\sqrt{10}$
$(0,-)$	$3/16$	$(\|0\rangle + \|1\rangle)/\sqrt{2}$
$(1,+)$	$3/16$	$(-\|0\rangle + \|1\rangle)/\sqrt{2}$
$(1,-)$	$5/16$	$(-\|0\rangle + 3\,\|1\rangle)/\sqrt{10}$

Each row's state is the normalised output of the *two* operations, not a state inferred from the second POVM alone. Summing either pair at fixed j gives $1/2$, and all four probabilities sum to one.

Now change only the first instrument to $\mathcal{J}_j(X) = K_j X K_j^\dagger$ with $K_j = |{+x}\rangle\langle j|$. Its effects are still $K_j^\dagger K_j = E_j$, so its first statistics agree for *every* input, not merely our chosen state. But both conditional outputs are $|{+x}\rangle$. After the same rotation their z mean is $-\sin\theta = -\sqrt{3}/2$, giving

$$p_{\mathrm{reset}}(j,k) = \frac{1}{4}\left(1 - k\frac{\sqrt{3}}{4}\right) = \frac{4 - k\sqrt{3}}{16}. \tag{10.7}$$

For either j, the final unit vector is

$$\frac{M_k U\,|{+x}\rangle}{\sqrt{(1 - k\sqrt{3}/4)/2}}, \qquad U\,|{+x}\rangle = \frac{(\sqrt{3}-1)\,|0\rangle + (\sqrt{3}+1)\,|1\rangle}{2\sqrt{2}}.$$

The comparison separates the first outcome's probability from its conditional output. Figure 10.1 keeps the later rotation and second operation fixed. Its two routes represent different apparatus procedures, not two simultaneous branches of one experiment. Each first apparatus records the label j. Forgetting that label would require summing the unnormalised operations over j.

These probabilities and states differ from the Lüders table even though the first POVM is identical. An observation's dynamical specification is its instrument. A second pure-ready meter can realise these operations by the isometry $\psi \mapsto \sum_k M_k\psi \otimes |k_M\rangle$. Retaining the two meter records coherently instead of conditioning them will be the comparison in L14.

A finite indirect measurement also permits a precise comparison between the input and output observables. Let A and B be Hermitian system matrices, let M be a Hermitian meter pointer, and let U_M be a unitary coupling. All expectations in this comparison use the initially prepared product state $\rho \otimes \tau_M$. Define the input operators and the output operators pulled back to that initial state by

$$A_{\mathrm{in}} = A \otimes \mathbb{1}, \qquad B_{\mathrm{in}} = B \otimes \mathbb{1},$$
$$M_{\mathrm{out}} = U_M^\dagger(\mathbb{1} \otimes M)U_M, \qquad B_{\mathrm{out}} = U_M^\dagger(B \otimes \mathbb{1})U_M.$$

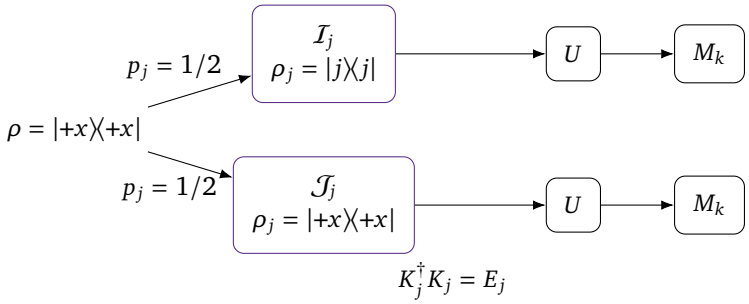


Figure 10.1: Two first instruments with effects E_j and the same first probabilities for the specified input. Their conditional outputs differ, so the common later operations U and M_k give different joint reading statistics.

The noise operator compares the indicated pointer with the input observable. The disturbance operator compares the final and initial versions of the other system observable:

$$N = M_{\text{out}} - A_{\text{in}}, \qquad D = B_{\text{out}} - B_{\text{in}}, \qquad \epsilon_A^2 = \langle N^2 \rangle, \qquad \eta_B^2 = \langle D^2 \rangle.$$

The roots satisfy $\epsilon_A, \eta_B \geq 0$. These root mean square quantities include any nonzero mean differences. They are defined on the joint prepared state, rather than inferred from the separate input and output distributions alone. They need not equal the standard deviations of the error and kick densities used for the Gaussian pointer.

Proposition 10.3 (Finite noise–disturbance bound)**.** *For this finite matrix model, with input system spreads ΔA and ΔB,*

$$\epsilon_A \eta_B + \epsilon_A \Delta B + \Delta A \eta_B \geq \frac{1}{2} |\langle [A, B] \rangle_\rho|.$$

Proof. The final pointer and final system observable commute because the uncoupled operators act on different factors. Unitary conjugation therefore gives $[M_{\text{out}}, B_{\text{out}}] = 0$. Substituting the noise and disturbance definitions yields

$$[N, D] + [N, B_{\text{in}}] + [A_{\text{in}}, D] = -[A_{\text{in}}, B_{\text{in}}].$$

Take expectations, use the triangle inequality, and apply the finite-dimensional Robertson inequality to each commutator. The input product state's system spreads are ΔA and ΔB. Consequently

$$\frac{1}{2} |\langle [A, B] \rangle_\rho| \leq \Delta N \Delta D + \Delta N \Delta B + \Delta A \Delta D \leq \epsilon_A \eta_B + \epsilon_A \Delta B + \Delta A \eta_B.$$

The last inequality uses $\Delta N \leq \epsilon_A$ and $\Delta D \leq \eta_B$, since a variance is the mean square minus the squared mean. ∎

The mixed terms cannot generally be omitted. Sheet 10 computes a CNOT example with zero pointer noise and nonzero disturbance that distinguishes this bound from the naive product bound. This finite proof uses only Hermitian matrices; an unbounded-operator extension needs its own domain and moment hypotheses. The quantities and bound follow M. Ozawa, *Phys. Rev. A* **67**, 042105 (2003), arXiv:quant-ph/0207121.

What is the state after a reading?

An observation is an instrument. Its POVM supplies probabilities, and its normalised positive-probability operations supply conditional states. Lüders' state is uniquely selected by conditioning every sharp sub-event. Every instrument has a pointer model, by the cited representation theorem. Two instruments with one POVM can give different later probabilities and states; resolution and unread disturbance also belong to the apparatus model rather than to a mean operator alone.

Summary. Sub-event conditioning proves Lüders' uniqueness without a dimension exception. Pointer projection sandwiches induce CP additive instruments. The Gaussian position instrument has a continuous Kraus density; ignoring its record adds a reflected momentum-kick convolution. The centred unchirped pointer obeys $\epsilon_Q \eta_P = \hbar/2$. The capstone computes how a changed first instrument alters the rest of an otherwise fixed sequence.

Exercises. Try Sheet 10 in the separate *Exercise sheets* booklet (English).

Sources. Busch–Lahti–Pellonpää–Ylinen, *Quantum Measurement*, §7.7 pp. 159–161, Theorem 7.14 p. 160; conditioning pp. 211–212; position model pp. 241–243, equations (10.32)–(10.33) p. 242; repeatability Theorem 10.4 p. 248. Busch–Lahti–Mittelstaedt, pp. 41–43, 84–86. The direct density-operator conditioning proof and the fully specified qubit sequence are supplied here.

Lecture 11

Decoherence: the unread environment

Objectives. *An environment can record alternatives without anyone reading its record. The resulting channel suppresses interference in a part while the combined state can remain pure. We compute this for a qubit and for position-dependent random kicks, then distinguish the small-separation and resolved-collision regimes before estimating two physical decoherence times.*

- Compute a dephasing factor from environmental record overlap and distinguish a reduced mixture from a selected outcome.
- Prove the Gaussian position-kick channel's kernel factor with the correct infinite-dimensional scope.
- Check the regime and units of a scattering estimate for a dust grain and for a model molecule in thermal radiation.

11.1 A record changes a part's interference

Let the system be a qubit and the environment have a fixed unit ready vector e. For an environment unitary U_E take the joint coupling

$$U = |0\rangle\langle 0| \otimes \mathbb{1} + |1\rangle\langle 1| \otimes U_E.$$

Its two orthogonal system sectors show that U is unitary. A pure input $\psi = a\,|0\rangle + b\,|1\rangle$ gives

$$\Psi = a\,|0\rangle \otimes e + b\,|1\rangle \otimes U_E e.$$

The total remains a unit vector. Put $c = \langle e, U_E e\rangle$. L6's local pairing gives the reduced operator

$$|a|^2\,|0\rangle\langle 0| + a\bar{b}\,\langle U_E e, e\rangle\,|0\rangle\langle 1| + \bar{a}b\,\langle e, U_E e\rangle\,|1\rangle\langle 0| + |b|^2\,|1\rangle\langle 1|\,.$$

In particular the $|0\rangle\langle 1|$ coefficient is $\bar{c}\,a\bar{b}$; the rank-one trace identity is $\mathrm{Tr}(|u\rangle\langle v|) = \langle v, u\rangle$. By linearity the reduced map is

$$\begin{pmatrix} \rho_{00} & \rho_{01} \\ \rho_{10} & \rho_{11} \end{pmatrix} \longmapsto \begin{pmatrix} \rho_{00} & \bar{c}\rho_{01} \\ c\rho_{10} & \rho_{11} \end{pmatrix}. \tag{11.1}$$

The conjugate on the upper entry follows from our antilinear-first inner product. Cauchy–Schwarz gives $|c| \le 1$. Orthogonal environmental records have $c = 0$ and remove the local coherence completely; parallel records have $|c| = 1$ and change only its phase. We call c the environmental overlap, or *decoherence factor*; the upper coherence uses $\bar{c}$ and its magnitude is multiplied by $|c|$. It measures how much interference survives when the record is ignored. The map is a channel by L7's initially separate environment construction. The unchanged diagonal entries give the same probabilities for reading the system in its recording basis.

The off-diagonal entries determine what happens when its two alternatives are recombined. Suppressing those entries changes interference without changing either recording-basis probability.

For a qubit environment initially $|0\rangle$, choose $U_E = e^{-i\theta\sigma_y/2}$. Then $c = \cos(\theta/2)$. At $\theta = \pi$ the records are orthogonal. For $a = b = 1/\sqrt{2}$ the joint output is a Bell vector, while the system becomes $\mathbb{1}_2/2$. Its purity is $\mathrm{Tr}\,\rho_1^2 = 1/2$ although the joint purity is one. No stochastic choice of 0 or 1 occurred in this unitary calculation. At $c = 0$ the map is $\rho \mapsto \sum_{j=0}^{1} |j\rangle\langle j|\,\rho\,|j\rangle\langle j|$, the unread Lüders channel of L10. Reading an environmental record and selecting its outcome would be the extra operation described by an instrument in L10.

The real environmental rotation in this example illustrates the overlap in Figure 11.1. These are vectors in the environment's real coordinate plane, not points on a Bloch sphere.

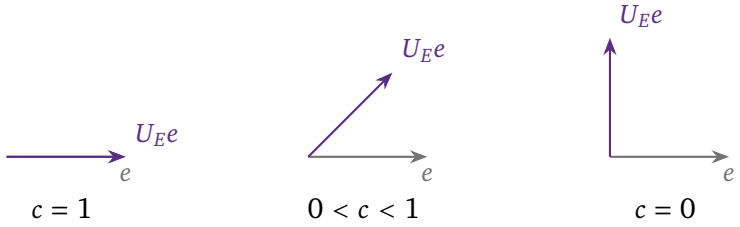


Figure 11.1: Environmental record vectors for the rotation example with $0 \le \theta \le \pi$. Coincident records preserve system coherence; orthogonal records remove it from the reduced state.

A mixture prepared by an ignored classical random label is often called *proper*. The same density operator obtained by restricting an entangled pure state is called *improper*, in d'Espagnat's terminology promised in QM-I, Lecture 11, after the half of a singlet as a mixture. These words concern the preparation and the larger state, not two kinds of density operator. Local bounded observations cannot distinguish them. Nor does a diagonal reduced operator identify one privileged ensemble: L2 already exhibited many preparations of one state. A global test can distinguish the entangled state from a global incoherent mixture; L12 computes it.

11.2 Position dephasing as a channel

On $L^2(\mathbb{R})$ take $\Lambda_{\mathrm{dec}} > 0$ with units length^{-2} time^{-1}. For $t > 0$ define the probability density

$$\gamma_t(k) = \frac{e^{-k^2/(4\Lambda_{\mathrm{dec}}t)}}{\sqrt{4\pi\Lambda_{\mathrm{dec}}t}}, \qquad V(\hbar k) = e^{ikQ}.$$

Here k has inverse-length units and $\hbar k$ is the momentum kick. The variance of k is $2\Lambda_{\mathrm{dec}}t$. Define the output by its bounded expectations,

$$\mathrm{Tr}(\Phi_t(\rho)A) = \int \gamma_t(k)\,\mathrm{Tr}(\rho e^{-ikQ}Ae^{ikQ})\,dk. \tag{11.2}$$

L7 proved that this scalar rule constructs a CP, trace-preserving map: this is (7.4) with its scalar density equal to γ_t and unitary family $\mathcal{W}(k) = V(\hbar k)$. At $t = 0$ define $\Phi_0 = \mathrm{id}$ separately; there is no ordinary density γ_0. Thus a symbolic integral of operators is not the existence argument for this channel.

Proposition 11.1 (Gaussian suppression of position coherence). *For $t \geq 0$, the kernel ρ_t of $\Phi_t(\rho_0)$ satisfies*

$$\rho_t(x, x') = e^{-\Lambda_{\mathrm{dec}} t (x-x')^2} \rho_0(x, x'). \tag{11.3}$$

For general trace-class inputs kernels are interpreted in $L^2(\mathbb{R}^2)$, hence almost everywhere, rather than as prescribed pointwise values. Here almost everywhere allows changes on sets of zero planar area. Such changes do not alter the kernel as an element of $L^2(\mathbb{R}^2)$ or the operator it defines.

Proof. We first compute the multiplier for one wave packet. We then identify it with the channel through bounded matrix elements. Finally we pass from finite spectral sums to all trace-class inputs by positive-tail bounds.

For a rank-one wave packet, conjugation by e^{ikQ} multiplies its kernel by $e^{ik(x-x')}$. The Gaussian Fourier integral is

$$\int \gamma_t(k) e^{ikd}\, dk = e^{-\Lambda_{\mathrm{dec}} t d^2}.$$

For completeness put $\zeta = \Lambda_{\mathrm{dec}} t > 0$ and call the integral $J(d)$. Since $k\gamma = -2\zeta\gamma'$, differentiation and integration by parts give $J'(d) = -2\zeta d J(d)$, with $J(0) = 1$; Gaussian tails justify the operations by finite cutoffs. Solving gives the displayed factor. The Fourier integral supplies the expected multiplier. It remains to show that this multiplier represents the channel already defined by its bounded expectations. To connect that integral with the scalar definition of the channel, first take input φ and testing vectors f, g in C_c^∞. Use $A = |g\rangle\langle f|$ in (11.2), cut k off at $\pm K$, and exchange the continuous iterated Riemann integrals on the resulting compact box. The omitted tail is at most $\|f\|\,\|\varphi\|^2\,\|g\| \int_{|k|>K} \gamma_t(k) dk$, by Cauchy–Schwarz. Taking $K \to \infty$ gives

$$\langle f, \Phi_t(|\varphi\rangle\langle\varphi|) g\rangle = \iint \bar{f}(x) e^{-\Lambda_{\mathrm{dec}} t (x-x')^2} \varphi(x) \overline{\varphi(x')} g(x')\, dx\, dx'.$$

S4 extends the tests and input to L^2: the left side is continuous in the input by the uniform bound $|\,\langle V\varphi, AV\varphi\rangle - \langle V\xi, AV\xi\rangle\,| \leq \|A\|\,(\|\varphi\| + \|\xi\|)\,\|\varphi - \xi\|$ for unitary V. The right side has the same continuity by the product-kernel L^2 bound. Matrix elements separate operators, proving the rank-one identification. Linearity now gives the kernel for finite spectral sums.

The finite-sum computation alone does not cover an infinite-rank state. To pass to every positive trace-class operator, its spectral kernels $\sum_j p_j \varphi_j(x)\overline{\varphi_j(x')}$ converge in $L^2(\mathbb{R}^2)$, since those product vectors are orthonormal and $\sum_j p_j^2 < \infty$. A square-integrable kernel defines a bounded operator of norm at most its L^2 norm, by Cauchy–Schwarz followed by integration. The operator defined by this kernel limit agrees with ρ by the operator-norm convergence of positive spectral tails. Multiplication by the Gaussian factor has norm at most one on kernel space, so the transformed kernels also converge there. The channel of a positive omitted input tail is positive with the same small trace, hence has operator norm at most that trace. Its operator limit is consequently $\Phi_t(\rho)$, identifying the kernel limit. Complex linearity extends the conclusion to L1's trace-class span. At $t = 0$ it is the identity. ∎

The kernel factors multiply, so $\Phi_s \Phi_t = \Phi_{s+t}$. This explicit infinite-dimensional channel semigroup did not use L8's finite GKSL theorem. Position probabilities stay unchanged: each e^{ikQ} commutes with $E_Q(I)$, so (11.2) gives $\mathrm{Tr}(\Phi_t(\rho) E_Q(I)) = \mathrm{Tr}(\rho E_Q(I))$. A diagonal of a general L^2 kernel is not determined by its equivalence class; this bounded-event identity is the general population statement. For smooth kernels one can also read it from $x = x'$ in (11.3).

For finite-rank operators with Schwartz eigenvectors, differentiating the explicit factor produces the formal expression

$$\dot{\rho} = -\Lambda_{\mathrm{dec}}[Q,[Q,\rho]], \qquad L_1 = \sqrt{2\Lambda_{\mathrm{dec}}}\,Q,$$

with free motion neglected. The double commutator's kernel is $(x-x')^2\rho(x,x')$. This identifies the differential rule on these test kernels; it is not a claim that every trace-class input is differentiable, or that this test class has been proved a generator core. The channel is defined on all trace class by (11.2). The spectral shift rule gives $e^{-ikQ}E_P(I)e^{ikQ} = E_P(I-\hbar k)$, so bounded momentum-event probabilities obey the convolution rule of (10.4). For finite input momentum variance its ordinary variance addition gives $2\hbar^2\Lambda_{\mathrm{dec}}t$, by the physical kick variance.

11.3 Which scattering regime does the Gaussian model describe?

The Gaussian channel is mathematically valid for every separation. Its identification with a scattering experiment has a restricted regime. The following kinetic derivation is *sketched*, following Schlosshauer, §3.1–3.4 pp. 119–138. It assumes a massive object with negligible recoil, independent stationary scattering events, and negligible free motion during the coherence comparison. A symmetric incident bath removes directed phase drift. These are physical assumptions, not consequences of CP.

Let Γ be the total collision rate and q the momentum-transfer component along the separation. In one event the coherence gains a factor $\chi(d) = \langle e^{iqd/\hbar}\rangle$ from the transfer distribution. During a short interval, the probabilities of no collision and one collision are $1-\Gamma\,dt$ and $\Gamma\,dt$ to first order. Thus

$$\dot{\rho}(x,x') = \Gamma(\chi(x-x')-1)\rho(x,x').$$

For a symmetric distribution with finite second moment and $|qd/\hbar| \ll 1$ at typical transfers,

$$\chi(d) \simeq 1 - \frac{\langle q^2\rangle d^2}{2\hbar^2}, \qquad \Lambda_{\mathrm{dec}} = \frac{\Gamma\langle q^2\rangle}{2\hbar^2}.$$

This is the quadratic small-separation law. If one uses the total three-dimensional transfer magnitude in an isotropic bath, its component second moment is one third of that total; the factor must be converted.

At separations resolved by the scattered particles, the overlap factor in the cited isotropic model averages close to zero. The decay rate then approaches Γ, rather than increasing without limit as d^2. The wavelength language is an order-of-magnitude criterion; the precise small parameter is transfer times separation divided by $\hbar$. Saturation is an asymptotic property of the specified scattering model, not a universal monotonicity theorem for every conceivable kick distribution. One must not extrapolate the quadratic coefficient to a macroscopically resolved separation just because its Gaussian channel remains positive. In applying a rate estimate, first compare separation with the bath resolution scale. Use the quadratic coefficient only for unresolved alternatives. Once collisions resolve the alternatives, estimate the collision rate itself. The two examples below use these different choices.

In the controlled-qubit model, the interaction preserves the system basis 0, 1 and makes its superpositions entangle with distinct records. A *pointer basis* here consists of vectors $|j\rangle$ for which $U(|j\rangle\otimes e) = |j\rangle\otimes e_j$: they remain products while their superpositions can entangle with records. Position-dependent scattering similarly favours localised packets or

coarse position sectors. Exact delta functions are not pointer states in this Hilbert space. The widths, stability and competition with free spreading depend on the interaction and timescale; decoherence does not select one universal basis for every Hamiltonian.

11.4 A dust grain in air, and a model molecule in light

Take air at 300 K, pressure $p \simeq 1.01 \cdot 10^5$ Pa, and mean molecular mass $m \simeq 4.8 \cdot 10^{-26}$ kg. The ideal-gas and Maxwell estimates give

$$n = \frac{p}{k_B T} \simeq 2.45 \cdot 10^{25}\,\mathrm{m}^{-3}, \qquad \bar{v} = \sqrt{\frac{8k_B T}{\pi m}} \simeq 470\,\mathrm{m\,s}^{-1}.$$

Specify the grain's *diameter* as 10 μm, so its radius is $a = 5\,\mu$m. With geometric effective cross section πa^2,

$$\Gamma \simeq n\bar{v}\pi a^2 \simeq 9 \cdot 10^{17}\,\mathrm{s}^{-1}, \qquad \tau_{\mathrm{dec}} \simeq \Gamma^{-1} \simeq 10^{-18}\,\mathrm{s}.$$

At a separation $d = 1\,\mu$m, the thermal wavelength $h/\sqrt{2\pi m k_B T} \simeq 1.9 \cdot 10^{-11}$ m is much smaller than d. This is the resolved, approximately saturated regime. The quadratic formula would give an unjustified much shorter time. The stated time is a characteristic stationary scattering-model estimate, not a measurement of its initial microscopic transient. It depends on ideal-gas flux, a heavy stationary grain and the chosen effective cross section; changing radius to 10 μm would multiply the rate by four. Units check directly: $n\bar{v}\pi a^2$ is inverse time.

For a contrasting quadratic-regime example, take Schlosshauer's model large molecule of radius 10 nm in room-temperature photon radiation. Table 3.1, p. 135, estimates

$$\Lambda_{\mathrm{dec}} \simeq 10^6\,\mathrm{cm}^{-2}\mathrm{s}^{-1} = 10^{10}\,\mathrm{m}^{-2}\mathrm{s}^{-1}.$$

Choose $d = 10$ nm. Then $\Lambda_{\mathrm{dec}} d^2 \simeq 10^{-6}\,\mathrm{s}^{-1}$ and $\tau_{\mathrm{dec}} \simeq 10^6$ s, about twelve days. Both the object and the separation are small compared with the roughly 10 μm thermal-photon wavelength. The *small separation* justifies the quadratic rate; the *small object* justifies the Rayleigh cross section used to estimate its coefficient. They are distinct conditions. The table uses a dielectric-sphere approximation with contrast of order one and a strong radius/temperature dependence $\Lambda_{\mathrm{dec}} \propto a^6 T^9$. It is not a measured universal coefficient for every molecule, nor the radius of a fullerene. These different reservoirs and objects need not have comparable times.

Why are position superpositions difficult to observe?

A part coupled to an initially separate, unread environment evolves by a channel. Records suppress its interference while the joint state can remain pure. In the stated scattering regime the small-separation decay rate is $\Lambda_{\mathrm{dec}} d^2$; resolved collisions instead give a rate near the total collision rate. Position probabilities stay fixed in this pure-dephasing approximation. The reduced mixture supplies no unique selected outcome; that remaining question is the measurement problem.

Summary. Environmental overlap determines the qubit's coherence. L7's scalar construction gives a genuine Gaussian kick channel, whose kernel factor is proved in the appropriate L^2 sense. Quadratic diffusion and resolved-collision saturation describe different physical regimes. The two estimates illustrate why assumptions and scales precede the numerical

rate. Proper and improper refer to preparation context; neither local density statistics nor diagonalisation by itself chooses an outcome.

Exercises. Try Sheet 11 in the separate *Exercise sheets* booklet (English).

Sources. Schlosshauer, *Decoherence and the Quantum-to-Classical Transition*, pp. 88–93, 119–138; saturation equations (3.45)–(3.48) p. 129, Rayleigh coefficient (3.66)–(3.67) p. 134 and Table 3.1 p. 135. The air calculation uses the saturated collision rate, not an extrapolation of Table 3.2's quadratic values. The explicit channel proof uses L1, L6 and L7.

Lecture 12

The measurement problem

Objectives. *An instrument gives the probabilities and conditional states of a reading. A unitary apparatus model explains those statistics, but does its global output describe one definite result? We prove a calibrated pointer cannot turn a superposition into one pointer sector or a global pointer mixture. A three-qubit test identifies what decoherence leaves open, and the premises of state completeness (C), universal unitary evolution (U) and one observer-independent global outcome (D) organise the possible responses.*

- Prove the pure-ready no-objectification theorem with complete calibration and orthogonal pointer sectors.
- Compute a global coherence test and explain the limits of moving the cut in a reading chain.
- State C/U/D and distinguish their roles from the separate assumptions of extended Wigner-friend arguments.

12.1 Calibration does not produce one global outcome

Work within one coherent sector with the full bounded-observable algebra, so that the global interference test below is admitted, rather than excluded by an additional restriction of the kind specified in L5. On $\mathcal{H}_S$ let (E_j) be a complete countable system PVM, with strong sum $\mathbb{1}$. Let (Π_j) be orthogonal pointer projections on a separable meter space $\mathcal{H}_M$, and let φ_0 be its fixed unit ready vector. A unitary U is *calibrated* if, for every unit $\psi_j \in \operatorname{ran} E_j$,

$$\chi_j = U(\psi_j \otimes \varphi_0) \in \operatorname{ran}(\mathbb{1} \otimes \Pi_j). \tag{12.1}$$

Thus every input already belonging to sector j gives the corresponding pointer reading with certainty. The output system need not remain ψ_j, and the measured sector may be degenerate. Calibration concerns all its vectors. Here calibration on an ONB already implies that by bounded linearity and the closed pointer range; calibrating only selected vectors without a spanning argument is insufficient.

Theorem 12.1 (No unitary objectification, pure ready state)**.** *Under this calibration, let the unit input ψ have at least two nonzero components $E_j\psi$. Then the global output $\chi = U(\psi \otimes \varphi_0)$ lies in no individual pointer sector. Its density operator is not any finite or countable convex mixture of density operators individually supported in pointer sectors.*

Proof. We follow each input component into its calibrated pointer range. Their superposition retains cross blocks between different ranges, whereas every mixture of definite pointer states has zero cross blocks.

Put $p_j = \left\|E_j\psi\right\|^2$, with $\sum_j p_j = 1$, and, when $p_j > 0$, $\psi_j = E_j\psi/\sqrt{p_j}$. Orthogonal strong completeness and boundedness of U give the vector-norm expansion

$$\chi = \sum_{j:p_j>0} \sqrt{p_j}\,\chi_j.$$

Calibration puts these unit vectors in orthogonal pointer ranges. Hence $\left\|(\mathbb{1}\otimes\Pi_j)\chi\right\|^2 = p_j$. At least two weights are positive, so none is one; no pointer sector contains χ. For $j \neq k$ with $p_j, p_k > 0$, the cross block is

$$(\mathbb{1}\otimes\Pi_j)\,|\chi\rangle\langle\chi|\,(\mathbb{1}\otimes\Pi_k) = \sqrt{p_j p_k}\,\left|\chi_j\right\rangle\left\langle\chi_k\right| \neq 0.$$

Every density operator supported in a single pointer sector has zero cross blocks between different sectors. Their convex mixtures retain that property, including countable mixtures: their positive omitted tail T_N obeys $\left\|(\mathbb{1}\otimes\Pi_j)T_N(\mathbb{1}\otimes\Pi_k)\right\| \leq \|T_N\| \leq \operatorname{Tr} T_N \to 0$. The displayed nonzero block therefore excludes all such mixtures. ∎

The proof uses only linearity, unitarity, a fixed product ready state and orthogonal calibration. It does not use a model of consciousness or a particular microscopic interaction. It also does not say that an input already in one sector fails to give a definite calibrated output. The obstruction is the coherent superposition of distinct sectors.

The proved statement uses a pure ready vector. Including an environment and using a mixed ready state require care about which total system the conclusion concerns.

An initially pure, separate environment can be included in the meter: replace $\mathcal{H}_M$ by $\mathcal{H}_M \otimes \mathcal{H}_E$ and include it in φ_0. Its unread records may hide all local interference, but the same global proof applies. For a mixed ready density $\tau_M = \sum_\ell s_\ell\,|m_\ell\rangle\langle m_\ell|$, suppose the mixed calibrated output for every sector input is supported in its pointer sector. Positivity then puts every $U(\psi_j \otimes m_\ell)$ with $s_\ell > 0$ in that sector. Purify the ready density by $\Phi_0 = \sum_\ell \sqrt{s_\ell}\,m_\ell \otimes f_\ell$. The extended unitary $U\otimes\mathbb{1}$ is calibrated with this pure ready vector and the enlarged pointer projections $\Pi_j \otimes \mathbb{1}$. The proved theorem therefore applies to the description including that purifying partner. Busch, Lahti, Pellonpää and Ylinen (BLPY), Theorem 22.2 p. 494, supply a different, *cited* statement about system plus probe without that partner. For a nontrivial measured observable and a nontrivial reading scale, their theorem rules out simultaneously satisfying their measurement assumptions, pointer value-definiteness, and the requirement that every initial system state yield a global system-plus-probe pointer-sector mixture. This does not say every individual output can never be pointer diagonal.

The robust version is also *cited*, Landsman, Definition 11.3 and Theorem 11.4 pp. 448–449. Let S_i be labelled sets of global density operators, with every pair from distinct sets at trace-norm distance at least $2\sqrt{1-\eta}$, $0 \leq \eta < 1/2$. For one fixed ready density τ_M and one unitary, its soundness hypothesis is explicitly

$$U(|\psi_i\rangle\langle\psi_i| \otimes \tau_M)U^\dagger \in S_i \quad \text{for every unit } \psi_i \in \operatorname{ran} E_i.$$

With this calibration into the corresponding sets and at least two nonzero input sectors, the output of some input belongs to none of the S_i. Here trace norm is the standard operator distance, normalised so density operators have distance at most two; its general construction and the robust proof are outside our standing facts. In finite dimension the trace norm of a Hermitian matrix T, with eigenvalues τ_r counted with multiplicity, is

$$\|T\|_1 = \sum_r |\tau_r|.$$

Thus computing this distance for two finite density matrices means subtracting them, diagonalising the Hermitian difference, and summing the absolute eigenvalues. This finite rule suffices for the two-state calculation on Sheet 12. The strict separation hypothesis, not merely the word “macroscopic”, is part of the result.

12.2 Three premises that do not fit together

For the ideal pointer in this argument, a definite outcome j means a state supported in its pointer sector; ignorance of an already definite outcome permits a mixture of such states. That is the state–outcome identification being used. A diagonal marginal alone is not the global mixture required by it.

Definition 12.2 (The measurement trilemma)**.** The three premises in this calibrated state-and-pointer description are:

- **C**: the global quantum state is complete; there are no further actual physical facts selecting a result beyond that state.
- **U**: the same universal unitary dynamics governs the system, apparatus and any included environment.
- **D**: each run ends with one unique, observer-independent *global* outcome, represented by the specified definite pointer alternatives.

Corollary 12.3. *For the calibrated pure-ready model, an input with two nonzero spectral components, and the stated pointer-sector identification of definite outcomes, C, U and D cannot all hold.*

With C and U, Theorem 12.1 leaves the global superposed state, not a state of one of those alternatives or its ignorance mixture. D demands an additional account. The definition is a conditional organisation of these premises, not a theorem refuting every interpretation of the word “outcome”. In particular, a definite outcome *relative to a branch* is not D’s unique global outcome.

The operational calculus remains usable. Given a recorded result, an instrument supplies its conditional state and later probabilities. The theorem asks how a description with universal U and complete C accounts for that unique record in the first place. Renormalising a projected branch is a probability-conditioning rule; it is not by itself a unitary derivation of a stochastic physical selection.

Part V follows the three responses discussed by Maudlin (1995) and Tumulka (pp. 93–94). Bohmian mechanics adds configuration facts beyond the wavefunction, giving up C in this sense. GRW changes the unitary dynamics, giving up U. Everett retains the unitary state and takes outcomes relative to branches, giving up D’s unique global outcome. These are the premises changed; their separate locality and empirical commitments must also be stated, not decided by this taxonomy. Other interpretive proposals make other specifications. We will compare the three on the same questions in L14.

A fundamental restriction of the observable algebra is another substantive specification, as L5 showed. Simply declaring pointer cross blocks unobservable would change the algebra under discussion; it is not a consequence of a small environmental overlap within the same full algebra. Practical inability to perform a global test and a fundamental prohibition of that test are different physical premises.

12.3 A diagonal marginal and a global test

Take system, meter and environment qubits, in that order. Begin with $|+x\rangle \otimes |0\rangle \otimes |0\rangle$. A CNOT with system controlling meter, then a CNOT with meter controlling environment, give

$$\Psi = \frac{|000\rangle + |111\rangle}{\sqrt{2}}. \tag{12.2}$$

The pointer and system each have marginal $\mathbb{1}_2/2$. The system–meter marginal is $(|00\rangle\langle 00| + |11\rangle\langle 11|)/2$. All of them agree with restrictions of the global incoherent mixture

$$\rho_{\text{mix}} = \tfrac{1}{2}\,|000\rangle\langle 000| + \tfrac{1}{2}\,|111\rangle\langle 111|\,.$$

But the bounded self-adjoint global operator

$$B = |000\rangle\,\langle 111| + |111\rangle\,\langle 000| \tag{12.3}$$

sends Ψ to itself, so $\langle \Psi, B\Psi\rangle = 1$. Each diagonal component in ρ_{mix} has zero expectation, so $\text{Tr}(\rho_{\text{mix}}B) = 0$. The coherence has not vanished from the total state; it moved beyond the observables of the retained part. The product $\sigma_x \otimes \sigma_x \otimes \sigma_x$ gives the same two means in this example and is another possible global correlation test. The distinction between global and restricted states is shown in Figure 12.1.

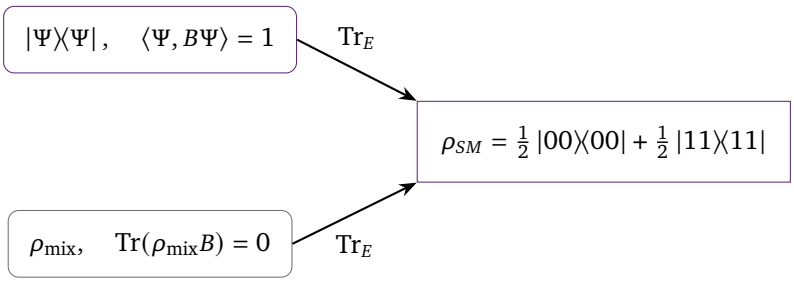


Figure 12.1: Two different global states have the same system–meter restriction. The global observable B distinguishes them, while any observable acting only on system plus meter has the same expectation.

Coherent access to every record is essential. If the environment record stays untouched and is traced out, no observable on system plus meter can recover the lost cross block. For a macroscopic environment that access may be extraordinarily difficult. Its difficulty explains the success of effective classical records, while leaving the theorem's logical distinction intact.

One can express the test as an erasure operation. Undo the two CNOTs in reverse order. The coherent state becomes $|+x, 0, 0\rangle$, whose system $+x$ probability is one. The mixture becomes $\mathbb{1}_2/2 \otimes |00\rangle\langle 00|$, whose system $+x$ probability is $1/2$. Erasing and recombining all records is a different subsequent experiment from reading and preserving their alternatives.

12.4 What moving the cut preserves

A von Neumann chain includes more parts in the unitary description: system, pointer, environment, further records. In the three-qubit example, reading the meter after the first CNOT gives j with probability $1/2$ and conditional system state $|j\rangle\langle j|$. Copying that retained record to the environment gives $|jjj\rangle$. Keeping both interactions coherent and conditioning

the final meter or environment record gives the same probabilities and conditional state. This is the restricted sense in which the cut can move without altering predictions.

More generally, write an instrument's record isometry (L10) as $J\psi = \sum_j K_j\psi \otimes |j\rangle$. Later controlled copying that leaves $|j\rangle$ unchanged preserves its sector weight $\left\|K_j\psi\right\|^2$ and its conditional system output. Orthogonality removes cross terms from every reading that ignores or preserves those records. Positive spectral sums give the same statement for density inputs. L14 proves this identity for the fully specified two-reading capstone.

The qualification about retained records matters. A projection update used to represent actual branch selection destroys global coherence; a coherent record-erasing unitary can detect its absence. Cut independence holds for the specified record-preserving operations. An early projection-and-mixture description and a coherent late description predict different results for a record-erasing test. Thus the claim that the cut can move always includes a restriction on what is done afterwards. The inverse-CNOT calculation explicitly distinguishes those operations. In ordinary reading chains the retained-record statistics agree; that agreement does not establish a unique global outcome.

12.5 Wigner's friend and the scope of further no-go claims

Take the meter in (12.2) as the friend's memory; Wigner's interference test is (12.3). Extend the chain to that memory. A unitary calibration gives a superposition of "system 0, record 0" and "system 1, record 1". Conditional on either record, the friend assigns a definite relative result. An external observer retaining coherent control of the entire isolated laboratory instead describes its global superposition and could in principle test its interference. The choice of a record reading versus a coherent global test must be included in the experimental description; they are not one unchanged experiment.

For example, conditional on memory record 0, the friend is certain that a retained system reading in the recording basis agrees with that record. Wigner is instead certain that a coherent test of B on Ψ returns +1. These certainties concern different continuations of the experiment. Transferring a prediction between agents requires specifying its observable, its conditioning record and whether that record is preserved or erased. This elementary distinction does not by itself prove any extended-friend no-go theorem.

The following named results are an optional outlook. They are not inputs to the theorem proved in this lecture. The model-relative independence conditions needed for a detailed locality comparison are defined in § 13.3.

Frauchiger–Renner (2018, arXiv:1604.07422) combine (Q) quantum probability-one inferences, (C) transfer of another agent's certainty, and (S) exclusion of contradictory single outcomes for an agent, in a scenario with coherent external control of entire labs. They argue that these premises are incompatible. Their inference-transfer condition labelled C is *not* our completeness C. This is a conditional argument about the stated inference rules and quantum-control scenario; no additional theorem is proved here and no unconditional inconsistency of quantum probability is inferred.

Bong et al. (2020, arXiv:1907.05607) formulate a different bundle called local friendliness: absoluteness of observed events, no superdeterminism of the later choices, and locality of the specified response probabilities. With suitable quantum control on the friend's scale, quantum predictions violate inequalities following from those premises. Observed-event absoluteness does not preassign every unperformed experiment; their locality premise is not Bell factorisation with outcome independence silently added. The two named arguments

and this lecture's C/U/D theorem have distinct hypotheses. Listing those hypotheses is more informative than calling each one "a proof that observers disagree".

Does a unitary pointer end in one reading?

An instrument predicts probabilities and conditional states along a specified calibrated chain, with retained-record statistics independent of the cut. A calibrated unitary with a pure ready state does not turn a superposition into one definite pointer sector or a global sector mixture. C, U and a unique observer-independent global outcome D cannot all hold under that definite-pointer description. Decoherence explains local loss of interference but does not supply the missing selection.

Summary. The superposition proof and its nonzero global cross block establish the pure-ready obstruction. Mixed-probe and robust extensions have additional cited hypotheses. A three-qubit global test distinguishes coherence from an incoherent mixture even with identical local marginals. Moving the cut preserves the stated retained-record statistics, not arbitrary record-erasing experiments. Part V compares specified changes to C, U and D.

Exercises. Try Sheet 12 in the separate *Exercise sheets* booklet (English).

Sources. BLPY, *Quantum Measurement*, Theorems 22.1–22.2 pp. 491–494; Landsman, *Foundations of Quantum Theory*, Theorems 11.2, 11.4 pp. 447–449; Tumulka, *Foundations of Quantum Mechanics*, pp. 93–94, 172–176 (the latter critically analyses the FR argument); Maudlin, *Topoi* **14**, 7–15 (1995). Von Neumann, *Mathematical Foundations of Quantum Mechanics*, pp. 272–273, 285, for the chain/cut discussion. Primary extended-friend papers are named in the text with their premises.

Part V

The three exits

The measurement problem brings three premises into conflict: completeness of the quantum state, universal unitary evolution and one definite outcome. Added variables, stochastic collapse and branching revise different premises. We compare what each proposal specifies and which questions it leaves open.

Lecture 13

Values added: contextuality, locality, Bohm

Objectives. *Giving up completeness C means adding physical facts to the state. Those facts cannot simply be noncontextual values of every observable. The Peres–Mermin square proves the obstruction. Bell locality separates into two model-relative conditions. Bohmian mechanics illustrates the cost: additional positions with contextual readings and, in equilibrium, the quantum position distribution preserved by the guiding flow.*

- Prove the square's value contradiction and the noncontextual bound four against the quantum statistic six.
- Prove Bell locality is parameter independence plus outcome independence, keeping setting independence separate.
- Derive the probability current and prove one-dimensional equivariance under the stated smooth-flow hypotheses.

13.1 Which rules should individual values obey?

Expectation values add: $\omega(A+B) = \omega(A) + \omega(B)$, even when A, B do not commute. That does not mean the outcome of reading $A+B$ is a sum of outcomes in one joint reading of A and B. For Pauli matrices σ_x, σ_z, each spectrum is $\{+1, -1\}$, but $(\sigma_x + \sigma_z)^2 = 2\mathbb{1}$, so their sum's spectrum is $\{+\sqrt{2}, -\sqrt{2}\}$. Individual spectral values of the two terms add to $-2, 0, 2$, none of these. A value rule additive on this noncommuting pair is therefore impossible.

Von Neumann's no-hidden-values argument assumes an additivity requirement beyond commuting families; Bell's critique targets that physical assumption (von Neumann pp. 201–202; Bell, ch. 1 p. 4). It does not invalidate the mathematics under its stated assumptions. L3's effect additivity likewise concerns probabilities, not a license to add deterministic eigenvalues of incompatible observables.

A *noncontextual value assignment* here chooses a spectral value $v(A)$ for the listed observables, independent of the commuting family in which it is read, and respects functional relations within each such family. In particular it respects its products. The same observable can belong to different commuting families. Noncontextuality requires its assigned value to remain the same when that family changes; it does not require all the observables to be read together. The usual condition $v(f(A)) = f(v(A))$ implies those rules for finite commuting matrices: assume the value rule also applies to their self-adjoint functions, including the auxiliary matrix below. Simultaneous diagonalisation gives common eigenspace projections (E_r) and a matrix $D = \sum_r d_r E_r$ with distinct d_r. Every member is a function of D, and $v(AB) = (fg)(v(D)) = f(v(D))g(v(D)) = v(A)v(B)$. No rule for a product of noncommuting operators has been introduced.

13.2 The Peres–Mermin square

In this section put $X = \sigma_x$, $Y = \sigma_y$, $Z = \sigma_z$. On two qubits use the nine observables

$$(A_{jk}) = \begin{pmatrix} \mathbb{1}\otimes Z & Z\otimes\mathbb{1} & Z\otimes Z \\ X\otimes\mathbb{1} & \mathbb{1}\otimes X & X\otimes X \\ X\otimes Z & Z\otimes X & Y\otimes Y \end{pmatrix}. \tag{13.1}$$

All are self-adjoint and square to $\mathbb{1}$, with eigenvalues ± 1. Each row or column is one commuting *context*. To check every product,

$$A_{11}A_{12} = A_{13}, \quad A_{21}A_{22} = A_{23}, \quad A_{11}A_{21} = A_{31}, \quad A_{12}A_{22} = A_{32}.$$

The first pairs act on different factors and commute. For the remaining row and column, $XZ = -iY$, $ZX = iY$ give

$$A_{31}A_{32} = (XZ)\otimes(ZX) = Y\otimes Y = A_{33}, \qquad A_{13}A_{23} = (ZX)\otimes(ZX) = -Y\otimes Y = -A_{33}.$$

Reversing either pair gives the same product, so these contexts commute as well; their third member is plus or minus that product. All three row products are $+\mathbb{1}$, the first two column products are $+\mathbb{1}$, and the third column product is $-\mathbb{1}$. The six compatible choices and their product signs appear in Figure 13.1.

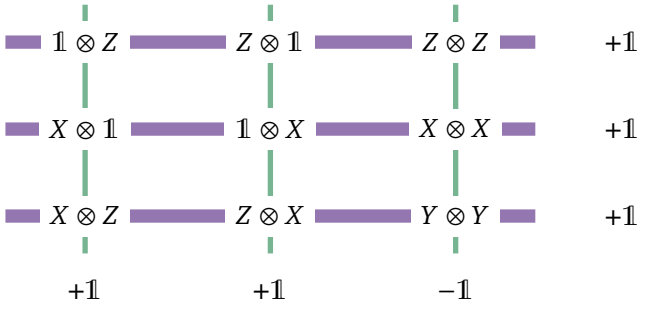


Figure 13.1: Each horizontal row and each vertical column is one compatible context. Product operators appear at the row ends and below the columns; the negative last column prevents one common assignment of nine values.

Theorem 13.1 (Kochen–Specker contradiction on two qubits)**.** *There is no assignment of context-independent values $v(A_{jk}) \in \{\pm 1\}$ respecting the six commuting-triple products of* (13.1).

Proof. Multiplying the three required row-value products gives $+1$. Multiplying the three required column-value products gives -1. Each expression is the product of the same nine scalar values, once each. They cannot differ. No noncommuting operator product was used in this scalar contradiction. ∎

This proof is in dimension four. Dimension three also has Kochen–Specker constructions, including Peres's 33 real rays; Cabello's 18 rays give another dimension-four construction (named only). In finite dimension at least three, the impossibility of assigning zero or one to every projection, adding to one on each orthogonal resolution, also follows from the cited Gleason theorem of L3 and L2's absence of a dispersion-free quantum state; that route depends on the cited theorem. The square is the independent finite proof supplied here.

A single qubit escapes this theorem: L3 constructed a two-valued projection assignment, and Bell gives a qubit hidden-value model (ch. 1 pp. 2–3). A statement about all quantum dimensions must retain this distinction.

The same square supplies a statistic. On different runs read each of the six contexts. Let R_j be its row outcome product and C_k its column outcome product, and define the sum of their separately sampled context means

$$\langle\chi\rangle = \langle R_1\rangle + \langle R_2\rangle + \langle R_3\rangle + \langle C_1\rangle + \langle C_2\rangle - \langle C_3\rangle. \tag{13.2}$$

For one noncontextual nine-value assignment write χ for the corresponding scalar sum. Then the product of these six *signed* terms is -1: each value occurs twice and the last term adds one minus sign. At least one term is -1, so $\chi \leq 4$. A probability mixture, with its assignment distribution independent of the context chosen, retains $\langle\chi\rangle \leq 4$. Quantum theory instead makes each signed context-product operator $\mathbb{1}$, giving $\langle\chi\rangle = 6$ for every density operator. The six contexts are separate experimental choices, not a single joint reading of nine noncommuting observables.

13.3 Locality is a property of a model

Take settings x, y, outcomes $\alpha, \beta = \pm 1$ and hidden state λ. Allow either countable Λ with preparation weights, or a subset of $\mathbb{R}^d$ with an ordinary probability density, extending the finite case and density remark in QM-I, Lecture 12. The distribution of λ is assumed independent of the later settings. This *setting independence* is a separate premise. A model specifies normalised nonnegative responses $p_\lambda(\alpha, \beta|x, y)$, averaged over that fixed distribution to give the observed probabilities. Conditions may fail on a set of hidden states of probability zero, which changes no averaged prediction; use one common such exception set for the finite CHSH choices. Define marginals by summing the other outcome.

Definition 13.2 (Parameter and outcome independence)**.** *Parameter independence* (PI) says the marginals are independent of the remote setting; denote their common values by $p_\lambda(\alpha|x, y) = p_\lambda(\alpha|x)$ and $p_\lambda(\beta|x, y) = p_\lambda(\beta|y)$. *Outcome independence* (OI) means

$$p_\lambda(\alpha, \beta|x, y) = p_\lambda(\alpha|x, y)p_\lambda(\beta|x, y).$$

Bell locality means there exist normalised local response distributions $r^{(1)}_\lambda(\alpha|x), r^{(2)}_\lambda(\beta|y)$ such that $p_\lambda(\alpha, \beta|x, y) = r^{(1)}_\lambda(\alpha|x)r^{(2)}_\lambda(\beta|y)$.

Proposition 13.3 (Bell locality $\Longleftrightarrow$ PI and OI)**.** *For the specified response probabilities, factorisation holds exactly when PI and OI both hold.*

Proof. Sum the factorisation over either outcome to identify its normalised local factors with the corresponding marginals and obtain PI; substituting those marginals gives OI. Conversely substitute the setting-independent marginals of PI into the OI product to obtain normalised local factors. ∎

The OI definition has no division by zero. When the conditioning marginal is positive, it equivalently says $p_\lambda(\alpha|\beta, x, y) = p_\lambda(\alpha|x, y)$, and similarly on the other side. A zero marginal already forces its associated joint probabilities to be zero by positivity. Setting independence was not proved by this equivalence and is still needed to infer the usual Bell bound. PI compares different remote settings at the same hidden state. OI instead compares outcomes

at the same pair of settings and hidden state. Neither comparison is the experimental averaging over hidden states. Jarrett's term "completeness" for OI is not L12's completeness C; we use PI/OI to keep that distinction visible.

For the ordinary quantum singlet model $\lambda = \psi^-$,

$$p(\alpha,\beta|x,y) = \tfrac{1}{4}(1 - \alpha\beta\,\mathbf{a}_x\cdot\mathbf{b}_y), \qquad p(\alpha|x,y) = p(\beta|x,y) = \tfrac{1}{2}.$$

Here $\mathbf{a}_x, \mathbf{b}_y$ are the two measurement axes of QM-I. Thus PI holds. At equal axes $p(+,+) = 0$, whereas the product of the marginals is $1/4$; OI fails. This is compatible with QM-I's proved no-signalling statement: the unconditional local marginal is unchanged by the remote choice. OI concerns conditioning on the remote outcome.

Recall the CHSH bound from QM-I, Lecture 12. For two settings on each side, define the observed correlations by

$$\mathcal{E}_{xy} = \sum_{\alpha,\beta=\pm1} \alpha\beta\; p(\alpha,\beta|x,y), \qquad |\mathcal{E}_{00} + \mathcal{E}_{01} + \mathcal{E}_{10} - \mathcal{E}_{11}| \le 2.$$

Bell locality and setting independence imply the displayed bound. The singlet has correlations $\mathcal{E}_{xy} = -\mathbf{a}_x\cdot\mathbf{b}_y$. For orthogonal unit coordinate axes, choose

$$\mathbf{a}_0 = \mathbf{e}_z, \quad \mathbf{a}_1 = \mathbf{e}_x, \qquad \mathbf{b}_0 = (\mathbf{e}_z + \mathbf{e}_x)/\sqrt{2}, \quad \mathbf{b}_1 = (\mathbf{e}_z - \mathbf{e}_x)/\sqrt{2}.$$

The four correlations are $-1/\sqrt{2}, -1/\sqrt{2}, -1/\sqrt{2}, +1/\sqrt{2}$, in that order. The absolute CHSH combination is therefore $2\sqrt{2}$.

A deterministic model may have outputs $A(x,y,\lambda), B(x,y,\lambda)$ depending on both settings. Its joint point distribution equals the product of its two point marginals, so OI holds automatically. If it also obeyed PI and setting independence, it would satisfy the CHSH bound proved in QM-I. Its pointwise bound averages unchanged over a convergent countable preparation sum, since the summands are bounded and the nonnegative weights sum to one. Hence a deterministic model reproducing the singlet violation must violate PI under those premises. Its averaged marginals can nevertheless be nonsignalling: PI conditions on a finer model state than the experimental average. "No Bell-local model" is not a theorem that one can send a controllable faster-than-light signal.

13.4 Bohm's additional positions and their current

For the spinless scalar theory, a wavefunction of N particles obeys

$$i\hbar\partial_t\psi = -\sum_k \frac{\hbar^2}{2m_k}\Delta_k\psi + V\psi, \qquad V \text{ real}, \quad m_k > 0.$$

It is accompanied by an actual configuration $\mathbf{q}(t)$, beyond the wavefunction. Where $\psi \neq 0$, the guiding equation is

$$\dot{\mathbf{q}}_k(t) = \frac{\hbar}{m_k}\,\mathrm{Im}\,\frac{\nabla_k\psi}{\psi}(t,\mathbf{q}(t)). \tag{13.3}$$

For smooth solutions, conjugating the Schrödinger equation and subtracting cancels the real potential and gives

$$\partial_t|\psi|^2 = -\sum_k \frac{\hbar}{m_k}\,\mathrm{Im}(\bar\psi\Delta_k\psi) = -\sum_k \nabla_k\cdot\mathbf{j}_k, \qquad \mathbf{j}_k = \frac{\hbar}{m_k}\,\mathrm{Im}(\bar\psi\nabla_k\psi).$$

The gradient-square term in the divergence is real and contributes no imaginary part. Therefore the velocity is $\mathbf{v}_k = \mathbf{j}_k/|\psi|^2$. This continuity equation suggests preservation of the $|\psi|^2$ distribution, called *equivariance*. We prove it on the line with explicit flow assumptions, rather than inferring global existence from a local differential identity. Electromagnetic vector potentials require the gauge-covariant current and are not included in this displayed model.

Theorem 13.4 (One-dimensional equivariance, conditional flow form)**.** *Let $\psi_s(x)$ be a normalised Schrödinger solution on $0 \le s \le t$, smooth enough for the continuity equation, with $\psi_s(x) \neq 0$ everywhere and jointly continuous $\partial_x v(s,x)$. Suppose there exists a family $X_s(a)$, continuous jointly in (s,a), solving $\dot{X}_s(a) = v_s(X_s(a))$, $X_0(a) = a$, throughout this slab for every $a \in \mathbb{R}$. Then a position initially distributed with density $|\psi_0|^2$ has density $|\psi_s|^2$ at each time s.*

Proof. We first show that the flow preserves the order of positions. Then the continuity equation preserves mass between two moving trajectories. Normalisation and strict positivity finally show that no final interval is omitted by the flow.

Trajectories cannot cross. On a compact tube up to a putative first meeting, $|\partial_x v| \le M$. Their positive gap $D_s = X_s(b) - X_s(a)$ obeys $\dot{D}_s \ge -MD_s$, so differentiation of $e^{Ms}D_s$ yields $D_s \ge (b-a)e^{-Ms} > 0$. Thus the flow is strictly ordered. For finite $a < b$, ordinary differentiation of the moving interval gives

$$\frac{d}{ds}\int_{X_s(a)}^{X_s(b)} |\psi_s(x)|^2 dx = -j_s(X_s(b)) + j_s(X_s(a)) + |\psi_s(X_s(b))|^2 v_s(X_s(b)) - |\psi_s(X_s(a))|^2 v_s(X_s(a)) = 0.$$

The endpoint terms cancel because $j = |\psi|^2 v$. Hence every such image interval carries exactly the initial mass between a, b. Let $a \to -\infty$, $b \to +\infty$. That initial mass tends to one, as does the total mass of the normalised ψ_s. The ordered image endpoints must therefore reach $-\infty, +\infty$: otherwise a nonempty excluded interval would have positive mass, since $|\psi_s|^2$ is strictly positive and continuous. The continuous ordered map is onto. For any final interval choose its endpoints' unique preimages. The moving-interval identity makes its probability under the transported initial position equal its $|\psi_s|^2$ integral. These interval probabilities determine the position distribution, proving equivariance. ∎

Global flow existence, nodes, singularities and the N-particle theorem are *cited*, not proved by the two-line continuity equation: Dürr–Teufel pp. 152–153; Tumulka, Theorem 1.3 pp. 28–29; Berndl et al., *CMP* **173**, 647–674 (1995), arXiv:quant-ph/9503013. Those results have regularity and potential hypotheses and generally assert almost-sure existence under quantum equilibrium, not existence for every arbitrary wavefunction or starting point. A stationary oscillator ground state gives a simple limiting case. Its spatial wavefunction is real and nowhere zero. Time evolution adds only a common phase, which cancels from the ratio of the spatial derivative to the wavefunction in the guiding equation. The velocity therefore vanishes. Each actual position stays fixed, and the stationary quantum position density is preserved. This does not say that every Gaussian packet has zero velocity.

The freely spreading Gaussian packet of QM-II supplies a nowhere-zero example with a linear velocity and global flow; its trajectory calculation is an example for Sheet 13, not an additional existence theorem assumed here.

13.5 What the theorems cost this theory

The additional position is an actual physical fact, so the wavefunction alone is not complete C in L12's sense. Quantum equilibrium specifies the initial configuration density $|\psi|^2$, and equivariance preserves it under the proved conditions. Extending agreement to general recorded observations requires modelling the apparatus configuration and its coupling; the scalar one-dimensional proof alone is not that general measurement theorem.

For a concrete pointer description, let x be the system coordinate and y a pointer coordinate. Suppose a specified apparatus interaction has produced the normalised outgoing wavefunction

$$\Psi(x,y) = a\psi_0(x)\varphi_0(y) + b\psi_1(x)\varphi_1(y), \qquad |a|^2 + |b|^2 = 1.$$

Here the system packets and pointer packets are unit vectors on the line. The pointer packets are smooth and have disjoint position supports. The actual apparatus has one pointer coordinate q_M. If it lies in the support of φ_0 with $\varphi_0(q_M) \neq 0$ and $a \neq 0$, the conditional system wavefunction $\Psi(\cdot, q_M)$ is proportional to ψ_0. The second term vanishes at that actual pointer position. A position in the other packet gives the other conditional branch.

In quantum equilibrium, integrating the joint position density over the first pointer support gives

$$\int_{\operatorname{supp}\varphi_0}\int_{\mathbb{R}} |\Psi(x,y)|^2 dx\, dy = |a|^2 \int_{\mathbb{R}} |\psi_0(x)|^2 dx \int_{\operatorname{supp}\varphi_0} |\varphi_0(y)|^2 dy = |a|^2.$$

Integrating over the second gives $|b|^2$. Thus the additional configuration identifies the actual pointer record, while equilibrium supplies its statistics. The full wavefunction still contains both terms and obeys the Schrödinger equation. This conditional branch description is effective while the pointer packets remain separated; it is not a physical GRW collapse. Real overlapping tails make the separation and the conditional branch identification approximate. The example explains the record mechanism for this assumed outgoing state, rather than proving that every apparatus produces it.

In Bohmian spin models, spin belongs to the spinor wavefunction and the apparatus interaction, not to an extra little vector attached to each particle. The velocity uses $(\hbar/m)\operatorname{Im}(\psi^\dagger\nabla\psi)/(\psi^\dagger\psi)$ in the corresponding model without a vector potential (cited: Dürr–Teufel, §8.4 pp. 158–161). A Stern–Gerlach outcome is a position record produced by that interaction. Different contexts can associate different outcomes to the same operator while preserving its quantum statistics. For two qubits, contextuality is required by the square's premises. For a single spin one half it is a feature of the Bohmian apparatus model, not forced by a nonexistent qubit KS contradiction. An optional cited illustration of the single-spin mechanism is Tumulka's inverted Stern–Gerlach example (§2.3.9 pp. 79–80, *cited*). It reverses the magnet and exchanges the output labels to preserve the spin calibration. The two arrangements can then assign opposite calibrated outcomes to the same initial wavefunction and configuration. This account uses its spinor apparatus model and approximately separated packets, with spatial/velocity assumptions and small tails neglected; it is not derived from the scalar flow theorem above.

For Bell's experiment the deterministic fine state includes wavefunction and configurations of both systems *and the relevant apparatus*. Settings remain independent experimental inputs in the Bell comparison. Averaging unknown apparatus positions without including them in λ need not leave a deterministic response model; whether it does depends on the

actual response functions. At the fully specified level OI holds, and reproducing the singlet violation forces PI to fail, while equilibrium averaged statistics remain nonsignalling. This is the precise model-relative locality cost, rather than a claim of controllable signalling or a premise shared by every proposed theory.

Pusey–Barrett–Rudolph's theorem is named only: under preparation independence for arbitrarily many independent copies and reproduction of the relevant quantum predictions, an ontological model (a hidden-state preparation/response model of the kind defined above) assigns non-proportional pure states non-overlapping hidden-state preparation distributions (ψ-onticity): in the countable case their overlap $\sum_\lambda \min\{\mu_\psi(\lambda), \mu_\varphi(\lambda)\}$ is zero, and in the density case the corresponding integral is zero (Tumulka, Theorem 5.6 pp. 212–213). It is not an assumption-free verdict about what a wavefunction must mean. The next lecture turns to changing U and changing D, on the same explicit-premise basis.

What does adding values cost?

Noncontextual values respecting commuting quantum products fail on the Peres–Mermin square. Bell locality is PI plus OI, with independent settings also needed in the Bell argument. Bohm adds configurations, uses contextual readings and reproduces equilibrium position distributions by equivariance; its deterministic Bell model violates PI while retaining nonsignalling averages. These statements specify the assumptions changed, not a verdict choosing one response to the measurement problem.

Summary. The square gives a finite state-independent contradiction and a statistic six against four. PI/OI separate the two parts of a factorised response without dividing by zero. The guiding current supplies a conditional one-dimensional equivariance proof with a genuine global distribution conclusion. Nodes and general global existence remain cited. Context and the full apparatus state matter when connecting those positions to readings.

Exercises. Try Sheet 13 in the separate *Exercise sheets* booklet (English).

Sources. Peres, *Quantum Theory: Concepts and Methods*, equation (7.1) p. 189, pp. 196–200; Bell, *Speakable and Unspeakable*, 2nd ed., ch. 1 pp. 2–4; Mermin, *PRL* **65**, 3373 (1990); Cabello, Estebaranz and García-Alcaine, *Phys. Lett. A* **212**, 183 (1996); Jarrett, *Noûs* **18**, 569–589 (1984), pp. 578–583; Dürr–Teufel, *Bohmian Mechanics*, pp. 152–153, 158–161; Tumulka, pp. 28–29, 79–80, 212–213. Experimental and global-existence primary sources are identified in the text.

Lecture 14

Collapse and branching: GRW and Everett

Objectives. *GRW changes universal unitary dynamics; Everett changes the requirement of one global outcome. We compute the exact and approximate claims behind a GRW localisation, and prove that relative branches reproduce L10's full reading-chain probabilities and conditional states. A common comparison then returns to the course's question: what data specify a quantum system, and which physical commitments remain open?*

- Derive GRW hit weights and a quantitative one-hit selection estimate, distinguishing exact expectations from approximate selection.
- Compute the labelled rigid-configuration damping rate and prove the finite reading-chain relative-state identity, including mixed inputs.
- Compare the three responses on equal questions and specify states, algebras, dynamics, composites and instruments consistently with superselection.

14.1 GRW: stochastic localisations without a meter

In the one-dimensional model, a wavefunction evolves by its Hamiltonian between randomly timed hits. Hits occur at Poisson rate λ_{GRW} and have operators

$$L_z = (\pi r_C^2)^{-1/4} e^{-(Q-z)^2/(2r_C^2)}, \qquad r_C > 0. \tag{14.1}$$

The continuous label z is a localisation centre, not an angular-momentum component. Gaussian integration gives $\int \|L_z\psi\|^2\, dz = \|\psi\|^2$. Indeed this is exactly L10's Gaussian pointer family with $g = 1$ and meter width $r_C/\sqrt{2}$, whose normalisation was proved there. The identity $\int L_z^\dagger L_z dz = \mathbb{1}$ means its positive scalar pairings, not an operator-norm integral.

GRW postulates the centre density $q(z) = \|L_z\psi\|^2$ and the selected state $L_z\psi/\sqrt{q(z)}$ where $q(z) > 0$, without a physical apparatus being present. Equality to a mathematical instrument family does not turn this stochastic law into a unitary measurement of an unobserved meter. Its nonunitary random hits are the change to U.

Proposition 14.1 (Exact branch weights). *Let $\psi = \sum_{j=1}^m \sqrt{p_j}\psi_j$, with unit branches having disjoint position supports, $p_j \geq 0$ and $\sum_j p_j = 1$. Put $a_j(z) = \left\|L_z\psi_j\right\|^2$. Expectations and probabilities below use the centre density $q(z)dz$. A hit gives branch weights*

$$w_j(z) = \frac{p_j a_j(z)}{\sum_k p_k a_k(z)}, \qquad \mathbb{E}_z[w_j] = p_j.$$

Proof. Multiplication by L_z preserves the support orthogonality, so $q(z) = \sum_j p_j a_j(z)$. The jth squared norm after normalising by q is the stated w_j. Averaging with the *centre density* gives $\int q(z) w_j(z) dz = p_j \int a_j(z) dz = p_j$. ∎

The post-hit branch shape is $L_z\psi_j/\sqrt{a_j(z)}$, not generally ψ_j. Preserving expected weights does not mean selecting one branch: when the packets are unresolved, the weights can remain nearly unchanged. The identity is an average over all hit centres. A particular centre can increase one branch weight and decrease another. To establish selection, most centres must strongly favour one branch. Here is the additional separation argument.

For one explicit hit, take two equal-weight packets with constant position densities on $[-\ell, \ell]$ and $[d-\ell, d+\ell]$, where $d > 2\ell$. Each nonzero density is $1/(2\ell)$. Discontinuous packet edges cause no difficulty for this bounded localisation operation. At centre $z = 0$, direct integration gives

$$a_0(0) = \frac{\operatorname{erf}(\ell/r_C)}{2\ell}, \qquad a_1(0) = \frac{\operatorname{erf}((d+\ell)/r_C) - \operatorname{erf}((d-\ell)/r_C)}{4\ell}.$$

Here $\operatorname{erf}(u) = 2\int_0^u e^{-v^2} dv/\sqrt{\pi}$. The posterior first-branch weight is

$$w_0(0) = a_0(0)/(a_0(0) + a_1(0)).$$

With $\ell = 0.01 r_C$ and $d = 3r_C$, it is approximately 0.9998765. The second branch remains present with a small positive weight. With the same width but $d = 0.1 r_C$, the first weight is only about 0.5024998. The same centre barely changes the unresolved equal weights. These are values of the continuous hit-response functions at the specified centre, not probabilities of the singleton event $z = 0$, which has probability zero. Nearby centres have nearby weights. Their occurrence probabilities must still be computed using the density $q(z)$. The comparison concerns a single hit; averaging all centres preserves each prior weight by the proposition above.

Suppose the finite branch supports lie in $[x_j - \ell, x_j + \ell]$, with centre separations at least d and $\ell < d/3$. Define disjoint centre regions $R_j = [x_j - d/3, x_j + d/3]$. The geometry is shown in Figure 14.1.

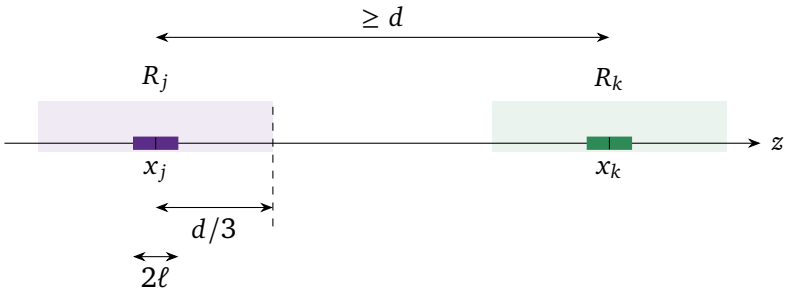


Figure 14.1: Thick segments contain the packet supports; shaded intervals are the centre regions. To leave its own region, a centre drawn from that packet must acquire Gaussian noise of magnitude at least $d/3 - \ell$. The drawing shows the support bound, not the full noncompact hit density.

Explicitly,

$$a_j(z) = \int |\psi_j(x)|^2 \frac{e^{-(z-x)^2/r_C^2}}{\sqrt{\pi} r_C}\, dx.$$

Conditional on the jth density a_j, the centre has the scalar distribution of a position in that packet plus Gaussian noise of density $e^{-u^2/r_C^2}/(\sqrt{\pi}r_C)$. Hence

$$\delta_j := \int_{R_j^c} a_j(z)dz \le \operatorname{erfc}\left(\frac{d/3-\ell}{r_C}\right), \qquad \delta := \sum_j p_j\delta_j \le \operatorname{erfc}\left(\frac{d/3-\ell}{r_C}\right).$$

Here $\operatorname{erfc}(b) = 2\int_b^\infty e^{-u^2}du/\sqrt{\pi}$ is just the Gaussian two-sided tail. These bounds follow from the support restriction: leaving R_j requires noise of magnitude at least $d/3-\ell$. For $0<\epsilon<1/2$ set $S_j = \{z \in R_j : w_j(z) \ge 1-\epsilon\}$. Then

$$p_j - \delta - \delta/\epsilon \ \le \Pr(S_j) \le p_j + \delta, \qquad \Pr\left(\bigcup_j S_j\right) \ge 1 - \delta - \delta/\epsilon. \tag{14.2}$$

We bound two ways for a hit to fail the selection criterion: its centre can lie outside all the regions, or it can lie in a region without giving that region's branch enough weight. To prove the estimates, since $R_j \subset R_k^c$ for $k \ne j$ and $q(1-w_j) = \sum_{k\ne j} p_k a_k$, leakage from all wrong packet densities into R_j is at most δ, so $p_j - \delta \le \Pr(R_j) \le p_j + \delta$. Also $\int_{R_j} q(1-w_j)dz$ is exactly that wrong-packet leakage. The probability in R_j where $1-w_j > \epsilon$ is at most this integral divided by ϵ. Summing over disjoint R_j gives total leakage at most δ, while centres outside their union have probability at most δ. These observations give both bounds in (14.2).

When the support gap satisfies $(d/3-\ell)/r_C \gg 1$, $\delta \to 0$: one branch has weight near one with high probability and its label probabilities approach p_j. No lower bound on the prior weights was assumed. The stronger sufficient condition $\ell \ll r_C \ll d$ also keeps a selected narrow packet nearly undistorted for typical hit centres; narrowness is not necessary for the weight-selection estimate alone. The exact statement requires disjoint support; overlapping wave-packet tails require additional errors. Since a Gaussian multiplier is strictly positive, a finite hit does not exactly delete every other nonzero branch. The treatment of those small tails and of physical records also involves the model's ontology.

14.2 The averaged channel and amplification

The unread one-hit map $\mathcal{T}(\rho) = \int L_z \rho L_z dz$ is L10's unread Gaussian pointer channel. Completing the square in its kernel gives

$$\mathcal{T}(\rho)(x,x') = e^{-(x-x')^2/(4r_C^2)}\rho(x,x').$$

Equivalently it is L11's proved random-unitary construction with kick density $r_C e^{-r_C^2k^2}/\sqrt{\pi}$. The identification follows from the same Gaussian Fourier integral, with the kernel/pairing scope proved in L10–L11. With free motion neglected the Poisson average is

$$\Phi_t(\rho) = e^{-\lambda_{\mathrm{GRW}}t}\sum_{n=0}^{\infty}\frac{(\lambda_{\mathrm{GRW}}t)^n}{n!}\mathcal{T}^n(\rho).$$

The nonnegative coefficients sum to one. Positive trace tails construct the sum, and the same construction after every finite spectator proves CP, as in L7. Thus it is a channel. Its kernel multiplier is the scalar exponential series

$$\exp\left[-\lambda_{\mathrm{GRW}}t\left(1-e^{-(x-x')^2/(4r_C^2)}\right)\right]. \tag{14.3}$$

It saturates at rate λ_{GRW} for resolved separations. It is not L11's purely quadratic diffusion kernel. Including a Hamiltonian gives the ensemble equation

$$\dot{\rho} = -\frac{i}{\hbar}[H, \rho] + \lambda_{\mathrm{GRW}}(\mathcal{T}(\rho) - \rho).$$

This full equation is *sketched*: unbounded Hamiltonian domains require additional work. The zero-free-motion channel and its kernel above were constructed directly, without invoking an unproved unbounded GKSL theorem.

For three spatial dimensions replace the prefactor in (14.1) by $(\pi r_C^2)^{-3/4}$ and the squared distance by $|Q - z|^2$. For N *labelled* constituents with independent rate λ_{GRW} per label, each hit changes only its coordinate. Multiplying their independent Poisson kernel factors gives damping rate

$$\Gamma(q, q') = \lambda_{\mathrm{GRW}} \sum_{i=1}^{N} \left(1 - e^{-|q_i - q_i'|^2/(4r_C^2)}\right). \tag{14.4}$$

Evaluating this continuous coefficient at matched rigid translations $q_i' = q_i + \mathbf{d}$ gives $N\lambda_{\mathrm{GRW}}(1 - e^{-d^2/(4r_C^2)})$, $d = |\mathbf{d}|$. Point configurations label the coefficient, not Hilbert-space states; using it for coherence between narrow normalised packets is an approximation. Deformations or rotations require the full sum of different displacements.

With conventional model choices $\lambda_{\mathrm{GRW}} = 10^{-16}\,\mathrm{s}^{-1}$, $r_C = 10^{-7}\,\mathrm{m}$, and a pointer with $N = 10^{23}$ labels at separation $d = 10^{-6}\,\mathrm{m}$, the total hit rate is $10^7\,\mathrm{s}^{-1}$ and the overlap factor is e^{-25}. The first-hit waiting time is about $10^{-7}\,\mathrm{s}$; it is also approximately the ensemble coherence time here. To apply the selection estimate in three dimensions, use for each label the slabs perpendicular to $\mathbf{d}$ in place of R_j. The Gaussian noise component along $\mathbf{d}$ has the same one-dimensional density, so the leakage proof applies to the full centre probabilities and global branch weights. For two rigid branches whose constituent parallel support half-width is $\ell = 0.01 r_C$, the present $d = 10 r_C$ gives $\delta \le \mathrm{erfc}(10/3 - 0.01) \simeq 2.60 \cdot 10^{-6}$. With $\epsilon = 0.01$, some branch has weight at least 0.99 with probability at least $1 - 101\delta > 0.9997$, after one hit, on the stated $10^{-7}\,\mathrm{s}$ mean first-hit scale. Compact support idealises real tails, which would add an error term. For $d \lesssim r_C$, the rate is smaller and a single hit need not resolve the alternatives.

14.3 Everett: relative states reproduce the reading chain

Everett retains unitary dynamics and treats outcomes as relative to branches, rather than D's unique global outcome. We prove the particular agreement between the descriptions: L10's probabilities and conditional states are already encoded in a unitary chain with retained records. The proof is finite dimensional, and does not assert a unique fundamental branch count.

Let the first instrument have finite Kraus operators $K_{j\alpha}$, the second $M_{k\beta}$, and let U be the intermediate system unitary. The labels j and k are the observed results. The labels α and β distinguish Kraus terms belonging to one result; these labels are also retained in the dilation probes. A history fixes j and k, while summing over α and β. Both Kraus lists have total completeness $\mathbb{1}$. L7's unitary dilations use fresh independently prepared pure probes, with orthogonal output labels $|j, \alpha\rangle$ and $|k, \beta\rangle$. These labelled probes include the dilation registers. The second coupling acts on the system and its fresh probe, leaving the first record untouched. Define the composed operation $\mathcal{J}_{jk} = \mathcal{I}_k' \circ \mathrm{Ad}_U \circ \mathcal{I}_j$, where $\mathrm{Ad}_U(X) = UXU^\dagger$.

Theorem 14.2 (Finite chain identity). *Immediately after these two readings, the global pure state has orthogonal history components Ψ_{jk} with weights $w_{jk} = \|\Psi_{jk}\|^2 = \operatorname{Tr} \mathcal{J}_{jk}(\rho)$. For positive w_{jk} its reduced relative system state is $\mathcal{J}_{jk}(\rho)/w_{jk}$. If ρ is mixed, purify it with a reference R; the partial trace defining that system state includes R as well as both probes and all dilation registers.*

Proof. We compute one history before normalising it. Orthogonal probe labels will turn its squared norm into the instrument probability and its partial trace into the instrument output. For pure input ψ the unnormalised history vector is

$$\Psi_{jk} = \sum_{\alpha,\beta} M_{k\beta} U K_{j\alpha} \psi \otimes |j,\alpha\rangle \otimes |k,\beta\rangle \,.$$

The total is $\sum_{jk} \Psi_{jk}$. Probe orthogonality gives $w_{jk} = \sum_{\alpha,\beta} \left\| M_{k\beta} U K_{j\alpha} \psi \right\|^2 = \operatorname{Tr} \mathcal{J}_{jk}(|\psi\rangle\langle\psi|)$. Tracing the probes removes their cross terms and leaves exactly

$$\sum_{\alpha,\beta} M_{k\beta} U K_{j\alpha} \, |\psi\rangle\langle\psi| \, K_{j\alpha}^\dagger U^\dagger M_{k\beta}^\dagger .$$

This is $\mathcal{J}_{jk}(|\psi\rangle\langle\psi|)$, proving the claim. For mixed input take a purification $\Omega \in \mathcal{H}_S \otimes \mathcal{H}_R$. Apply each Kraus product tensored with $\mathbb{1}_R$ to Ω in the displayed sum. Its squared norm is the same trace by the purification pairing. The partial trace over R and both probes yields $\mathcal{J}_{jk}(\rho)$, by that pairing or a finite spectral expansion. Tracing only the probes would instead leave an operator on $S + R$. Normalise only when $w_{jk} > 0$. Completeness gives $\sum_{jk} w_{jk} = 1$. ∎

For L10's actual capstone there is one Kraus operator per outcome: $K_j = E_j$ and the second operator is M_k. The four weights are $5/16, 3/16, 3/16, 5/16$, and the relative system vectors are exactly its four tabulated vectors. Replacing the first instrument by $K_j = |{+x}\rangle\langle j|$ instead gives $(4 - k\sqrt{3})/16$ and the reset final states. The theorem therefore reproduces both full sequences, not just ideal repeatable measurements.

Let Π_h project onto each history's orthogonal record sector. For any bounded A commuting with every Π_h, cross terms vanish and

$$\langle \Psi, A\Psi \rangle = \sum_{h : w_h > 0} w_h \left\langle \Psi_h/\sqrt{w_h}, A\Psi_h/\sqrt{w_h} \right\rangle .$$

A subsequent unitary commuting with these record projections preserves the weights, though it can evolve the within-branch states. The theorem's conditional-state identity refers to the end of the specified chain; further dynamics must be included in the composed operation if compared. Actual macroscopic decoherence gives approximately robust records, not a uniquely specified exact branching decomposition for every situation.

A record-erasing unitary is outside that record-preserving premise. L12's inverse CNOTs restore system σ_x mean one from the coherent three-qubit state and zero from its incoherent mixture. Unitarity cannot merge the orthogonal total states: the inverse CNOTs send $|000\rangle$ to $|000\rangle$ and $|111\rangle$ to $|100\rangle$. The record factors reset while the two system alternatives stay orthogonal, so the incoherent mixture retains zero σ_x mean. Every surviving external copy must also be coherently controlled to recover that interference.

14.4 Weights, probabilities and the same comparison questions

The identity uses squared norms and the instrument probabilities already defined by Born's rule. It does not derive the interpretation of those weights as probabilities from unitary dynamics alone, or select one global branch. Wallace's decision-theoretic programme is a named representation argument with additional assumptions: sufficiently rich available acts, coherent preferences across decisions, continuity, preferences depending only on the resulting state (state supervenience), macrostate indifference, and indifference to branch splitting that leaves rewards unchanged (§5.6 pp. 174–181; theorem discussion §5.7 p. 182 onward). Its conclusion is reported under those axioms, not as a proof from U alone. Here rich available acts mean that the compared decisions can actually be implemented within the theorem's class of transformations. Coherent preferences require consistent choices when the same decision is divided into stages. Continuity controls what happens under small changes of an act. State supervenience says that two acts with the same resulting state have the same preference. Macrostate indifference removes a preference for microscopic differences within one rewarded macrostate. Branch-splitting indifference removes a preference for extra branches when their rewards are unchanged. These are decision assumptions in addition to unitary dynamics, rather than consequences of the chain identity. Counting branches is not supplied as an alternative probability law.

Locality also needs the model specified. GRW does not inherit PI merely from calling its wavefunction complete. For separate noninteracting labs, averaging a remote hit gives a local TP map; its centre probability cancels the conditional normalisation, leaving the other lab's marginal unchanged (cited: Tumulka, §5.5.9 pp. 232–234). At the predictive wavefunction-state level this gives PI under the stated setting assumptions; Bell-violating correlations then require OI to fail. Conditioning on future realised collapse centres would define a different fine description. The selected wavefunction changes nonlocally even when averaged marginals do not signal. Everett likewise retains the Bell correlations; rejecting a unique global outcome is not construction of a classical factorising single-outcome model (Wallace, §8.7 p. 310).

Question	Bohm	GRW	Everett
State	Wavefunction plus configuration	Stochastic wavefunction	Unitary state with relative records
Ontology	Configuration and wavefunction	GRWm matter density or GRWf flashes	Universal wavefunction
Dynamics	Schrödinger and guiding flow	Unitary intervals and random hits	Universal unitary dynamics
Records	Actual apparatus configuration	Localisation with small tails; GRWm/f account	Branch-relative records
Probabilities	Quantum equilibrium, equivariant	Postulated centre law	Norm weights; interpretation needs further argument
Locality	Fine deterministic PI fails; equilibrium nonsignalling	Noise-averaged nonsignalling; conditional changes nonlocal	Usual Bell correlations; no unique-global-outcome model assumed
Tests	Equilibrium quantum predictions	Additional isolated noise and coherence loss	Standard unitary predictions
Main change	C	U	D

The table compares the stated models, not every possible modification. GRW's record ontology and residual tails need their own account; a large weight is the proved approximate selection criterion, not exact deletion. Bohmian nonequilibrium variants and modified unitary theories would have additional empirical claims beyond this comparison.

14.5 The quantum-system specification

Definition 14.3 (The course's system card)**.** A quantum-system specification gives a nonzero separable Hilbert space, its admitted observable algebra and normal states, allowed dynamics and symmetries, the composite rule, and the instruments used for observations.

For the full algebra $\mathcal{B}(\mathcal{H})$, states are exactly L1's density operators and pure states are L2's rays. With the two-sector algebra of L5, states instead have the unique block-diagonal representatives proved there; its pure states are single-sector rays. Unbounded observables must have spectral projections in the admitted algebra. State and purity are relative to that algebra, not to a choice of Hilbert-space representative alone.

Allowed symmetries must respect this specification. Within a coherent sector Wigner supplies unitary or antiunitary ray implementations. Sector-preserving ones act within sectors; more general algebra-preserving discrete symmetries may permute equivalent sectors. For example, swapping two isomorphic blocks sends $A_1 \oplus A_2$ to $A_2 \oplus A_1$ and preserves the block algebra without fixing its sector projections. Which such maps are physical symmetries is additional data. The baseline closed evolution is a sector-compatible unitary group under L5's continuity and lift premises; GRW deliberately changes that dynamical law.

Composites use L6's Hilbert tensor product. Their observable algebra must also be specified: imposing each part's sector restriction and imposing only a total conserved charge are different choices. For systems without superselection the course used the full composite algebra. A part initially prepared independently of its environment has the channel description proved in L7; the finite semigroup model has L8's GKSL structure.

For a worked L5-type card, take $\mathcal{H} = \mathbb{C} \oplus \mathbb{C}^2$, $\mathcal{A} = \mathbb{C} \oplus M_2$, $\rho = 1/3 \oplus (2/3)\,|+x\rangle\langle+x|$ and $H = 0 \oplus (\hbar\nu/2)\sigma_z$. Time evolution is the corresponding block unitary. The sector instrument $\mathcal{I}_j(\rho) = E_j \rho E_j$ has effects $E_j \in \mathcal{A}$, probabilities $1/3, 2/3$, and admissible conditional states 1 and $|+x\rangle\langle+x|$ at this initial time. With an ordinary qubit partner, choose composite algebra $M_2 \oplus M_4$ on the two composite sectors. This specifies the algebra as well as the tensor space.

Effects alone do not ensure that an operation descends to admitted states. On the simpler algebra $\mathbb{C} \oplus \mathbb{C}$, the ray $|+x\rangle\langle+x|$ and $\mathbb{1}_2/2$ represent the same restricted state. The one-outcome channel with $K_0 = |0\rangle\langle+x|$, $K_1 = |1\rangle\langle-x|$ has total effect $\mathbb{1}$ in that algebra, yet sends those representatives to $|0\rangle\langle0|$ and $\mathbb{1}_2/2$. Their later sector probabilities differ. Hence an instrument must act unambiguously on the admitted states, with admissible outputs, not merely have effects in the algebra. Operations defined directly on block-diagonal states avoid that representative dependence; permitted sector transfers are additional physical data, not a requirement that each Kraus operator commute with the sector sign.

This common operational structure leaves L12's C/U/D question open. Bohm adds configurations; GRW changes dynamics and specifies record ontology; Everett changes the account of global versus relative records. The card states what each proposal must specify and what its predictions can be compared against, without declaring one interpretation a theorem.

What is a quantum system, and what remains open?

The observable algebra and its normal states specify expectations and events; tensor products specify composites, compatible dynamics specify change, and instruments specify observations. On the full algebra the states are density operators; superselection changes their identification and purity. The formalism predicts complete reading-chain statistics, but universal unitarity, completeness and one global outcome cannot all hold in L12's calibrated model; which premise to change is a further physical or interpretive choice. The three responses were compared by the same mathematical and experimental questions.

Summary. GRW's expected branch weights are exact, whereas one-hit selection needs separation and a tail estimate. Its Poisson jump kernel gives labelled amplification, with explicit modelling limits. Everett's finite branch weights and reduced relative states reproduce the full capstone, including a mixed input's reference trace. That identity leaves the probability interpretation as a further argument. The system card keeps states, algebras, dynamics and instruments together and makes the remaining commitments explicit.

Exercises. Try Sheet 14 in the separate *Exercise sheets* booklet (English).

Sources. Tumulka, *Foundations of Quantum Mechanics*, pp. 104–109, 124–129, 232–234 (his width convention has $r_C = \sqrt{2}\sigma$; our GRW/Bassi choice $r_C = 10^{-7}$ m corresponds to $\sigma \simeq 7.1 \cdot 10^{-8}$ m); Ghirardi–Rimini–Weber, *PRD* **34**, 470 (1986); Bassi et al., *RMP* **85**, 471 (2013). Wallace, *The Emergent Multiverse*, pp. 94, 174–182, 308–311; Griffiths, *Consistent Quantum Theory*, pp. 137–147, 216–219. The explicit tail bound and finite capstone identity are supplied here.